Tools to Design a Biomimetic Humanoid Robot Actuated by Pneumatic Artificial Muscles and a Synthetic Neural Network

by

Ben Bolen

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in
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Dissertation Committee:
Alexander Hunt, Chair
Bin Jiang
Radhika Reddy
David Turcic

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Abstract

Biomimetic robots offer a platform for studying neuromechanical control through experiments that would be impractical or unethical with living subjects. This dissertation advances a long-running effort at Portland State University to build a humanoid robot, from the trunk down, actuated by pneumatic artificial muscles and controlled by a synthetic neural network. The isometric force of Festo braided pneumatic actuators (BPAs) was characterized over a wide range of resting lengths, contractions, and pressures using a custom test jig. The analysis produced novel equations for the maximum isometric force as a function of resting length and a three-coefficient nondimensional force surface that predicts isometric force from relative strain and relative pressure. The actuator model was carried to the joint level through isometric knee torque experiments on two test stands: a simplified pinned knee and a biomimetic knee with a migrating instantaneous center of rotation. This approach isolated the sources of prediction error, revealing constant length offsets, series compliance in brackets and artificial tendons, and a loss of usable actuator length when BPAs wrap around the joint. The biomimetic knee driven by optimized BPA placements was shown to meet or exceed human isometric knee torque over most of the range of motion. Tools were created for curating relevant neuromechanical research, as well as simulating neuromechanical models based on human and robot biomechanical models.

Dedication

*

Darian

azīzam, jānam,

nūr-e cheshmam — the light of my eyes

qorbānat beram

Bābā kheyli dūset dāre

↘ ↘ ↘

"Los corazones obreros que laten

por nuestra causa, felices serán.

Si entusiasmados y unidos combaten,

de la victoria, la palma obtendrán."

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Finally, I thank my family for their support throughout this long process.

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1 Introduction

1.1 Motivation

Biomimetic robots provide a platform for investigating how mechanical structure and control interact to generate movement, offering complementary insights to simulation-based studies [Andrews, 1985, Millard et al., 2013, Arnold et al., 2010]. Wheeled robots remain the platform of choice in structured environments, where their speed, efficiency, and payload capacity are difficult to match; in the unstructured terrain of the natural and built world, legged machines hold the advantage, and biomimetic legs that reproduce human force capabilities have the potential to excel where wheels cannot [Frey et al., 2026, Ha et al., 2025]. Musculoskeletal humanoids and limb systems driven by tendon-like actuators have been shown to support biologically meaningful experiments on motion generation, internal force transmission, and neural control hypotheses [Asano et al., 2017, 2019, Hitzmann et al., 2018, Liu et al., 2018, Chen et al., 2019]. The dual perspective of using robots to learn about biology, and using biology to design more adaptive robots, has been emphasized in both biomimetic robotics and biomechanics [Ijspeert, 2020, Asano et al., 2017]. With such robotic systems, experiments can be performed that would not be practical or ethical with human test subjects, such as removing a specific muscle from the system or blocking a sensory feedback pathway. Experiments of this kind allow direct testing of how the components of the neuromechanical system interact, and they offer new opportunities for testing neural control theories that are faster and cheaper than human

observation studies [Shin et al., 2018, Goldsmith et al., 2021, Schilling et al., 2022].

Artificial muscles are a key enabling technology for this class of robot. Among the available actuation technologies, McKibben-style braided pneumatic actuators (BPAs) combine low mass, high force-to-weight ratio, and intrinsic compliance with a force–length curve that qualitatively resembles that of skeletal muscle [Liang et al., 2020, Hunt et al., 2017a]. BPAs have been used in quadrupedal robots [Aschenbeck et al., 2006, Hunt et al., 2017a, Scharzenberger, 2019], musculoskeletal humanoids [Asano et al., 2017, Hitzmann et al., 2018], and dexterous manipulation systems [Chen et al., 2019]. However, BPAs also differ from biological muscle in important ways: their maximum tensile force occurs at resting length rather than at an optimal fiber length, their maximum contraction ratio is substantially lower than that of human muscle, and their force output is a highly nonlinear function of internal pressure, contraction, and loading state [Hunt et al., 2017a, Sárosi et al., 2017, Liang et al., 2020]. The author's experience with 10 mm actuators made this gap concrete: manufacturer specifications overstate the force and contraction available at the short resting lengths a leg design requires, and designs based on those specifications delivered less range of motion and joint torque than intended. Nor could the published empirical models repair the discrepancy, because each is fitted to measurements at a single resting length [Sárosi et al., 2017, Martens and Boblan, 2017, Hunt et al., 2017a]; none accurately predicts how force changes when a designer chooses a different actuator length. Because muscle resting length is one of the primary degrees of freedom available to the robot designer, the absence of a force model valid at arbitrary resting length blocked rational actuator selection. Designing a biomimetic robot around BPAs therefore requires quantitative, predictive models of the actuator and of the joint torques it can produce.

Actuator force alone, however, does not determine the torque a joint produces. The robot knee transmits BPA force to the skeleton through mounting brackets, artificial tendons, and paths that wrap over the joint, and each of these elements departs from the rigid-body idealization: brackets and fixtures deflect under load, tendons stretch, and a wrapping actuator loses usable length as the knee flexes. At the start of this work it was unknown how much these transmission effects reduce the torque delivered at the knee relative to the rigid-body prediction, and no method existed for correcting that prediction in a form that would transfer between joint configurations. The biomimetic knee's optimized muscle routing [Morrow et al., 2020] compounded the problem: it had never been validated experimentally or compared against human isometric torque data, so even the nominal torque envelope of the robot leg rested on unverified calculations.

The remaining gaps concern the controller. No simulation pipeline existed for closing the loop between a musculoskeletal robot model of this kind and a biologically realistic neural controller. The neuromechanical modeling literature provides the ingredients, including two-level spinal central pattern generator architectures [Rybak et al., 2006, McCrea and Rybak, 2008], neuromechanical simulation frameworks [Markin et al., 2010], and synthetic nervous system tooling [Nourse et al., 2023], but it does not provide their assembly for a BPA-driven bipedal robot with human-like muscle routing. Wiring realistic feedback into such a controller exposes a fourth gap, this one in the literature rather than in methods: the physiological properties of the group Ia, group Ib, and group II afferents that should inform the feedback design are scattered across hundreds if not thousands of primary papers and reviews [Van de Crommert et al., 1998, Windhorst, 2007, 2022], with no structured, model-oriented resource from which a robot designer could draw them.

Four gaps thus stood between the existing robot design and a robot that can walk under neural control: no actuator force model valid at arbitrary resting length, no quantified correction for the torque lost through brackets, tendons, and wrapping, no closed-loop simulation pipeline joining the robot's musculoskeletal model to a biologically realistic neural controller, and no consolidated resource for the afferent data that such a controller requires.

1.2 The Biomimetic Humanoid Robot Project

This dissertation is part of a long-running effort at Portland State University to build a biomimetic humanoid robot from the trunk down, actuated by pneumatic artificial muscles and controlled by a synthetic neural network (SNN). The project is deliberately iterative. Knowledge from biomechanics and neuroscience guides the design of robots intended to perform complex tasks in unstructured environments with improved efficiency, balance, adaptability, and resilience. Those robots then become physical models with which biological hypotheses can be tested, validated, and refined through controlled changes that would be impractical or unethical in living subjects. The resulting biological insight informs the next generation of robot design, and lessons from both directions can be translated into improved prosthetic and orthotic devices.

The present work builds directly on a series of studies by the author and collaborators. In the author's master's project, the kinematics and dynamics of a biomimetic robot leg were matched against the Gait2392 OpenSim model, artificial muscle attachment and wrapping locations were determined for the robot legs, and MATLAB tooling was developed for

evaluating muscle lengths, moment arms, and isometric joint torques over the full range of motion [Bolen, 2019, Bolen and Hunt, 2019]. That work showed that simply transplanting human muscle attachment locations onto a BPA-actuated robot does not reproduce human torque profiles; the placement optimization that closes this gap was completed during this dissertation and is summarized in Section 8.1. In parallel, a biomimetic four-bar knee linkage with a migrating instantaneous center of rotation was designed and validated by Steele [Steele et al., 2017, Steele, 2018, Steele et al., 2018], and a canine-inspired quadruped platform, Muscle Mutt, formerly called DoggyDeux, was developed with antagonistic BPA pairs and synthetic neural network control [Scharzenberger, 2019]. That platform extended an earlier AARL demonstration of central pattern generator control of a pneumatically actuated robot [Hunt et al., 2017b].

1.3 Research Objectives

This dissertation addresses the first two gaps experimentally, resolves the fourth by consolidating the afferent literature into the Sensory Afferent Database [Bolen et al., 2023, Bolen and Hunt, 2024], and develops the tools and framework that close the third. The objectives are:

1. **Characterize the isometric force of Festo BPAs** with uninflated diameters of 10 mm and 20 mm over a wide range of resting lengths, contractions, and pressures; derive improved maximum-force and normalized force–strain–pressure models; and compare them against existing models (Chapter 3).

2. **Validate and improve the prediction of isometric knee torque** produced by BPAs

on pinned-joint and biomimetic four-bar knees; identify and model losses caused by constant length offsets, series compliance, and actuator wrapping; and compare the resulting torque envelopes with human capabilities (Chapter 4).

3. **Consolidate the sensory afferent literature into a structured, model-oriented resource**, the Sensory Afferent Database, by auditing and curating the published record of afferent properties across animal species and their role in motor control, and by evaluating AI-assisted literature triage for expanding the corpus (Chapter 6).

4. **Develop tools and a framework for neuromechanical control** of the humanoid robot by converting its OpenSim musculoskeletal model to MuJoCo, implementing an SNS-Toolbox two-level spinal CPG with proprioceptive and heel/toe contact feedback, and establishing a staged tuning and verification workflow (Chapters 7 and 8).

1.4 Dissertation Organization

Chapter 2 establishes the technical and scientific context for the dissertation. It reviews biomimetic and musculoskeletal robots, robot actuation technologies with an emphasis on braided pneumatic actuators and published models of their force–length–pressure behavior, and the human lower-limb biomechanics and OpenSim benchmark models that serve as design targets. It then covers the computational neuroscience of locomotor control, from two-level central pattern generator architectures and sensory afferent feedback to the reference circuits from which the dissertation's controller architecture is drawn. Two further topics frame the simulation work: neuromechanical simulation as a

concept, and the tooling used to execute it, from OpenSim and its SimTK ecosystem to MuJoCo, SNS-Toolbox, and AnimatLab. The chapter closes with the ethical case for testing neuromechanical hypotheses on robots rather than living subjects.

Chapter 3 develops and validates the actuator force models. It presents the isometric force characterization of Festo BPAs with 10 mm and 20 mm uninflated diameters over a wide range of resting lengths, contractions, and pressures; derives improved maximum-force and normalized force–strain–pressure models; and compares them against the manufacturer's tool and published models. Later sections examine the effects of resting length, the implications of the validated models for real-time control and simulation, and the biomimetic analogy between braided pneumatic actuators and skeletal muscle. The chapter's text and figures are adapted from the author's publication of this work [Bolen et al., 2026].

Chapter 4 carries the actuator models to the joint. It describes the robot leg model, the pinned-joint and biomimetic knee test stands, and the equipment used to collect isometric torque data; formulates the hybrid torque calculation method and the improved torque model whose correction terms account for a constant length offset, series compliance, and actuator wrapping; and documents the optimization algorithms used both to identify those terms and to redesign actuator routes. The chapter then reports the measured torque on both knees, the follow-up identification, the redesigned routes evaluated on the Gait2392 robot-body model, and the comparison of the resulting torque envelopes with human capabilities, before closing on the physical sources of prediction error and the transmission-design implications of mono- and bi-pulley routing.

Chapter 5 reports preliminary work on balance. It documents the inverted-pendulum

testbeds, which reduce the robot to a single actuated ankle joint: a first generation with a hinge ankle and a reverse-pulley transmission driven by an antagonist BPA pair, a second generation with a ball-and-socket ankle and a three-muscle arrangement, and a final platform that oscillates the support surface beneath the foot. It then summarizes the balance demonstrations performed under pseudo-vestibular inertial feedback, on flat ground and on the moving support, and their role as a baseline for the full-body control work of Chapter 8.

Chapter 6 presents the Sensory Afferent Database, the structured, model-oriented resource built to answer the afferent-literature gap. It describes the audited corpus and its curation coverage, in which papers carry structured curation notes, afferent classifications, and notes translating each sensory pathway toward robot or simulation use; the packaging of the corpus with its citation graph as a self-contained laboratory tool; and the evaluation of AI-assisted literature triage for expanding the database.

Chapter 7 assembles the neuromechanical simulation capability. It opens with the preliminary work that motivated its verification-first approach, including the author's prior AnimatLab bipedal walker, whose hand-tuned two-layer CPG produced suspended stepping but whose afferent feedback did not generalize to weight-bearing motion. It then documents the OpenSim-to-MuJoCo conversion of the robot's musculoskeletal model with its synthetic BPA knee, the Simulink SNS library built to prototype reflex topologies, and the staged tuning and evaluation workflows. The results sections report cross-platform verification of the contact-driven rhythm core, the BPA-actuated walker in MuJoCo, a gait-validation sweep against a locomotion reference library, a comparison of the production walker variants, and spiking-neuron mirrors of the non-spiking models. The chapter closes

with an analysis of what spiking does and does not change.

Chapter 8 discusses the work as a whole and the programs it enables. It summarizes the completed foundation studies, synthesizes the actuator, transmission, joint, and controller results into a unified actuator-to-controller design methodology, and presents the assembled simulation pipeline together with the two-layer CPG and its full sensory feedback. Building on that foundation, it defines the future work of the program: extension of the sensorium with vestibular, ocular, and cerebellar contributions; physical realization on a treadmill robot with embedded real-time neural control; and progression toward dynamic whole-body maneuvers. Chapter 9 consolidates the principal contributions of the dissertation.

Appendix A documents the muscle force formulations and their cross-platform comparison, the BPA force model implementation, the muscle-path length and moment-arm calculations, and the compliance and wrapping-loss implementations. Appendix B provides the neuron and synapse equations, spiking and non-spiking alike, used by the synthetic nervous system tools, relates those equations to their SNS-Toolbox implementation, and records the spiking-mirror calibration and the connectomes of the spinal-network models. Appendix C records the static BPA pathways, joint transformation matrices, bracket reference frames, stiffness and compliance parameters, adopted correction-term values, and the redesigned extensor route geometry.

2 Background

This chapter reviews the established knowledge against which the dissertation's contributions are defined: biomimetic and musculoskeletal robotics, the actuation technologies available to them, the human lower-limb biomechanics they imitate, the neuroscience of locomotor control they implement, and the simulation and ethical context in which they are tested. Work from the author's own program that goes beyond this background, including the prior AnimatLab bipedal walker, the Sensory Afferent Database, and the neuromechanical simulation pipeline, is presented as dissertation work in Chapters 6 and 7 rather than repeated here.

2.1 Biomimetic and Musculoskeletal Robots

Biomimetic robots offer many potential applications, ranging from assisting in factories and environmental exploration to serving as physical models of human movement. By developing high-fidelity biomimetic robots, we can perform tests that help improve our understanding of both biomechanics and the underlying neuromechanical systems that control movement [Shin et al., 2018, Asano et al., 2019, Ijspeert, 2020, Goldsmith et al., 2021, Hunt et al., 2017b, Schilling et al., 2022]. A few such biomimetic robots exist, such as the impressive Kengoro and Kenshiro robots of Asano et al. [2017] and Asano et al. [2019], which have proportionally human bodies, anatomically correct musculotendon arrangements, rolling-sliding joints, and joint ranges and output power comparable with human-measured values; their artificial muscles are implemented with brushless DC

motors, wires as tendons, gears, and integrated tension measurement. Kengoro, the more recent of the two, stands 1670 mm tall, weighs 56.5 kg, and carries 174 degrees of freedom when the end effectors are included [Asano et al., 2017]. A dog-like robot developed by Aschenbeck et al. [2006] and actuated with Festo BPAs was designed from a planar kinematic model scaled to an adult greyhound, with three sagittal degrees of freedom per leg driven by antagonistic BPA pairs; its front-leg pair alone lifted a 13.5 kg payload, almost double the complete robot's predicted mass. That platform was used to show how central pattern generators provide robust locomotion patterns that are adaptive to perturbations in ground speed [Hunt et al., 2017b].

Robots built with artificial muscles thus provide opportunities to investigate how muscles are controlled to produce desired motions; however, artificial muscles do not exactly replicate muscle properties, and existing robots do not capture all of the degrees of freedom and joint torques of the animal limbs they mimic. The biomimetic leg developed and tested in this dissertation sits squarely in this lineage: a human-proportioned skeleton, muscle-like actuators, and the goal of human-matched joint torques.

Figure 2.1 illustrates the human-scale skeletal structure and tendon routing of this approach.

2.2 Robot Actuation Technologies

2.2.1 Overview

A wide range of actuation technologies have been explored in robotics, including electric motors, hydraulic systems, smart-material actuators, and artificial muscles [He and Gao,

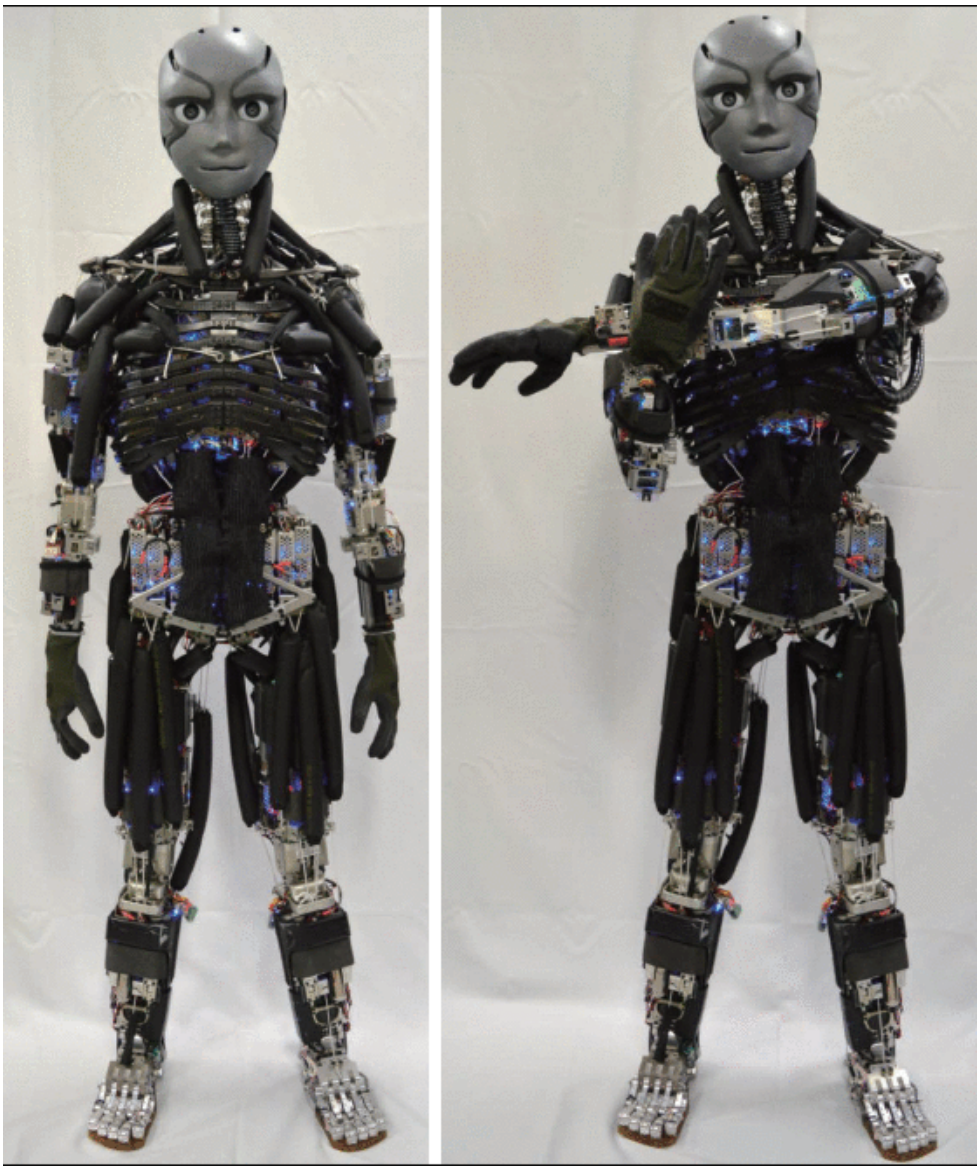


Figure 2.1: Kengoro human-mimetic musculoskeletal humanoid in neutral and upper-limb poses, illustrating its human-scale skeletal structure and densely routed tendon-driven musculature [Asano et al., 2017].

2020, Liang et al., 2020]. Electric motors are widely used due to their controllability and maturity, but they are limited by thermal constraints, gearing requirements, and reduced compliance compared to muscle-like actuators [Veale and Xie, 2016, Suzumori and Faudzi, 2018]. Hydraulic actuators offer exceptionally high power density and have enabled highly dynamic legged systems, but they remain relatively heavy, less efficient than biological muscle, and require a dedicated fluid infrastructure [Suzumori and Faudzi, 2018].

Among smart-material actuators, dielectric elastomer actuators (DEAs), a class of electronic electroactive polymers, contract by a different mechanism: an applied voltage squeezes a soft polymer film between compliant electrodes through electrostatic stress, and the thinning and radial spreading of the film produces strains that can exceed biological muscle [Madden et al., 2004]. Comparative studies report high specific work and good bandwidth for DEAs [Liang et al., 2020], but kilovolt-range driving voltages, electromechanical instability, and the compliant-electrode requirement have kept them out of legged robots. Thermally activated artificial muscles reach the most extreme performance figures of any technology surveyed by Liang et al. [2020]: twisted-fiber muscles support loads more than one hundred times those of a biological muscle of the same length and weight, with output stresses of 22 MPa. Their activation route, however, is shared with shape-memory alloys, whose heating is fast but whose cooling is limited by heat transfer away from the element, and that asymmetry sets the cycle time.

Closer to muscle, series-elastic and pneumatic actuators trade peak force and bandwidth for compliance, a trade that wearable and rehabilitation robotics exploit because force control and backdrivability matter most when a machine interacts with a human [Aksöz et al., 2019, Leibach, 2020, Li et al., 2020]. A recent review of pneumatic artificial

muscle technology documents continued progress in actuator design and in assistive applications [Zhagiparova et al., 2025]. None of these technologies yet combines muscle-like force, strain, and compliance in one actuator, which is the combination a biomimetic humanoid leg demands and that braided pneumatic actuators approximate most closely.

Figure 2.2 compares the operating mechanisms of these actuator technologies.

2.2.2 Braided Pneumatic Actuators

In contrast to the technologies above, McKibben-style braided pneumatic actuators (BPAs) combine low mass, high force-to-weight ratio, and intrinsic compliance with a force–length curve that qualitatively resembles that of skeletal muscle [Liang et al., 2020, Hunt et al., 2017a]. A BPA consists of an inflatable bladder inside a braided sleeve; pressurizing the bladder expands the bladder radially and contracts the sleeve axially, producing tensile force. As shown by Chou and Hannaford, the force–length relationship of McKibben actuators is governed by geometric constraints imposed by the braided shell, leading to a nonlinear dependence of force on actuator length through the braid angle [Chou and Hannaford, 1996]: actuator force is proportional to pressure and to a nonlinear geometric term $(3\cos^2\theta - 1)$, which governs both the maximum contraction limit and the curvature of the force–length relationship. Predicting their force as a function of length and pressure is therefore the first improvement of the torque tools developed in this dissertation, and the state of that prediction is the subject of the next subsection.

Figure 2.3 shows the commercial actuator assembly used in the laboratory.

However, BPAs exhibit several important differences from biological muscle. First,

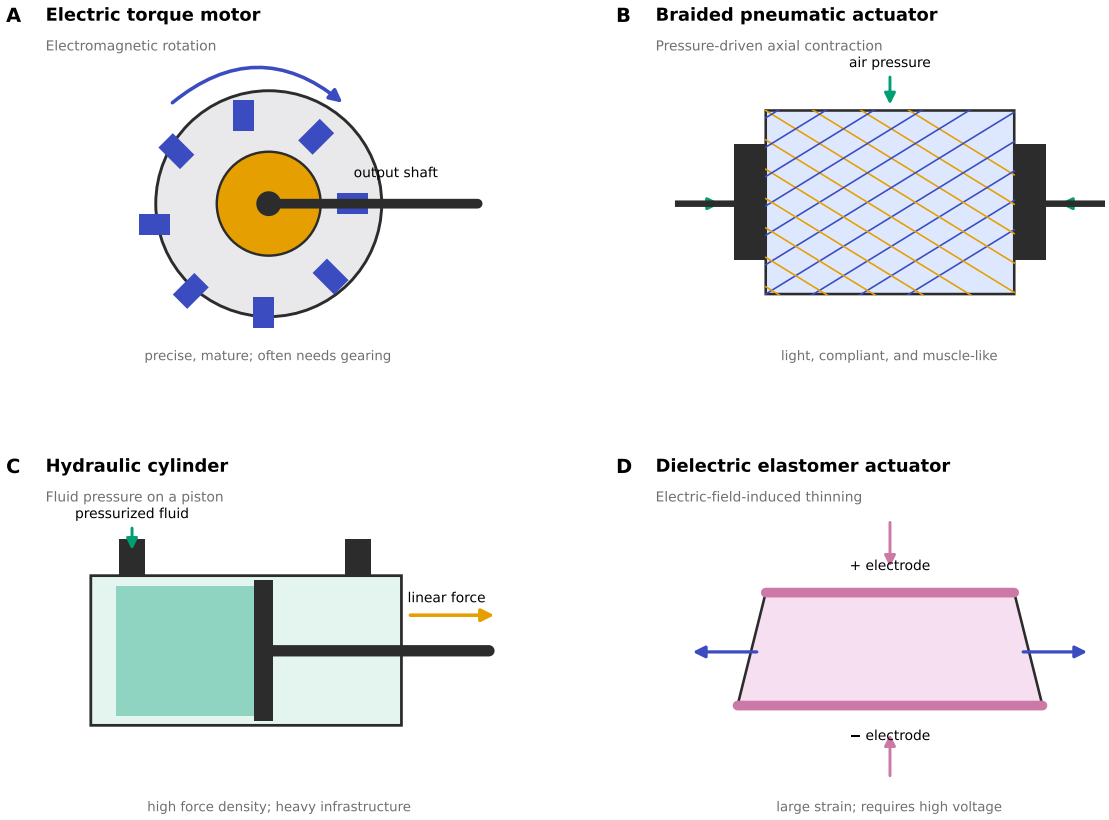


Figure 2.2: Representative operating mechanisms for robotic actuators. **(A)** An electric torque motor produces controlled shaft rotation. **(B)** A braided pneumatic actuator shortens as internal pressure expands its sleeve radially. **(C)** A hydraulic cylinder converts fluid pressure into linear piston force. **(D)** A dielectric elastomer thins and expands laterally under an applied electric field. The panels use a consistent schematic style to emphasize the distinct energy-conversion mechanisms and design tradeoffs summarized in the text [Suzumori and Faudzi, 2018, Liang et al., 2020].

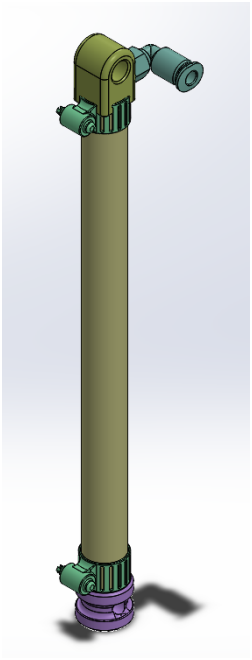


Figure 2.3: CAD representation of a Festo DMSP braided pneumatic actuator with the mounting and pneumatic connections used in the laboratory assembly [Festo AG & Co. KG, 2026]. The long cylindrical body contains the bladder and braided sleeve; pressurization through the end fittings converts radial expansion into axial contraction.

BPA maximum tensile force occurs at resting (i.e., uninflated) length (l_0) rather than at an optimal fiber length [Liang et al., 2020]. Second, their maximum contraction ratio is substantially lower than that of human muscle, presenting challenges for achieving biomimetic joint ranges of motion [Hunt et al., 2017a, Sárosi et al., 2017, Liang et al., 2020]. Third, BPAs have a highly nonlinear dependence on internal pressure (P), contraction (ϵ), and loading state [Hunt et al., 2017a, Martens and Boblan, 2017], which complicates control and motivates high-fidelity force–length–pressure modeling. Despite these differences, their muscle-like properties have made them attractive for biomimetic quadrupeds [Aschenbeck et al., 2006, Scharzenberger, 2019], humanoids [Asano et al., 2017, Hitzmann et al., 2018], and musculoskeletal arms [Kurumaya et al., 2016, Chen et al., 2019].

2.2.3 BPA Force–Length–Pressure Models

Previous studies have generated empirical models of the force–length–pressure relationship of BPAs. Sárosi et al. [2017] discuss several high-fidelity BPA force models. They present a static force model with a 21-coefficient polynomial function for a Festo MAS-20-200N (i.e., $\varnothing$20 mm, $l_0 = 200$ mm) with an impressive $R^2 = 0.9994$, as well as Sárosi's static force model for a Festo DMSP-20-400N-RM-RM (i.e., $\varnothing$20 mm, $l_0 = 400$ mm), with six coefficients and a $R^2 = 0.9995$ [Sárosi, 2012, Sárosi et al., 2013]. Martens and Boblan present an even more accurate model, in terms of absolute error, using five coefficients for a DMSP-10-250 (i.e., $\varnothing$10 mm, $l_0 = 250$ mm) and a DMSP-20-300 (i.e., $\varnothing$20 mm, $l_0 = 300$ mm) [Martens and Boblan, 2017]. However, as demonstrated in this work, different resting lengths produce different force–length curves, preventing the use of these single-length

models for designing systems with different resting lengths.

Hunt et al. [2017a] looked at six resting lengths of ∅10 mm Festo BPAs, accounting for differences in maximum contractile percentages, and elucidated the force-length-pressure relationship. In particular, for a given contraction percent and force F (in Newtons), the scalar pressure P (in kPa) required to produce this force for a ∅10 mm Festo artificial muscle can be determined by solving the equation:

$$P = P_0 + k_F F + k_S S + P_t \tan\left(c_a \left[\frac{\epsilon}{\epsilon_{\max} - k_E F} - c_b \right] \right) \tag{2.1}$$

where F is the required force, ϵ is the contraction, S encodes hysteresis ($S = 1$ shortening, $S = -1$ lengthening, $S = 0$ static), and the seven coefficients are defined in Table 2.1. The coefficients have been updated with corrected values, as those reported in Hunt et al. [2017a] contained typographical errors. The contraction ϵ and the maximum free-load contraction at 620 kPa, which appears as $\epsilon_{\max}$ in Equation (2.1), are defined in Section 3.1, together with the relative contraction $\epsilon^* = \epsilon / \epsilon_{620}$ on which the force surface developed in this dissertation is built. Because Equation (2.1) expresses pressure in closed form as a function of force and contraction, it can also be solved onto a grid of pressures and contractions to build a force–pressure–relative-contraction lookup table for design studies.

Festo provides a manufacturer tool, MuscleSIM, for predicting force as a function of pressure and contraction [Lang, Robert, 2005], but experimental comparisons indicate that it does not adequately account for variation in l_0 or actuator-specific contraction limits. Comparisons with experimental data indicate that none of the existing models effectively take into account differences in maximum force due to initial actuator length, and the Hunt

Table 2.1: Coefficients of the 10 mm Festo BPA pressure model of Equation (2.1), corrected from Hunt et al. [2017a].

Symbol	Value	Role
P_0	254 kPa	pressure offset
k_F	1.23 kPa N^{-1}	linear force-to-pressure gain
k_S	15.6 kPa	hysteresis offset (sign of S)
P_t	192 kPa	tangent-term amplitude
c_a	2.03	contraction-shape coefficient
c_b	0.46	contraction-shift coefficient
k_E	0.331×10^{-3} N^{-1}	force-dependent reduction of usable contraction

model was taken with low external forces ($\leqslant$106 N), so it is unclear how well it captures actuator behavior at higher forces. Chapter 3 compares these models with data collected for this dissertation and develops the improved characterization on which the torque and controller work of the later chapters depends.

2.3 Human Lower-Limb Biomechanics and the OpenSim Benchmark

Designing a biomimetic leg requires a quantitative benchmark of human muscle–joint behavior. This dissertation uses OpenSim [Delp et al., 2007, Seth et al., 2011, 2018] and the Gait2392 model, which descends from the interactive lower-extremity model of Delp et al. [1990] and includes musculoskeletal geometry from Hoy et al. [1990], planar knee kinematics from Yamaguchi and Zajac [1989], and a lower-limb model with 23 degrees of freedom and 54 musculotendon actuators [Arnold et al., 2010]. Muscle moment arms, force-generating properties, and functional roles in such models have been characterized extensively [Zajac and Gordon, 1989, Brand et al., 1982, Menegaldo et al., 2004, Millard et al., 2013, Sherman et al., 2013, Suderman and Vasavada, 2012], and standard reporting

conventions for kinematics and joint mechanics are established by the International Society of Biomechanics [Wu and Cavanagh, 1995, Wu et al., 2002, 2005, Derrick et al., 2020]. The human knee is itself a complex joint: its instantaneous center of rotation migrates posteriorly with flexion, a behavior captured by four-bar linkage descriptions of the joint [Romero et al., 2020, Steele, 2018]. Throughout this dissertation, Gait2392 serves as the reference against which the robot's muscle routing, moment arms, and torque capabilities are designed and judged.

2.4 The Biomimetic Knee and the Laboratory's Leg Model

Steele's biomimetic knee joint supplies the physical knee used in this dissertation [Steele et al., 2017, Steele, 2018, Steele et al., 2018]. It is a one-degree-of-freedom four-bar linkage whose crossed links reproduce the posterior migration of the human knee's instantaneous center of rotation (ICR) during flexion, so the rolling-and-sliding behavior of the human joint emerges from the link geometry itself rather than from cams or gear profiles; the design was calculated to reproduce an average sliding-to-rolling ratio of 0.079 over a 151° range of motion [Steele et al., 2018]. Because the ICR migrates, muscle moment arms about the knee change with flexion angle in a human-like way, which is exactly the behavior a torque prediction tool must capture. The biomimetic knee tests of Chapter 4 therefore exercise the corrected torque model against this geometry, and Figure 2.4 summarizes the mechanism.

In the author's master's work, muscle lengths, moment arms, and isometric joint torques were computed over the range of motion for both the Gait2392 model and a PAM-

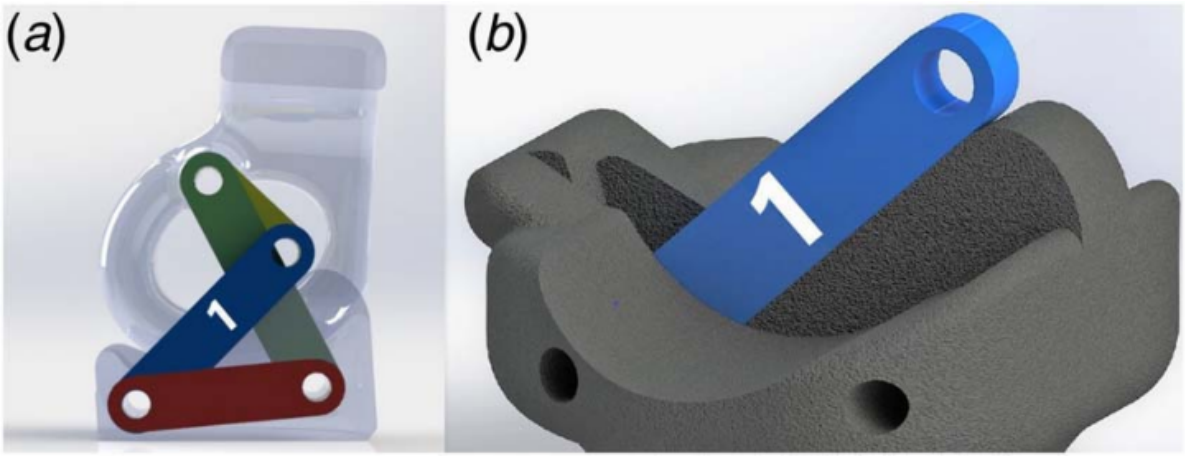


Figure 2.4: Biomimetic knee mechanism. **(a)** Sagittal cutaway at full extension showing the crossed four-bar linkage. **(b)** Exploded view of the mechanical stop. Reproduced from Steele et al. [2018, Fig. 6] under CC BY 4.0.

actuated robot leg model, demonstrating that several robot muscle torque profiles could closely match the human model while many fell far short for part of the RoM [Bolen, 2019, Bolen and Hunt, 2019]. Hand placement of muscle origins and insertions was a painstaking and iterative task, and the placement optimization that replaced it is dissertation work [Morrow et al., 2020], discussed in Chapter 4. Figure 2.5 places the laboratory's bipedal skeleton beside the OpenSim muscle-routing reference developed for it.

2.5 Neural Control of Locomotion: Central Pattern Generators

Locomotion in mammals is driven by spinal central pattern generators (CPGs), neural circuits that can produce alternating rhythmic activity of flexor and extensor motoneurons in the absence of rhythmic input and proprioceptive feedback [Brown, 1911, Rybak et al., 2006]. The modern architectural understanding of mammalian locomotor CPGs comes

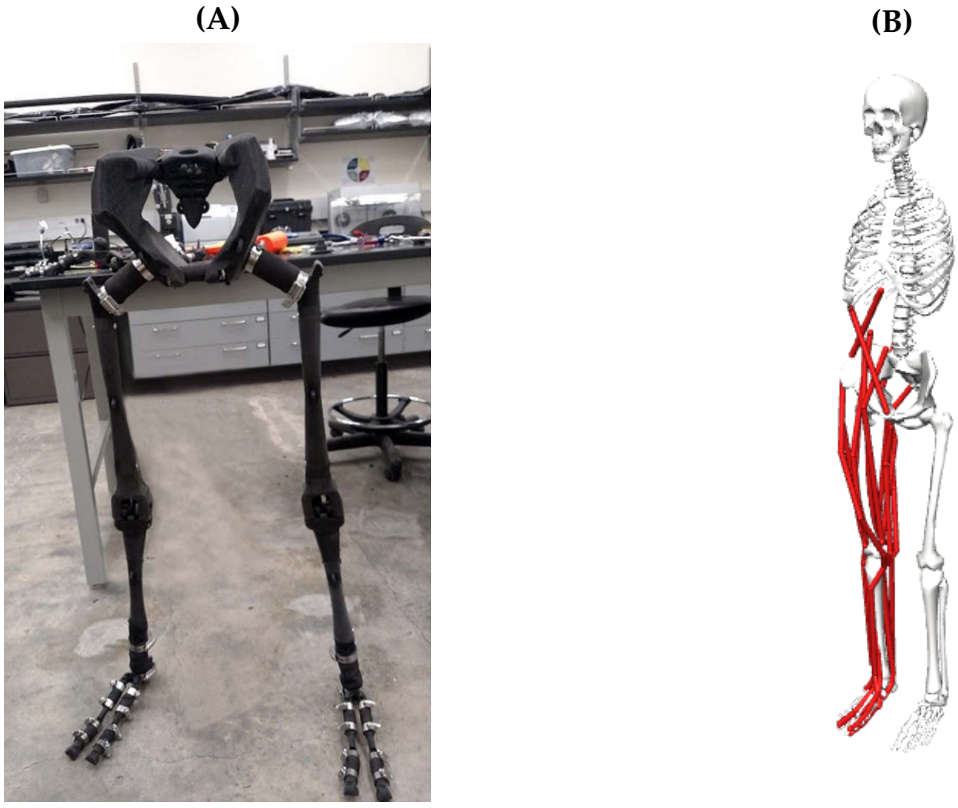


Figure 2.5: **(A)** The laboratory biped skeleton (actuators not installed in this view). **(B)** The Gait2392-derived OpenSim model used to develop its muscle routing; the view exposes the retained lower-limb musculotendon paths used as biomechanical design references [Bolen and Hunt, 2019, Morrow et al., 2020].

largely from the modeling work of the Rybak group and collaborators. Rybak et al. [2006] proposed that the locomotor CPG consists of two levels: a half-center rhythm generator (RG) and a pattern formation (PF) network, with reciprocal inhibitory interactions between antagonist neural populations at each level. This two-level organization explains, among other phenomena, spontaneous deletions of motoneuron activity during fictive locomotion in which the locomotor period and phase are maintained, and it permits independent control of the level of motoneuron activity and of step-cycle timing. McCrea and Rybak [2008] reviewed the electrophysiological support for this organization, and Rybak et al. [2015] related the model architecture to genetically identified spinal interneuron classes. Left–right coordination is provided by commissural pathways between the two half-centers [Molkov et al., 2015, Danner et al., 2016], and the rhythmogenic core of the CPG continues to be refined experimentally [Rancic and Gosgnach, 2021]. How this architecture is expressed as simulable circuit models is the subject of Section 2.7; the locomotor controller developed in Chapter 7 is built directly on this two-level organization.

2.6 Sensory Feedback in Locomotion

Sensory feedback is not a passive correction layer but an integral part of locomotor pattern generation. Reviews of afferent control of the CPG [Van de Crommert et al., 1998] and of muscle proprioceptive feedback and spinal networks [Windhorst, 2007, 2022] summarize the classical picture: group Ia afferents from muscle spindles signal muscle length and velocity, group Ib afferents from Golgi tendon organs signal force, and group II afferents signal length, with load-related feedback playing a role distinct from length-related feed-

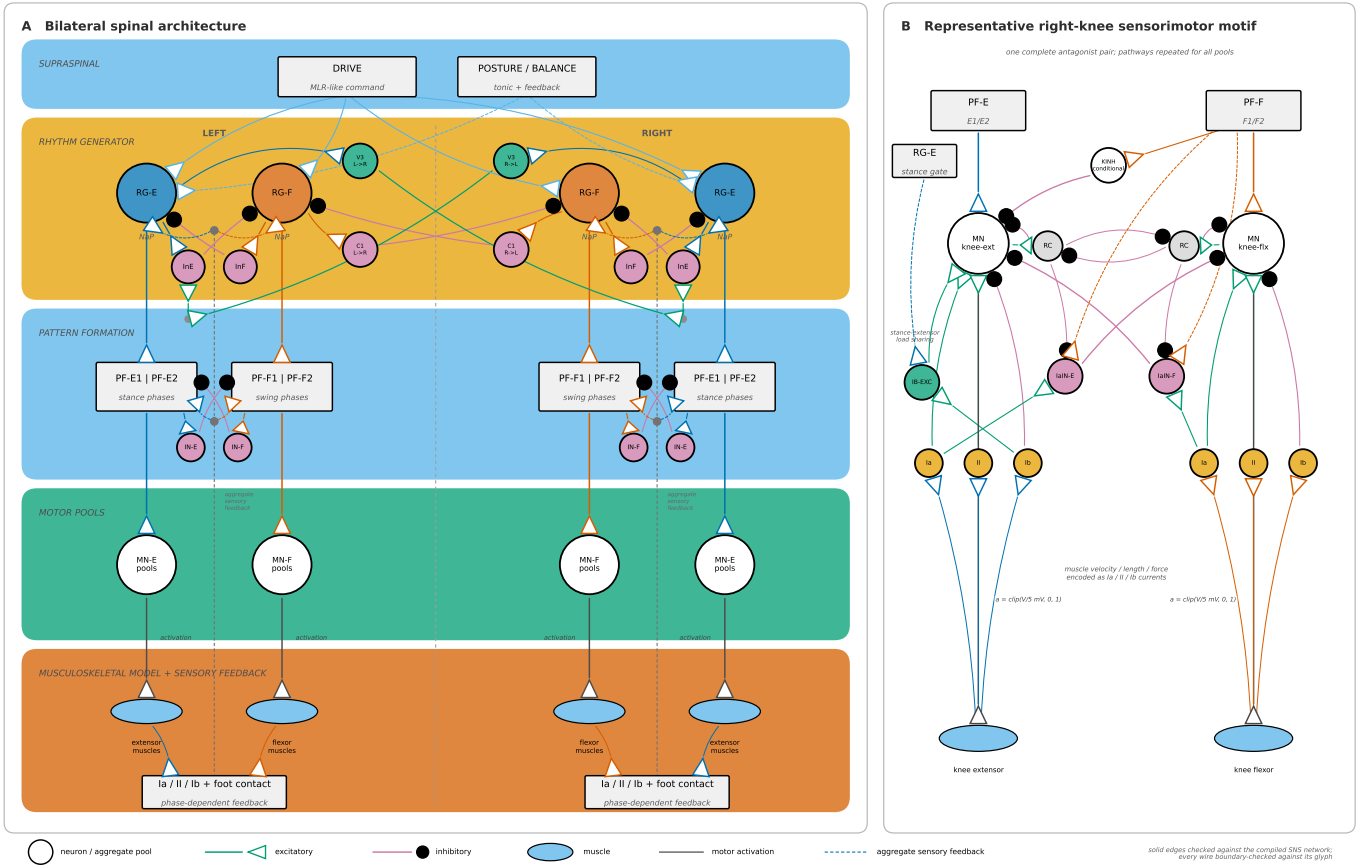


Figure 2.6: Locomotor circuit architecture synthesized from the literature. The bilateral organization combines rhythm-generation, pattern-formation, motoneuron, commissural, and proprioceptive pathways used across two-level CPG and neuromechanical models [Rybak et al., 2006, 2015, Shevtsova et al., 2016, Deng et al., 2019]. The detailed knee-control motif makes the reciprocal antagonist and sensory pathways explicit.

back during locomotion [Dietz and Duysens, 2000]. Removing proprioceptive feedback degrades locomotor patterns in mice [Akay et al., 2014]; proprioceptive input regulates and adapts locomotor activity [Lam and Pearson, 2002]; and the relative contributions of proprioceptive and vestibular systems to locomotion are the subject of ongoing study [Akay and Murray, 2021, Akay, 2020]. Of particular relevance to the tests planned in Chapter 8, tonic central and sensory stimuli have been shown to facilitate involuntary air-stepping in humans [Selionov et al., 2009], demonstrating that simple graded stimulation of motoneuronal and rhythmic circuits can elicit stepping-like output, an experiment readily reproduced in simulation by tonic stimulation of pattern-formation and rhythm-generator populations. The proprioceptive and contact feedback channels designed in Chapter 7 follow this afferent organization.

2.7 Computational Neuroscience of Locomotor Circuits

Computational neuroscience expresses hypotheses about locomotor circuitry as executable models. A locomotor model is a system of coupled differential equations in which every element of the schematic corresponds to an identified neural population or pathway; the wiring diagram is not an illustration of the model, it is the model. The founding hypothesis is Brown's half-center: alternating flexor and extensor activity arises from two reciprocally inhibitory central populations with opposing bias [Brown, 1911]. Modern instantiations of the half-center endow each neuron with persistent sodium channels, whose slow inactivation terminates each burst so that reciprocal inhibition switches the populations without any phasic input [Deng et al., 2019]. Rhythmogenesis, in this tradition,

is a property of the intrinsic membrane dynamics, not of a clock elsewhere in the network.

The two-level organization described in Section 2.5 is the field's current working architecture, and its models account for specific electrophysiological observations: the fictive-locomotion deletions reproduced by Rybak et al. [2006], the genetically identified interneuron classes mapped onto the model by Rybak et al. [2015], and the organization of circuits controlling two-joint muscles reviewed by Shevtsova et al. [2016]. Coupling such circuits to musculoskeletal models shows how afferent feedback shapes CPG output [Markin et al., 2010] and how biomechanics and sensory feedback regularize the behavior of different CPG architectures [Deng et al., 2022]. Physiologically plausible models of human locomotion use the same machinery to dissect the relative contributions of reflexes and CPGs, with locomotor primitives, Ia, II, and Ib pathways, and Renshaw cells represented explicitly [Di Russo et al., 2023], and recent models address interlimb coordination under abrupt perturbations such as the sudden loss of ground support [Shinohara et al., 2025].

For a mechanical engineer, the productive way to read this literature is as a controls tradition in which the block diagram is the controller itself: each block is an identifiable neural population with conductance-based dynamics, and each connection is a synaptic conductance with a physiologically interpretable weight. The connection schematics published with these models are therefore worth reading as carefully as their equations. The controller developed in Chapter 7 comes from this tradition: its two-layer organization follows the rat hindlimb model of Deng et al. [2019], and its verification tests are drawn from the deletion and stimulation phenomena that the models of Rybak et al. [2006] and the air-stepping experiments of Selionov et al. [2009] explain.

2.8 Synthetic Nervous Systems

For robots, these biological architectures can be implemented as synthetic nervous systems (SNSs): networks of conductance-based spiking and non-spiking neurons that are small enough to run in real time yet biologically grounded enough to be compared against animal data. The approach has been pursued most persistently by the insect-biorobotics lineage of the Quinn laboratory at Case Western Reserve University and by laboratories closely connected to it, the author's own included. Works in this lineage span several platforms: CPG-based neural control of a pneumatically actuated dog robot [Hunt et al., 2017b], analytically designed dynamic neural networks for human balance control [Hilts et al., 2019], the full-muscle rat hindlimb model and its moment-arm analysis [Young et al., 2019], the two-layer CPG neuromechanical walker built on it [Deng et al., 2019], which supplies this dissertation's controller architecture, and the DoggyDeux quadruped, a canine-inspired BPA robot designed at the author's laboratory, the Agile and Adaptive Robotics Laboratory (AARL) at Portland State University, for synthetic neural network control [Scharzenberger, 2019]. Related controllers from the wider neuromechanical community include sensory-entrained CPG control of bipedal walking [Li et al., 2017], inter-leg coordination analyses in stick-insect-inspired controllers [Nourse et al., 2019], reinforcement-reflex hybrids in insects [Goldsmith et al., 2021], and adaptive hindlimb walking controllers [Schilling et al., 2022].

The SNS-Toolbox [Nourse et al., 2023] makes such networks practical: it is an open-source Python package that simulates hundreds to thousands of spiking and non-spiking neurons in real time or faster, supports multiple software and hardware backends including

GPUs and embedded platforms, and couples directly to physics simulators such as MuJoCo for musculoskeletal control. The field it serves is active and growing: the perspective review of Szczecinski et al. [2023] traces neuromorphic legged-locomotion control from its insect-robot beginnings through current platforms to open problems, and physical testbeds in this lineage now extend from insect-scale robots to a robotic cat hindlimb [Armstead et al., 2025].

The boundary between background and contribution runs through this section as well. The prior AnimatLab walker with which the author's program first applied these architectures to a biped [Bolen and Morrow, 2020] is presented as prior dissertation work in Section 7.1, the Sensory Afferent Database that organizes the afferent data such networks require [Bolen et al., 2023] is the subject of Chapter 6, and the simulation pipeline that couples the controller to the robot model is built in Chapter 7. None of it is repeated as background here.

2.9 Neuromechanical Simulation

Neuromechanical simulation, the coupling of a musculoskeletal model to a model of the neural controller that drives it, has advanced along three partially overlapping lines. The first line embeds reflex physiology directly into the controller. The muscle-reflex walking model of Geyer and Herr [2010] demonstrated that positive force, length, and velocity feedback around each muscle, organized with simple interlimb coordination, is sufficient to produce self-stabilizing human-like walking dynamics and realistic muscle activations without any central pattern generator. That model became the reference point

for a generation of reflex-based neuromechanical controllers, including the cat hindlimb models of Section 2.7 [Markin et al., 2010, Shevtsova et al., 2016] and their extensions. Hybrid work makes the complementarity of the two strategies explicit: Haeufle et al. [2018] combined feed-forward muscle activations with neuronal reflex feedback in simulated human walking and found that the robustness benefit of each depends on which muscle carries it.

The second line replaces hand-designed circuitry with machine learning. Song et al. [2021] trained deep reinforcement learning policies on a physiologically realistic musculoskeletal model in OpenSim, reproducing human kinematics and muscle activations across locomotion tasks, with the learned policy re-usable for experiments that would be difficult in the laboratory. This approach inherits the fidelity of the musculoskeletal model but encodes control in opaque function approximations: the trained policy can reproduce behavior, but the individual "neurons" of the controller do not correspond to identifiable physiological circuits, so the class of scientific questions it can answer, such as lesions, deletions, and pathway stimulation, is narrower than its behavioral repertoire. Recent perspectives on predictive musculoskeletal simulation [De Groote and Falisse, 2021] place both learned and optimal-control approaches in the same frame: the models are mature, and the open problems are the cost of solving them and the biological interpretability of the control they find.

The third line, in which this dissertation sits, is biologically grounded synthetic nervous systems: controllers whose neurons and synapses are explicit, conductance-based, and mapped to identified physiological circuits, simulated at speeds sufficient for real-time robotics [Nourse et al., 2023]. Hierarchical CPG control is meanwhile spreading

to bipedal-capable musculoskeletal models, including motor-module bipedal walkers [Ichimura and Yamazaki, 2019], two-layer rhythm-generator and pattern-formation control of a musculoskeletal macaque model [Nishizaki et al., 2026], and two-level CPG control of muscle-driven robotic hind legs [Fukuoka et al., 2021], though always in simulation with generic musculature or on non-humanoid platforms; a two-layer SNS driving a custom-anatomy, artificial-muscle humanoid leg remains unclaimed. What distinguishes the present program from all three lines is the combination it attempts: an anatomically grounded controller (two-layer CPG with Ia, Ib, II, and contact feedback), a biomechanically validated pneumatic model (Chapter 4), and an eventual physical robot.

2.9.1 Simulation Tooling

All three lines depend on a software substrate that is itself part of the method: musculoskeletal geometry too elaborate to integrate by hand, joined to neural models too large to wire by hand. The tools used in this dissertation, and the ones adjacent to it in the same ecosystem, are the following.

OpenSim [Delp et al., 2007, Seth et al., 2018], distributed through the SimTK (Simulation Toolkit) repository, provides the biomechanical modeling side: musculoskeletal models with anatomically accurate geometry, muscle routing, and force-generating properties, along with tools for analyzing moment arms and isometric torque. The same SimTK infrastructure hosts SCONE, an open-source package for predictive simulation and optimization of neuromuscular control of biological motion [Geijtenbeek, 2019]. AnimatLab [Cofer et al., 2010] provides a graphical environment in which physical bodies and synthetic neural

systems are constructed and co-simulated; the AARL's prior walker (Section 7.1) was built in it. MuJoCo [Todorov et al., 2012] provides the physics-engine side: fast, accurate simulation of rigid-body dynamics with contacts, along with muscle-like actuators that include activation dynamics. MyoConverter bridges the biomechanical and physics-engine worlds by converting OpenSim musculoskeletal models to MuJoCo format with optimized muscle kinematics and kinetics [Caggiano et al., 2022, Wang et al., 2022]. Combined with the SNS-Toolbox's MuJoCo backend [Nourse et al., 2023], these tools were used to convert the laboratory's modified OpenSim leg model to MuJoCo, couple it to a synthetic nervous system with Ia, Ib, and II proprioceptive channels plus contact sensors, and run the full neuromechanical loop faster than real time (Sections 7.2–7.3). Simulation speed is the practical enabler: a single overnight run can sweep thousands of parameter combinations or execute hundreds of simulated gait cycles per condition, throughput that no human-subject or animal protocol could match, and that transforms lesion and stimulation studies from bespoke demonstrations into statistically powered experiments.

2.10 Testing Neuromechanical Hypotheses on Robots: The Ethical Case

The experiments this research program is designed to enable, including deleting a muscle from a limb, blocking a sensory pathway, lesioning part of a rhythm generator, and applying tonic stimulation to spinal circuits, cannot be performed on healthy humans. Invasive intervention on the human neuromuscular system is limited to clinical populations, is tightly constrained by ethics review, and permits neither the within-subject control nor the reversibility that neuromechanical hypotheses demand. The same experiments in

animals also carry real moral cost, and their justification is governed by the Three Rs framework: replacement of animal subjects with non-animal methods, reduction of the numbers required, and refinement of the remaining procedures [Russell and Burch, 1959, Tannenbaum and Bennett, 2015]. A robot that implements a validated neuromechanical hypothesis is precisely such a replacement method: every deletion, lesion, and stimulation experiment that runs on the synthetic nervous system is an experiment that does not require a living subject, and the robot population can be duplicated exactly, so that "subjects" are statistically identical across conditions, a form of reduction no breeding colony can offer. Refinement follows as well: the failure modes of these experiments damage hardware, not tissue.

Figure 2.7 shows what that accounting leaves out. The image is a frame from footage of a real decerebrate-cat experiment, the class of preparation on which the field's understanding of spinal locomotion has been built since its beginnings [Brown, 1911, WhiteCoatWaste, n.d.]: a living cat, its forebrain disconnected by a brainstem transection, walks on the treadmill because the neurons distal to the laceration are still living, still connected, and still firing action potentials. The published record of this work is drawn almost exclusively in schematic, labeled boxes for half-centers and color-coded reflex arcs that are honest about the circuitry and silent about the animal. The figure is included as the counterweight, documenting what the experiments looked like, and therefore what a replacement method stands in for.

The procedures behind that record are documented in films that remain publicly available. A 50 min classroom recording posted by the UTHealth Houston School of Dentistry first shows an intact cat playing with its handler, righting itself when lowered

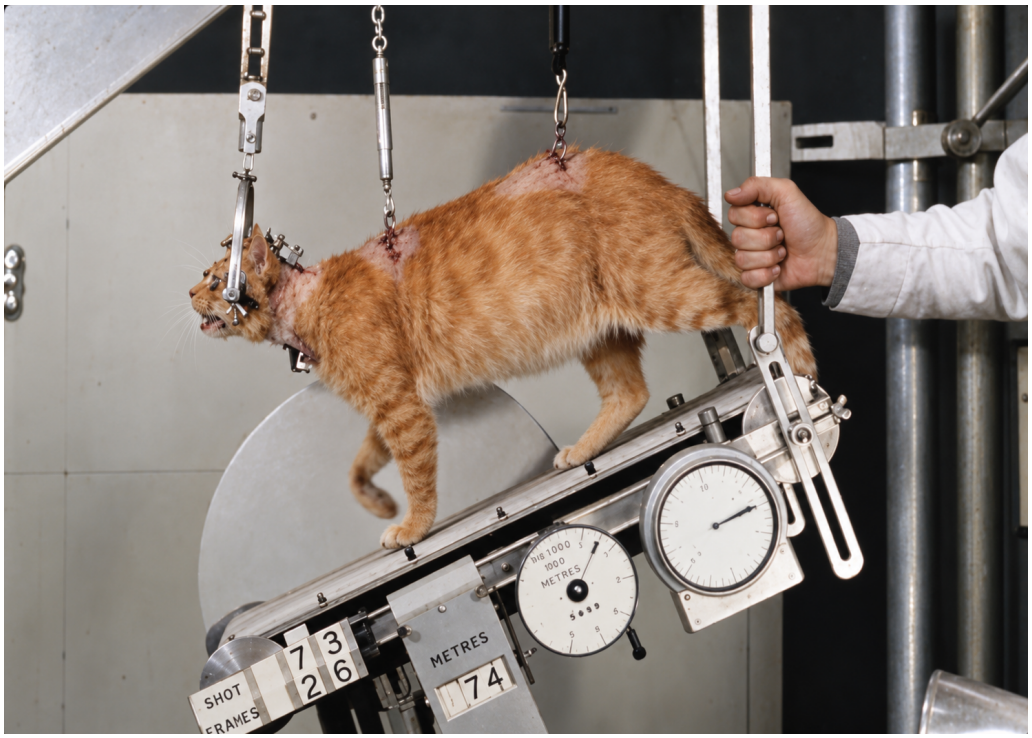


Figure 2.7: Frame from archival footage of a decerebrate-cat treadmill experiment [WhiteCoatWaste, n.d.], restored and machine colorized; color and fine detail are inferred by the restoration rather than recovered from the source. The cat walks on the treadmill belt while held in a stereotactic head holder whose fixation pin passes through or beside the eye orbit, with skull clamps and slung support completing the frame. Shown as documentary context for the replacement argument of this section, not as gait evidence.

and placing its paws accurately on the edge of a table, and then performs the decerebration for the class [UTHealth Houston School of Dentistry, 2025]: the animal is anesthetized with open-drop ether, more than the horse in the same demonstration series received, the carotid arteries are ligated and a tracheal cannula installed, and the skull is opened with a trephine before the brainstem is transected between the superior and inferior colliculi. The treadmill preparations that established locomotor neurophysiology are likewise preserved in the historical literature: Stuart [2007] reproduce the arrangement of the high-decerebrate cat treadmill experiments, with electromyographic recording from the hindlimb muscles, from the original Russian-language reports [Severin et al., 1967].

The honest accounting is murkier than the Three Rs framing alone suggests, because the models a robot implements are themselves products of animal experimentation. The half-center organization, the two-level rhythm and pattern architecture, and the length, force, and load signaling of the afferents were each established in preparations like the ones described above [Brown, 1911, Severin et al., 1967, Rybak et al., 2006], and the databases of afferent properties that parameterize sensory feedback are compiled from animal recordings [Bolen et al., 2023]. A synthetic nervous system therefore replaces future experiments; it does not replace, and cannot repay, the animal work on which its architecture is built. Recognizing this inheritance sharpens rather than undermines the case for robotic subjects: the knowledge purchased at that cost is precisely what makes a robotic replacement scientifically meaningful, and the robot is the instrument that keeps that knowledge productive without adding to the ledger.

For a robot to serve as a replacement subject, however, it must first earn standing as a scientific model. Webb's influential analyses of bio-robotic methodology [Webb, 2001,

2002] argue that a robot is a good model of biological behavior when the mechanism it implements is tested at the same level of organization as the target phenomenon, matching dynamics, control structure, and sensory ecology rather than surface appearance. The validation chain of this dissertation is constructed to that standard: the actuator force model is validated against bench data (Section 3.3), the joint torque model against isometric measurements and human torque envelopes (Section 4.6), and the neuromechanical controller can be verified against published tonic stimulation and deletion phenomena [Selionov et al., 2009, Rybak et al., 2006] before any locomotion tuning is attempted. Only then can a robot experiment be read as evidence about the hypothesized biological circuit rather than about the robot.

The argument is ultimately bidirectional, completing the loop described in Section 1.2. Experiments that are unethical, impractical, or statistically unreachable in humans and animals become routine on the robot; the resulting insights into neuromechanical control feed back into biology, into the design of the next robot iteration, and directly into prosthetic and orthotic devices that face the same problem of controlling muscle-like actuators for human users, whose central pattern generator controllers are an active topic in lower-limb exoskeleton research [Duan et al., 2024, Pery et al., 2026]. The ethical case and the scientific case are the same case: the robot is the instrument that makes these experiments possible.

3 Artificial Muscles

Publication note: Sections 3.1–3.2, Section 3.3, and Sections 3.4–3.8, with their associated figures and tables, appeared in Bolen et al. [2026] (Bolen, Elzein, Pang, and Hunt, *Actuators*, 2026). The knee-torque sections of Chapter 4 are also the subject of an unpublished manuscript in preparation by Bolen, Burgess, Morrow, Elzein, Pang, Scharzenberger, and Hunt [Bolen et al., 2025]; that manuscript has not been submitted for publication. For both works, B.B. performed the conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and original draft preparation; M.E., L.P., L.B., and C.M. contributed methodology, software, investigation, and data curation; and A.H. supervised the work, administered the projects, and acquired funding. All co-authors contributed to writing, review, and editing.

This chapter develops and validates the actuator force models on which the rest of the dissertation builds, and it reads as one self-contained study, from methods through results to discussion. Sections 3.1 and 3.2 present the methods: the force equations used throughout this work, and the test jig and measurement procedure used to characterize Festo BPAs with 10 mm and 20 mm uninflated diameters over a wide range of resting lengths, contractions, and pressures. Section 3.3 reports the resulting maximum-force and normalized force–strain–pressure models and compares them with the manufacturer's tool. The remaining sections discuss the effects of resting length, the nondimensionalized force prediction, the comparison with other published BPA models, the implications for real-time

control and simulation, and the analogy between these artificial muscles and biological muscle. The validated force model is the one used for the knee-torque predictions of Chapter 4 and for the BPA actuation of the neuromechanical simulations in Chapter 7.

3.1 Actuator Force Model

The actuator force model used throughout this dissertation is described here. First, the amount of contraction ε in a BPA is calculated as

$$\varepsilon = \frac{(l_0 - l_{\mathrm{m}})}{l_0} \tag{3.1}$$

where l_0 is uninflated resting length of the BPA measured from hose clamp to hose clamp. Current muscle length is l_{m}.

Similarly, ε_{620} is the amount of contraction in a BPA without external load when inflated at 620 kPa, calculated as

$$\varepsilon_{620} = \frac{(l_0 - l_{620})}{l_0} \tag{3.2}$$

where l_{620} is the muscle length measured at 620 kPa and free from external loads.

The amount of relative contraction of a BPA ε^* can then be calculated by dividing equation (3.1) by (3.2), such that

$$\varepsilon^* = \frac{\varepsilon}{\varepsilon_{620}} \tag{3.3}$$

describes the BPA's current contraction as a normalized percentage of its total.

The amount of normalized isometric force F^* in a Festo BPA is then calculated as

$$F^*(\varepsilon^*, P^*) = \begin{cases} c_0 \cdot \left(\exp(-c_1 \cdot \varepsilon^*) - 1\right) + P^* \cdot \exp\left(-c_2(\varepsilon^*)^2\right) & \text{for } F^* > 0 \\ 0 & \text{for } F^* \leqslant 0 \end{cases} \tag{3.4}$$

where P^* is pressure P normalized by 620 kPa. BPA diameter dependent coefficients are determined by the actuator characterization experiments reported in Section 3.3 (Table 3.2).

The maximum force F_{max} in a BPA is calculated as

$$F_{max}(l_0, P_{\max}) = a_1 \cdot P \cdot \arctan\left(a_2 \cdot P \cdot (l_0 - 0.0075)\right) \tag{3.5}$$

with coefficients determined by the actuator characterization experiments reported in Section 3.3 (Table 3.1). When maximum system pressure is set to $P = 620$ kPa, $F_{\max}(l_0, 620) = F_{620}(l_0)$. Multiplying (3.4) by (3.5) yields

$$F(\varepsilon^*, P^*, l_0) = F^*(\varepsilon^*, P^*) \cdot F_{620}(l_0) \tag{3.6}$$

as the equation for isometric force F in a Festo BPA.

3.2 BPA Isometric Force Characterization Experiment

To measure artificial muscle force as a function of length and pressure, we built a test jig from extruded aluminum and 3D printed parts. Festo BPAs with 10 mm and 20 mm diameter were tested with various resting lengths and different amounts of contraction. These data are used to develop an improved model that more accurately predicts maximum

muscle force based on l_0, current muscle length l_m, and pressure P. The results are compared with existing models and the new data are used to create an updated model.

A test jig was made of 80/20 brand 1515 series extruded aluminum (Figure 3.1). Artificial muscles were placed vertically in the jig one at a time. The upper end was attached to a load cell. Force data was collected using a CALT DYLY-103 100 kg S-shaped load cell for the ∅10 mm BPAs and a Loadstar RAS1-2HKS-S for the ∅20 mm BPAs. The load cell was used in conjunction with a HX711 Load Cell Amplifier. Pressure data was obtained from a Freescale MPX5700 GP 5 V pressure sensor. Building air supply pressure was controlled with two pressure regulators in series. The first was a Parker model 20R113GC $0 - 120$ psi pressure regulator. The second was a Husky 3/8 in. High Performance Air Regulator HDA72200. A Festo VTUG-10-MRCR-S1T-26V20-T516LA-UL-532S-8K valve manifold VTUG-G was used to deliver air from the pressure regulator to the actuator. This manifold is comprised of eight two-in-one bidirectional normally closed Festo VUVG-S10-T32C-AZT-M5-1T1L valves.

Pressure and load cell amplifier data were sent to MATLAB 2021b via an Arduino Uno style Sparkfun BlackBoard C microcontroller. The computers that used MATLAB were running either Windows 10 or Windows 11. During phases when the Arduino was collecting force and pressure data to send to MATLAB, the Arduino would also trigger (via MATLAB) an Onsemi 2N4401 NPN transistor to make the valve manifold open or close the valve. For other data collection, the valve was manually opened and closed.

The inner distance between the hose clamps on each BPA was measured with a FANUC tape measure to determine the muscle's length l and resting length (l_0). This is how Festo AG & Co. KG [2026] defines l_0, although in Hunt et al. [2017a] it was measured

to also include end cap length. We then inflated each BPA to $P_{620} = 620\,\text{kPa}$, with one end allowed to move freely in the axial Degree of Freedom (DoF), and measured the length l_{620} to calculate maximum contraction at 620 kPa using Equation (3.2). The distance between the crossmembers was then controlled to obtain different amounts of contraction, the muscles were inflated to various pressures, and the contractile force was recorded at the pressure–contraction pairs. This was done for ∅10 mm BPA resting lengths (l_0) of 112 mm–518 mm. For ∅20 mm BPAs, resting lengths (l_0) of 300 mm–509 mm were used.

3.3 BPA Force Characterization

This section reports the results of the isometric force characterization experiments described in Section 3.2. First, maximum isometric force at 620 kPa is modeled as a function of resting length. The remaining data are then normalized to build a predictive force surface, and the resulting model is compared with the manufacturer's tool and with other published BPA models.

3.3.1 Maximum Force at 620 kPa

Data of the maximum force from BPA characterization tests of the 10 mm and 20 mm diameters at 620 kPa show a dependency on resting length (Figure 3.2). This is a previously unreported characteristic of these artificial muscles. Detailed analysis of the Festo tool [Festo AG & Co. KG, 2026] does in fact predict a change in maximum force with the resting length; however, the Festo tool predicts increased force with shorter lengths, while our collected data indicate decreasing force with shorter lengths. The data show a force

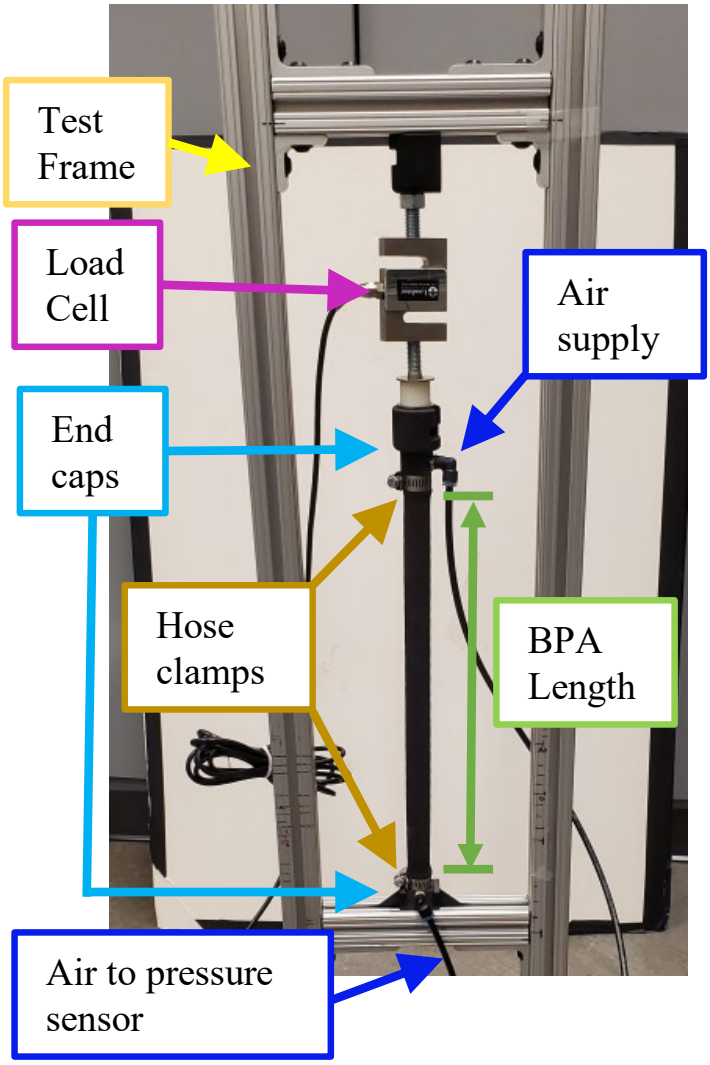


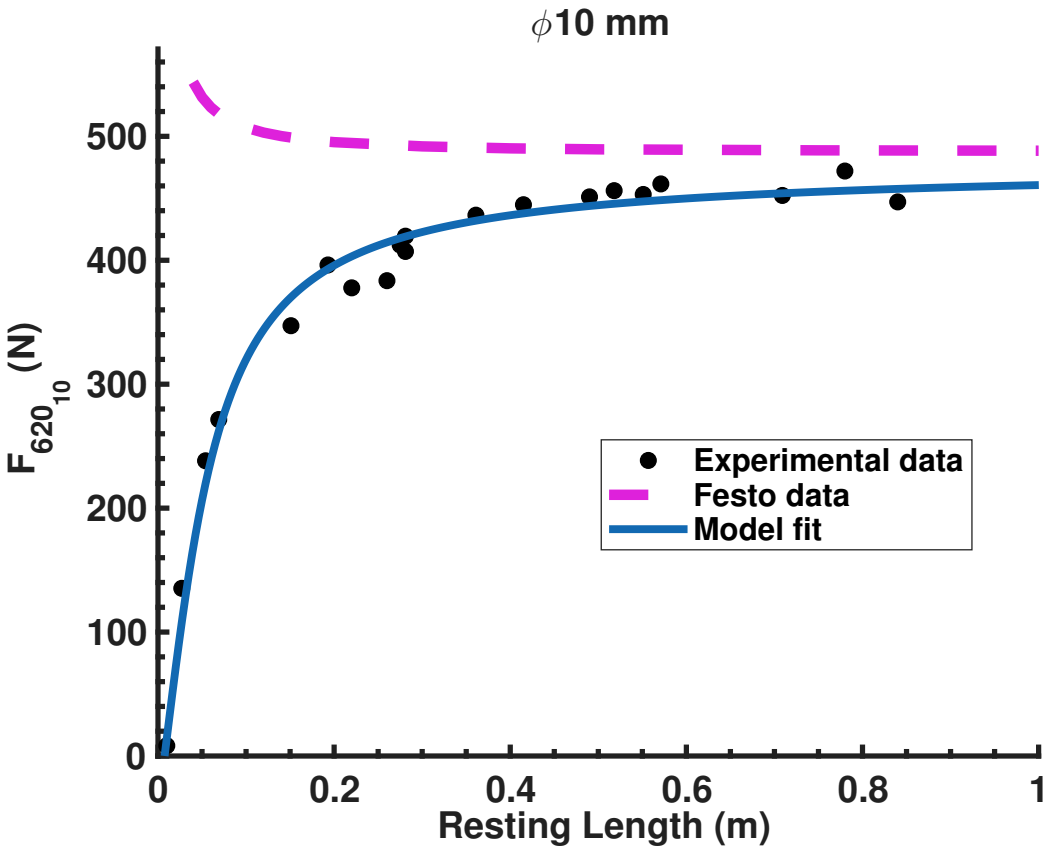
Figure 3.1: A picture of a BPA in the isometric force test stand with components labeled.

response resembling an arctan curve along the l_0 dimension. Using the Nonlinear Least Squares method and a Least Absolute Residual robustness, we fit an arctan curve to the data to get the maximum force at 620 kPa given a resting length as:

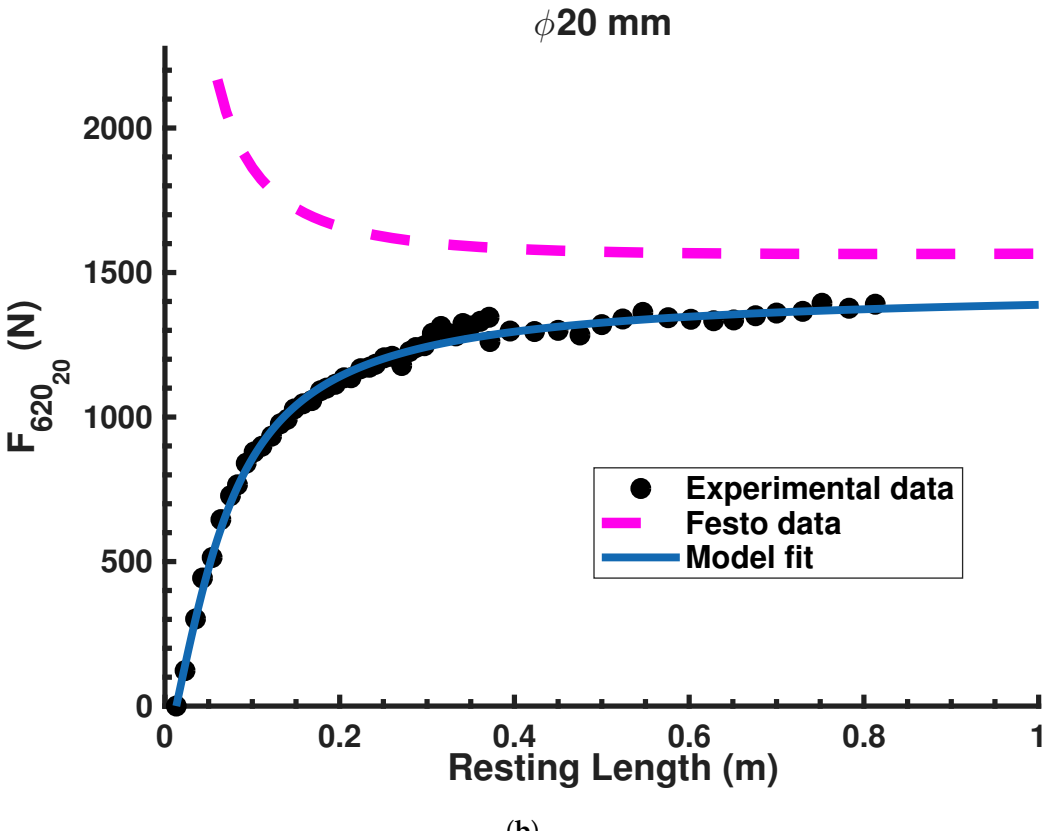
$$F_{620_{10}} = 303.5\,\text{N} \cdot \arctan\left(19.03\,\text{m}^{-1} \cdot (l_0 - 0.0075)\right) \tag{3.7}$$

$$F_{620_{20}} = 922.4\,\text{N} \cdot \arctan\left(15.37\,\text{m}^{-1} \cdot (l_0 - 0.013)\right) \tag{3.8}$$

The length is offset by 0.0075 m and 0.013 m because solid modeling showed that the end caps contact each other at these lengths. At these lengths, the actuator would not be able to contract to produce force.


ϕ10 mm
500
400
300
200
100
0
$F_{620_{10}}$ (N)
0
0.2
0.4
0.6
0.8
1
Resting Length (m)
Experimental data
Festo data
Model fit

(a)


ϕ20 mm
2000
1500
1000
500
0
$F_{620_{20}}$ (N)
0
0.2
0.4
0.6
0.8
1
Resting Length (m)
Experimental data
Festo data
Model fit

(b)

3.3.2 Force as a Function of Pressure and Resting Length

Data from BPA characterization tests of the 10 mm muscles resulted in force–pressure pairings for different muscle resting lengths (l_0) (Figure 3.3a). Similar to the maximum force data, these data show a force response resembling an arctan curve along the l_0 dimension with a more linear response to changes in pressure. Using the Nonlinear Least Squares method and a Least Absolute Residual robustness, we fit an arctan curve to the data to get the maximum force given a resting length and pressure as described by the model form given in Section 3.1, where $a_1 = 0.4895\,\mathrm{N\,kPa^{-1}}$ and $a_2 = 0.030\,68\,\mathrm{kPa^{-1}\,m^{-1}}$ for the 10 mm actuator, and $a_1 = 1.49\,\mathrm{N\,kPa^{-1}}$, $a_2 = 0.0248\,\mathrm{kPa^{-1}\,m^{-1}}$ for the 20 mm actuator. Goodness-of-fit measures are given in Table 3.1; throughout this dissertation, goodness of fit is reported as the root-mean-square error (RMSE), the adjusted fraction of variance unexplained (FVU, for which $R^2 = 1 - \mathrm{FVU}$), and the maximum absolute residual, as defined alongside the reference implementation in Appendix A.

3.3.3 Maximum Contraction at 620 kPa

Figure 3.3(b) shows an attempt at a linear fit for maximum contraction at 620 kPa (ϵ_{620}) as a function of resting length (l_0). Error bars have been included to show the effect of ± 1 mm accuracy with measurements of l_0 and l_{620}. There was a large amount of variance in the data, with the linear fit giving an adjusted $\mathrm{R}^2 = 0.4124$ and an RMSE $= 0.0083$. Since there is no direct, predicable relationship between maximum contraction and l_0, this value should be recorded in each muscle used on a robot to best predict the force it may produce at different pressures and contraction.

Table 3.1: Maximum force equation coefficient values with confidence intervals (CI) and goodness-of-fit measures. Equation (3.5) is compared to data taken at 620 kPa. Equations (3.7) and (3.8) are compared against maximum force from using the Festo tool. It was not necessary to fit an adjusted R^2 in these cases.

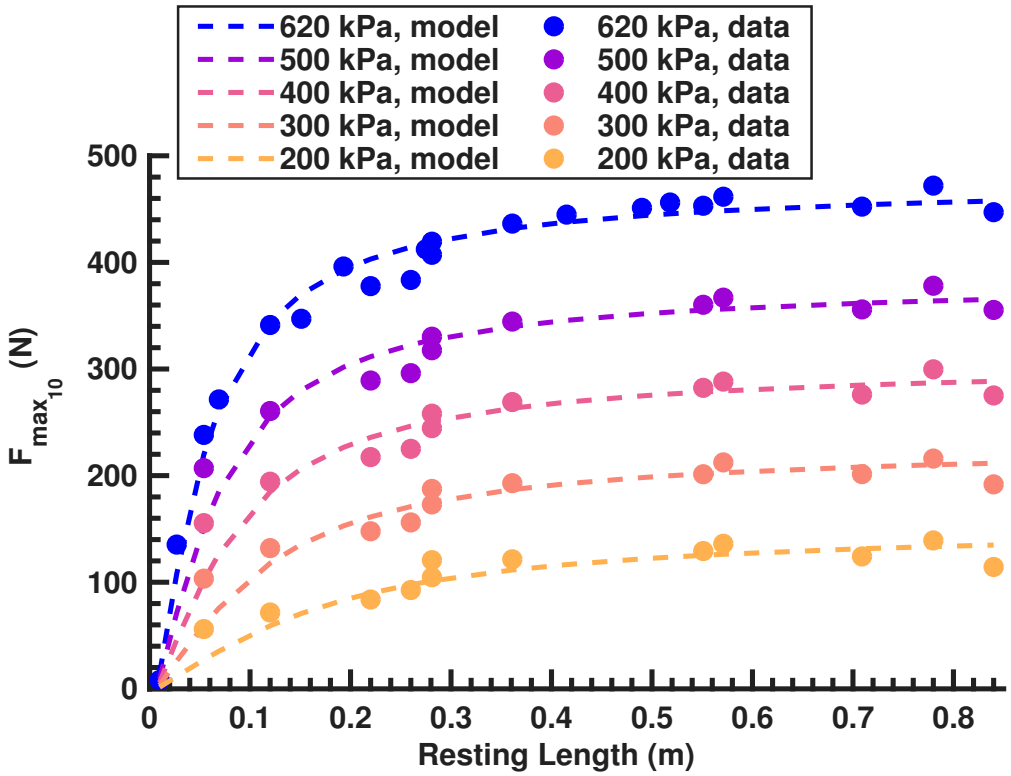
Equation	Coefficient	CI (95%)	Our Model			Festo Tool		
			Adj. R^2	RMSE	Max. Error	Adj. R^2	RMSE	Max. Error
(3.5)	$a_1 = 0.4895\,\mathrm{N\,kPa^{-1}}$ $a_2 = 0.030\,68\,\mathrm{kPa^{-1}\,m^{-1}}$	(0.4822, 0.4968) (0.0282, 0.03317)	0.9945	11.61 N	55.3 N	0.9854	14.7 N	30.9 N
(3.7)	$b_1 = 303.5\,\mathrm{N}$ $b_2 = 19.03\,\mathrm{m^{-1}}$	(300, 308) (17.48, 20.57)	0.9854	14.72 N	30.9 N	–	189.9 N	375.6 N
(3.8)	$b_1 = 922.4\,\mathrm{N}$ $b_2 = 15.37\,\mathrm{m^{-1}}$	(914.2, 930.7) (14.75, 15.98)	0.9945	23.83 N	62.1 N	–	668.4 N	1590.1 N

Table 3.2: Normalized isometric BPA force equation (Equation (3.4)) coefficient values and goodness-of-fit measures.

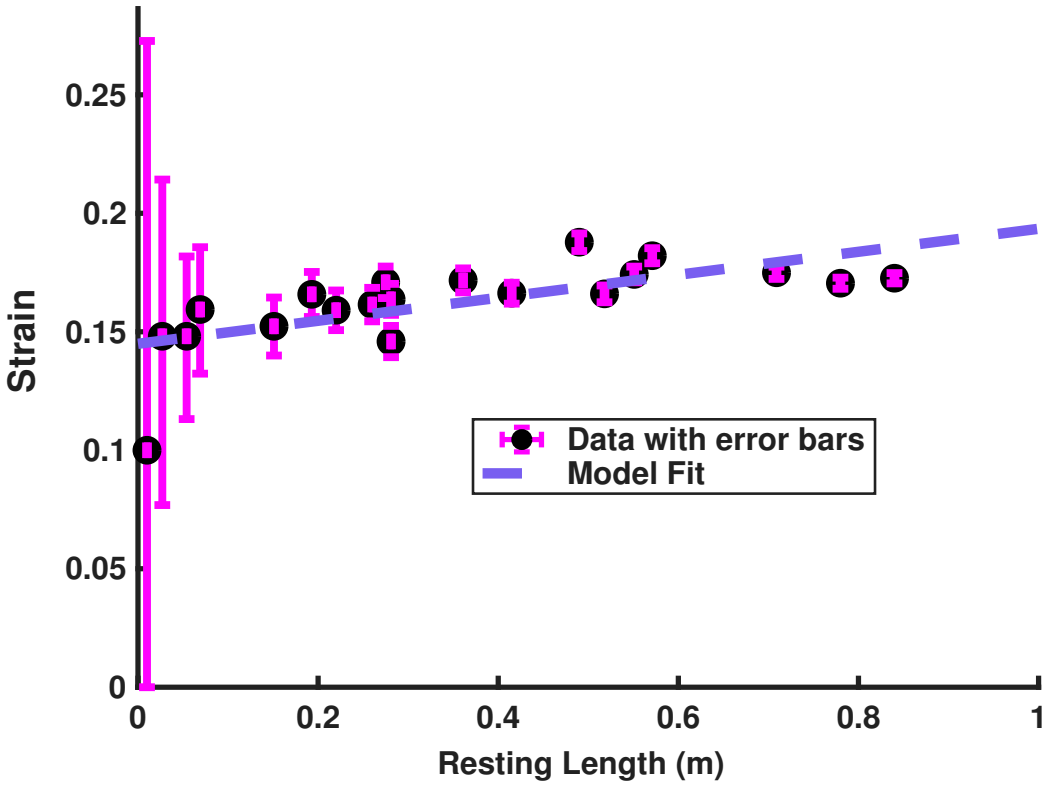
BPA	Coefficient	CI (95%)	Model			Validation		
			Adj. R^2	RMSE	Max. Error	Adj. R^2	RMSE	Max. Error
∅10 mm	$c_0 = 0.5682$	(0.5584, 0.578)	0.9998	0.005118	10.3%	0.9994	0.0245409	10.6%
	$c_1 = 4.254$	(4.126, 4.383)						
	$c_2 = 0.5597$	(0.5429, 0.5766)						
∅20 mm	$c_0 = 0.2579$	(0.2401, 0.2756)	0.992	0.02294	6.8%	0.9943	0.0231303	5.7%
	$c_1 = 6.477$	(5.558, 7.396)						
	$c_2 = 1.321$	(1.239, 1.403)						

3.3.4 Force as a Function of Pressure and Contraction

In addition to being a function of the pressure and resting length, the force produced by the actuator is also a function of the amount of actuator contraction, with less force being applied as the actuator contracts. To build this model, the collected data of force, pressure, and contraction are normalized by dividing by F_{620}, P_{620}, and ϵ_{620}, respectively. This had the effect of compressing the data into a 3D surface ranging from 0 to 1 on all axes. Normalized data are compared with pressure isoclines of force predicted by the Festo tool divided by the maximum force equation described in the previous section in Figure 3.4. With this comparison, it is clear that the Festo tool over-predicts the expected force, especially at lower pressures, contraction, and resting lengths.

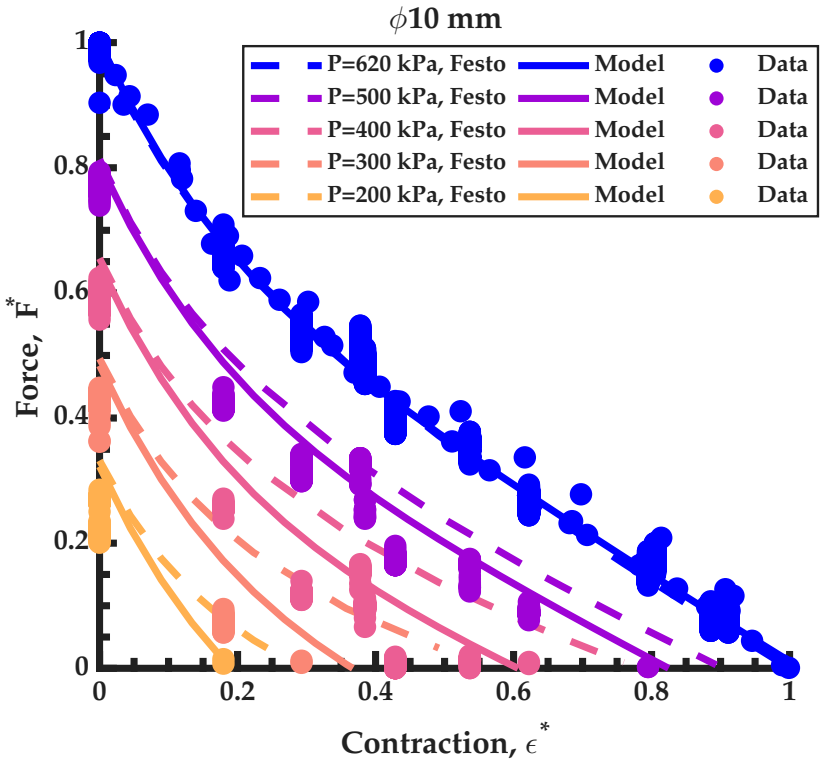


(a)

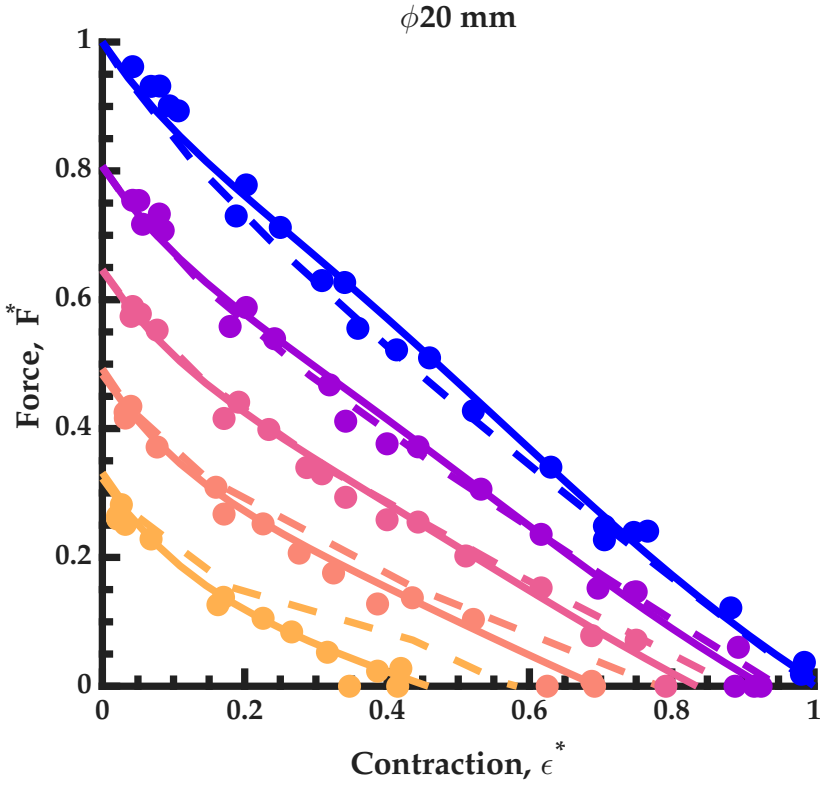


(b)

Figure 3.3: Results for finding the relationship between F_{620}, l_0, P_{620}, and $\epsilon_{620_{10}}$. (**a**) Isoclines



(a)



(b)

Figure 3.4: Surface fit for $F^*(\epsilon^*, P^*)$. Solid lines are from our model. Dashed lines from

We therefore derived our own equation for isometric force in the BPA as a function of pressure and contraction. Visual analysis of the experimental data shows an exponential relationship between ϵ^* and F^*, and a linear relationship between P^* and F. We fit a surface to the data using Nonlinear Least Squares and a Least Absolute Residual robustness such that it is described by Equation (3.4). For the ∅10 mm BPAs, the result of the improved fit can be seen in Figure 3.4a. The adjusted $R^2 = 0.9998$, a RMSE $= 0.004537$, and a maximum absolute residual of 10.6%. Solving (3.4) for the ∅20 mm BPA, yields different coefficients, and results are seen in Figure 3.4b. Coefficient values and goodness-of-fit statistics are found in Table 3.2.

3.4 Effects of Resting Length

We have developed new F_{620} equations for 10 mm and 20 mm diameter Festo BPAs that capture the change in maximum force produced by the actuators at 620 kPa as a function of their resting length. When examining Equation (3.7) and measured data in Figure 3.2b, it can be seen that as l_0 goes to infinity, $F_{620_{10}}$ goes to 470 N. When the Festo tool [Lang, Robert, 2005] was queried at $l_0 = 1$ m as an approximation of infinity, it predicts the ∅10 mm BPA can produce $F = 490$ N at P_{620}. This is within the 10% variability that Festo states may occur in manufacturing tolerances. Similarly, the Festo data predict a ∅20 mm BPA will produce $F = 1570$ N. Our results for Equation (3.8) show that $F_{620_{20}}$ goes to 1460 N as l_0 approaches infinity. Although the difference is much larger, it is also within the 10% manufacturing tolerance.

Our measured data differ significantly from the Festo model as l_0 goes to zero. For the

measured data, as the length gets smaller, the force does as well, whereas the Festo tool predicts exponential force increase (see Figure 3.2). As a specific example, when specifying $l_0 = 112\,\text{mm}$ for the 10 mm diameter BPAs, the Festo predicted F_{620} is 498.6 N. However, using Equation (3.7), F_{620} is calculated to be 335.3 N. Actual force in the $l_0 = 112\,\text{mm}$ was measured at 350.9 N. Therefore, the error in predicted F_{620} for the ∅10 mm BPA is 4.5% for our model and 42.1% using the Festo tool. Researchers using BPA resting lengths under 300 mm should take note of these results.

Festo AG & Co. KG [2026] (p. 12) states that force will be reduced by approximately 10% for a minimum nominal length. In Festo AG & Co. KG [2018] (p. 14) they say the force can deviate by 10% based on factors such as production variation, material variation, and deviation from nominal length (l_0). Festo AG & Co. KG [2013] (p. 13) notes that testing was done with BPAs that are ten times the uninflated diameter, and that theoretical force can increase by up to 10%. In each datasheet, the characteristic curves appear to be the same. As demonstrated above, the 100 mm ∅10 mm BPA should have much less force than what is shown in the curve. This is our experience and the reason for the testing. It is unclear why the MuscleSIM tool [Lang, Robert, 2005] then predicts higher force as resting length decreases. It could be a reversed sign coefficient in their internal curve-fit model. We have shown here that as the nominal length goes to zero, so does force.

BPA models commonly predict higher forces than are measured experimentally because they do not account for end effects in regions of the pressure vessel where those effects dominate. Nominally, the clamped ends fix the inner and outer diameter while allowing axial and radial displacement. In the isometric case, there is only radial displacement. The boundary conditions force steep gradients in stress and strain near the ends. Near

the boundary conditions, axial, radial, and circumferential strains deviate from analytic solutions for a cylindrical model. For a constant pressure, the end effect region appears to have the same length. Therefore, in shorter BPAs, the end effects dominate. The axial length scale of the boundary section is comparable to l_0, and the active force differs substantially from that of an assumed perfect cylinder.

Manufacturing variations seem to play an important role in the behavior of these actuators. It remains to be seen how ϵ_{620} can be known a priori. We still suspect that it is a function of l_0, but it might also be a function of other factors such as product batch number or UV light exposure [Festo AG & Co. KG, 2026] (p. 15); [Festo AG & Co. KG, 2018] (p. 6); [Festo AG & Co. KG, 2013] (p. 13). We found there to be larger variations in the 10 mm BPAs than in the 20 mm BPAs. Modeling work on ideal McKibben actuators describes how the relationship between contraction and initial length is a function of the total number of twists and the angle of twist in the usable muscle area [Chou and Hannaford, 1996]. It is unclear if there are variations in the angle and number of twists in the Festo BPAs due to manufacturing. Uncovering the relationship between ϵ_{620} and l_0 for the Festo DMSP artificial muscles, if it exists in a meaningful way, will require additional controlled tests.

Similarly, it is not clear how the maximum force might change with manufacturing variations. This was reasonably corrected for by normalizing the equations to each individual's maximum contraction, but more nuanced details might emerge with more data collection. Not knowing these relationships a priori means there is still a need for robots and other systems to go through an iterative design stage if it is discovered that the system does not produce the torque or have the RoM that the design team expects. However, the data and models provided here should enable faster redesigns with fewer

additional measurements.

3.5 Nondimensionalized Force Prediction

We have also introduced the concept of nondimensionalized isometric force that is a function of relative strain and relative pressure (Equation (3.4)). This elegant equation has only three coefficients with low error (see Table 3.2). In the work presented here, we introduce the relative pressure term, $P^* = P/P_{620}$. Our data show that by normalizing force, contraction, and pressure, we are able to create a simplified force equation as a function of contraction and pressure that scales well with initial actuator length, as expressed by Equation (3.6) in Section 3.1.

For example, given $l_0 = 54\,\text{mm}$ and $P = 300\,\text{kPa}$, the measured force was $F_{actual} = 103\,\text{N}$. Force prediction using only the Festo tool predicts the force to be $F_{predict} = 270\,\text{N}$. This is 161% greater than the measured force. Using Equations (3.7) and (3.4) with Equation (3.6) gives $F_{predict} = F^*(0, 0.48) \cdot F_{620}(0.054) = 0.49 \cdot 230\,\text{N} = 112\,\text{N}$. This is an error of 8%, which is much more accurate than using the Festo tool only.

The presented work is capable of predicting force output for 10 mm and 20 mm muscles with different initial muscle lengths, internal pressure, and contraction amount with minimal initial data collection. However, it remains to be seen if these relationships are maintained over the full lifecycle of the muscles. There are not yet any peer reviewed studies investigating the long-term behavior of these artificial muscles, though the Festo datasheet provides some indication of what can be expected [Festo AG & Co. KG, 2026] (p.19). They report that the 40 mm BPA can complete 1 million cycles at 6000 N (maximum

rated load), or 10 million cycles at 4000 N. The service life of the other size BPAs is not reported; however, it is mentioned that thermal stressing may reduce the lifespan and they should be inspected every 500,000 cycles for signs of aging [Festo AG & Co. KG, 2018] (pp. 10–11). Future research should look at the behavior of these muscles over time.

In our work, $P_{620} = 620\,\text{kPa}$ is the maximum supply pressure for our system, and other users of this actuator may use a different supply pressure. However, this supply pressure is not required, and normalizing by this value was chosen as a method for improving the optimization techniques by reducing the sensitivity of the term associated with pressure. This equation will work for any pressure supplies below 620 kPa. We also anticipate the equation will also work for pressures up to 800 kPa (the maximum rated pressure for Festo ∅10 mm BPAs); however, we do not have the data to confirm this.

3.6 Comparison with Other Published BPA Models

The proposed model was compared with the formulations by Sárosi [2012], Sárosi and Fabulya [2012], and Martens and Boblan [2017], to evaluate their predictive performance across actuator sizes and operating conditions (see Table 3.3 and Figure 3.5). For the ∅20 mm BPA, parameters from Sárosi [2012] were used, corresponding to actuators with resting lengths of 200 mm and 400 mm. These were compared against the experimental data at $l_0 = 300\,\text{mm}$ and $l_0 = 450\,\text{mm}$, respectively. For the ∅10 mm BPA with resting lengths of 257 mm and 233 mm, parameters from Sárosi and Fabulya [2012] were applied. In each case, the maximum contraction parameter (k_{max}) was determined numerically by solving $F(6.2\,\text{bar}, k_{max}) = 0$, ensuring consistency with the experimental maximum

pressure condition.

Table 3.3: Goodness-of-fit comparison for the Bolen, Sárosi, and Martens and Boblan models at different resting lengths (in mm) for 10 mm and 20 mm BPAs. RMSE and maximum error are reported in newtons, and FVU denotes the fraction of variance unexplained.

Diameter	Length	Bolen			Sarosi			Martens and Boblan		
		RMSE	FVU	Max. Error	RMSE	FVU	Max Error	RMSE	FVU	Max. Error
10 mm	257	22.9	0.0353	46.0	143.8	1.3903	235.8	73.8	0.3668	122.9
	233	16.8	0.0187	30.3	132.7	1.1655	222.6	62.6	0.2590	101.2
20 mm	300	35.6	0.0144	114.0	132.1	0.1992	313.4	72.8	0.0605	137.8
	450	67.6	0.0258	166.6	177.7	0.1785	294.8	862.1	4.2022	1321.1

While the Sárosi [2012] and Sárosi and Fabulya [2012] models capture general trends in force–contraction behavior, discrepancies were observed when applying parameters derived from different actuator lengths. In particular, the model tends to over-predict force at higher pressures and low amounts of contraction, suggesting sensitivity to geometric scaling effects not explicitly accounted for in the formulation.

The Martens and Boblan [2017] model is derived from a geometric and material-based framework and provides a more physically grounded representation of actuator behavior. However, its accuracy depends strongly on precise geometric parameters, including braid angle, wall thickness, and fiber length. Small deviations in these parameters can lead to significant differences in predicted force, particularly at higher pressures.

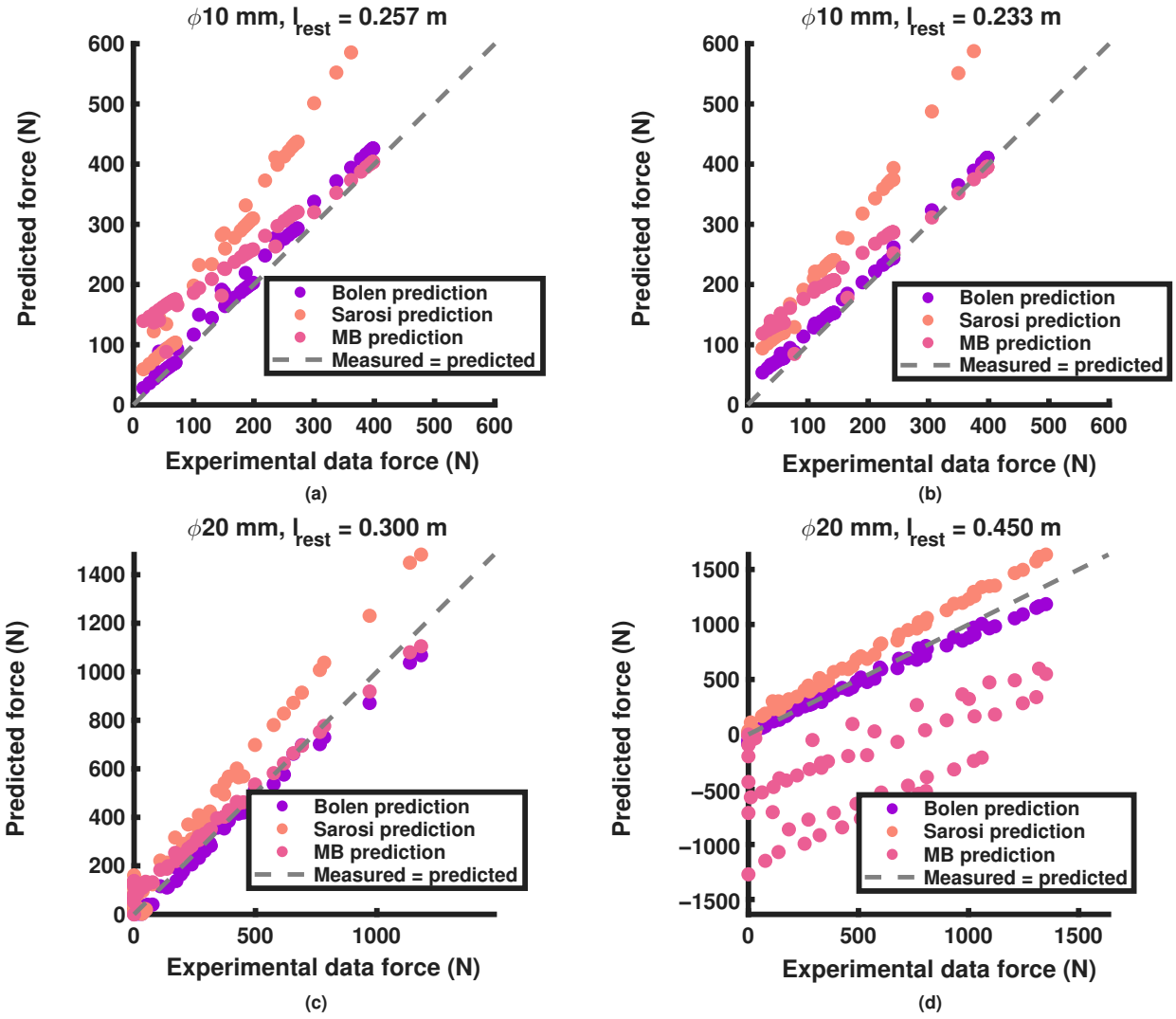


Figure 3.5: Comparison between measured and predicted BPA force at varying amounts of contraction and pressure. A $y = x$ perfect fit line is added for reference. The BPAs compared are: (**a**) ∅10 mm, $l_0 = 257$ mm, (**b**) ∅10 mm, $l_0 = 233$ mm, (**c**) ∅20 mm, $l_0 = 300$ mm, and (**d**) ∅20 mm, $l_0 = 450$ mm.

The Martens and Boblan model was first evaluated at the same pressure–contraction points as in their experimental data [Martens and Boblan, 2017] (Tables A1 and A2),

indicating that it reproduces the general published force map shape but is sensitive to the assumed initial geometry. Examining their ∅20 mm BPA, for example, using the explicitly stated $l_0 = 300\,\text{mm}$ and initial diameter $D_0 = 20\,\text{mm}$ significantly under-predicted the results compared to using the implied $l_0 = 296\,\text{mm}$ ([Martens and Boblan, 2017] (Table A1)) and $D_0 + H_0 = 21.8\,\text{mm}$, where $H_0 = 1.8\,\text{mm}$ is the material thickness. The diameter adjustment can be inferred from [Chou and Hannaford, 1996]. This adjusted diameter was also included for the ∅10 mm BPA to give a better fit. As can be seen in Figure 3.5 and Table 3.3, the highest accuracy is observed when ∅20 mm and $l_0 = 300\,\text{mm}$, which is at or very near their experimental measurements. For the ∅10 mm BPAs the accuracy decreases even though their experimentally measured $l_0 = 250\,\text{mm}$ is close to our $l_0 = 257\,\text{mm}$ and $l_0 = 233\,\text{mm}$. The largest error occurs at ∅20 mm and $l_0 = 450\,\text{mm}$ and the model starts to predict negative force. We have calculated that the $F_{PE} \cdot \frac{dD}{DL}$ term becomes very large compared to the others [Martens and Boblan, 2017] (p. 5). This suggests that the model is sensitive to geometric parameter selection and interpretation, particularly with respect to initial length and diameter definitions.

In comparison, the present normalized model demonstrates improved consistency across actuator lengths by incorporating experimentally derived scaling through F_{620} and ϵ_{620}. This normalization reduces sensitivity to geometric variation and enables a unified representation of actuator behavior, while maintaining predictive accuracy over the tested range of pressures and contractions.

3.7 Implications for Real-Time Control and Simulation

Although pneumatic muscle actuators exhibit hysteresis, thermodynamic effects, and other dynamic behavior, accurate and high-performance controller designs can still be achieved using a static force map approach when the underlying static force model is accurate. Martens and Boblan explicitly note that BPA dynamics are often dominated by the air dynamics inside the actuator and the mass flow behavior of the valve, such that precise modeling of the static force characteristic remains crucial for torque control accuracy [Martens and Boblan, 2017]. This view is consistent with more recent BPA control work, including a sensor-less torque control interface for BPA-driven joints [Martens et al., 2018], stable force/position/stiffness control of antagonistic pneumatic-muscle joints [Ugurlu et al., 2019, Hunt et al., 2017b], and sensor-less angle stiffness control of antagonistic BPA systems [Shin and Kogiso, 2022].

In this context, the present model is well suited for real-time use because it reduces the static isometric force prediction to a low-order normalized surface, $F^*(\epsilon^*, P^*)$, scaled by the resting-length-dependent term $F_{620}(l_0)$. This structure is computationally inexpensive to evaluate online and is also suitable for lookup-table implementation. While the present work does not model the full actuator–valve dynamics, it provides the static force relationship needed as a feedforward or inner-model component in real-time torque control, simulation, and preliminary design workflows. It is anticipated that improved real-time control for high-speed and high-precision systems will benefit more from improved modeling of airflow dynamics than from modeling actuator hysteresis, damping, and other dynamic behavior. Characterizing the actuators under dynamic operating conditions,

such as isokinetic (constant-velocity), isobaric, and isotonic (constant-load) contractions, is a natural next step beyond the isometric characterization reported here, for which the pressure–force dynamics model of Elzein et al. [2025] provides the modeling foundation.

3.8 Biomimetic Artificial Muscle Analogy

BPA artificial muscles are often said to be analogous to biological muscles because they have force–length curves and can produce force only in tension. This analogy is worth closer inspection. For example, the passive force–length curve has been described as an exponential function arising from connective tissue elements, while the active force–length relationship is typically modeled as a bell-shaped curve reflecting actin–myosin overlap [Winters, 1990]. These formulations emphasize that muscle force generation is governed by nonlinear geometric and material interactions rather than simple linear elasticity. McKibben-type pneumatic artificial muscles (of which Festo is an example) exhibit analogous nonlinear behavior, though arising from fundamentally different physical mechanisms. As shown by Chou and Hannaford, the force–length relationship of McKibben actuators is derived from geometric constraints imposed by the braided shell, leading to a nonlinear dependence of force on actuator length through the braid angle Chou and Hannaford [1996]. In their formulation, actuator force is proportional to pressure and a nonlinear geometric term $(3\cos^2\theta - 1)$, which governs both the maximum contraction limit and the curvature of the force–length relationship. The work presented here demonstrates that this nonlinear curve can be well approximated by an exponential as well.

However, the behavior of these artificial muscles differs distinctly from that of biological muscles. For example, the optimal fiber length of a biological muscle, l_{OFL}, is not equivalent to l_0 in an artificial muscle. Artificial muscles produce their maximum active force at their resting length, which is also their maximum length. In contrast, the maximum active force that a biological muscle can produce is somewhere in the middle of the total contractile length of the muscle.

Furthermore, biological muscles can adapt the optimal operating length through structural remodeling (e.g., sarcomere addition or tendon compliance) while BPAs have a fixed geometric configuration. This allows biological systems to self-optimize for the particular tasks the animal is subjected to. In order to re-optimize a robot for new tasks, new actuators would have to be swapped into the system. This distinction highlights a key limitation of BPAs in biomimetic applications, while also reinforcing the importance of accurate nonlinear modeling for control and design.

The analogy also suggests two design observations. First, the force–length penalty is shared by both families: force falls as the actuator operates away from its optimum, so a limb design must route each actuator such that its highest-force operating range coincides with the joint angles where torque is needed most, because the force lost at the extremes of the range of motion cannot be recovered by pressure alone. Second, the two families gain moment arm in different ways. Skeletal muscle attaches to the skeleton itself, through tendons or directly over broad areas of bone, so the moment arms of large animals follow from skeletal geometry. The robot's BPAs instead act through brackets that protrude from the limb segments; that hardware is mechanically exposed, and its series compliance is one of the loss mechanisms quantified in Chapter 4.

The analogy finally raises a question that this dissertation can pose but not settle. Every muscle-like actuator loses force as it operates away from its optimal length, and a soft actuator working against soft structure has no rigid skeleton from which to gain a moment arm. Given those two facts, it may be no accident that nearly all large land animals are built on rigid skeletons and that large soft-bodied land animals are essentially absent. Whether the force–length penalty contributes to that absence, or whether it reflects other constraints such as support and protection, is a question for comparative biomechanics rather than for a single actuator study. For the designer, the lesson holds regardless of the answer: some moment arm must come from somewhere, and on this robot it comes from protruding brackets whose behavior must be modeled rather than ignored.

4 Isometric Knee Torque

Publication note: Sections 4.1–4.4, Section 4.6, and Section 4.8 are also the subject of an unpublished manuscript in preparation by Bolen, Burgess, Morrow, Elzein, Pang, Scharzenberger, and Hunt [Bolen et al., 2025]. This manuscript has not been submitted for publication. Author contributions for both works are stated in Chapter 3.

4.1 Robot Leg Model and Knee Test Stands

The primary components of the robot leg are the knee joint, femur, tibia, BPA assemblies, and artificial tendons (Fig. 4.1**A–C**). The artificial bone components are 3D printed using a combination of Onyx, carbon fiber, and high-strength high-temperature fiberglass on Markforged Onyx One and Mark Two printers, as well as machined links for the knee mechanism. Onyx is a proprietary Markforged material that consists of chopped carbon fiber in nylon. Artificial tendons are made with Shimano bicycle brake cable.

The leg assembly geometry and dimensions are based on bones scans of a 6 ft (1.83 m) tall person [Bolen and Hunt, 2019, Seth et al., 2018, Delp et al., 1990, 2007]. Two knee joints were tested, one pure rotational joint, and one biomimetic joint with rotation and translation. Each joint is driven by two antagonistic Festo actuators. Both robot joints have the same center of rotation in the femur frame at $\theta_k = 0\,\text{deg}$.

Details on the biomimetic robot knee used in this study can be found in Steele [2018], Steele et al. [2017, 2018]. In brief, this 1-DoF joint (Fig. 4.2) uses a four-bar linkage to change the effective moment arm during rotation. The intersection point of the internal

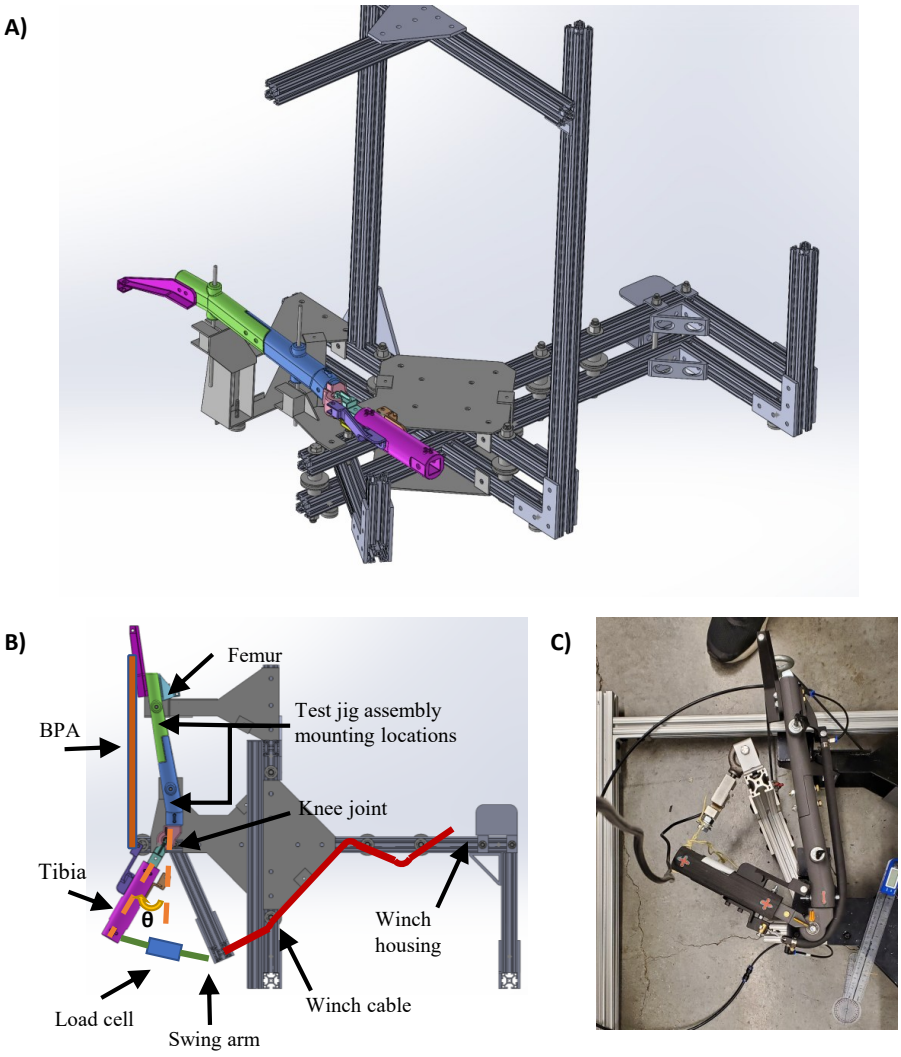


Figure 4.1: **(A)** CAD solid model assembly of the revolute joint knee in the test stand. **(B)** 1-DoF pinned-joint robot knee in the test apparatus with important components labeled. Leg is shown with 30 degrees flexion, i.e. $\theta_k = -30^\circ$. **(C)** Picture from isometric torque testing. Setup shows the revolute knee configured for a test of the extensor BPA. The knee is flexed with $\theta_k = -120^\circ$. Also note the compressed shape of the inflated BPA on the knee during this high degree of flexion.

crossed links define the instantaneous center of rotation (ICR). This causes the joint ICR location in the femur frame to translate posteriorly in the X direction and distally in the Y direction except for a brief proximal move near full flexion. Reference coordinate frames are given by Wu and Cavanagh [1995], Wu et al. [2002]. This is similar to how the human knee behaves [Bolen and Hunt, 2019, Morrow et al., 2020, Seth et al., 2018, Delp et al., 2007, 1990]. The correct location of the ICR is important for both theoretical and experimental torque calculations.

Uniarticular knee extensor and knee flexor are used to actuate the joint. Each knee joint type is tested using ∅10 mm Festo BPAs. With this size BPA the muscles were used in a singular configuration. The biomimetic knee joint is tested using a ∅20 mm Festo BPA. The extensor is actuated with two BPAs in parallel, and the flexor is actuated with two BPAs in with pathways mirrored about the Z axis. Each artificial muscle is pinned at the muscle origin location on the femur while the other end is either pinned or attached to the muscle insertion location on the tibia via an artificial tendon.

Torque about the knee is generated by the moment of the BPA force about the joint's instantaneous center of rotation (ICR). Two vectors enter that moment, and each is a quantity with a magnitude and a direction, not a direction alone: the position vector $\mathbf{d}$, drawn from the ICR to the point where the actuator force is applied, and the force vector $\mathbf{f} = \|\mathbf{f}\|\hat{\mathbf{u}}$, whose direction $\hat{\mathbf{u}}$ is the line of action of the actuator. The moment of the force about the joint is $\mathbf{m} = \mathbf{d} \times \mathbf{f}$, and its magnitude is the force magnitude multiplied by the moment arm r, the perpendicular distance from the joint axis to the line of action,

$$r = \|\mathbf{d} \times \hat{\mathbf{u}}\| \tag{4.1}$$

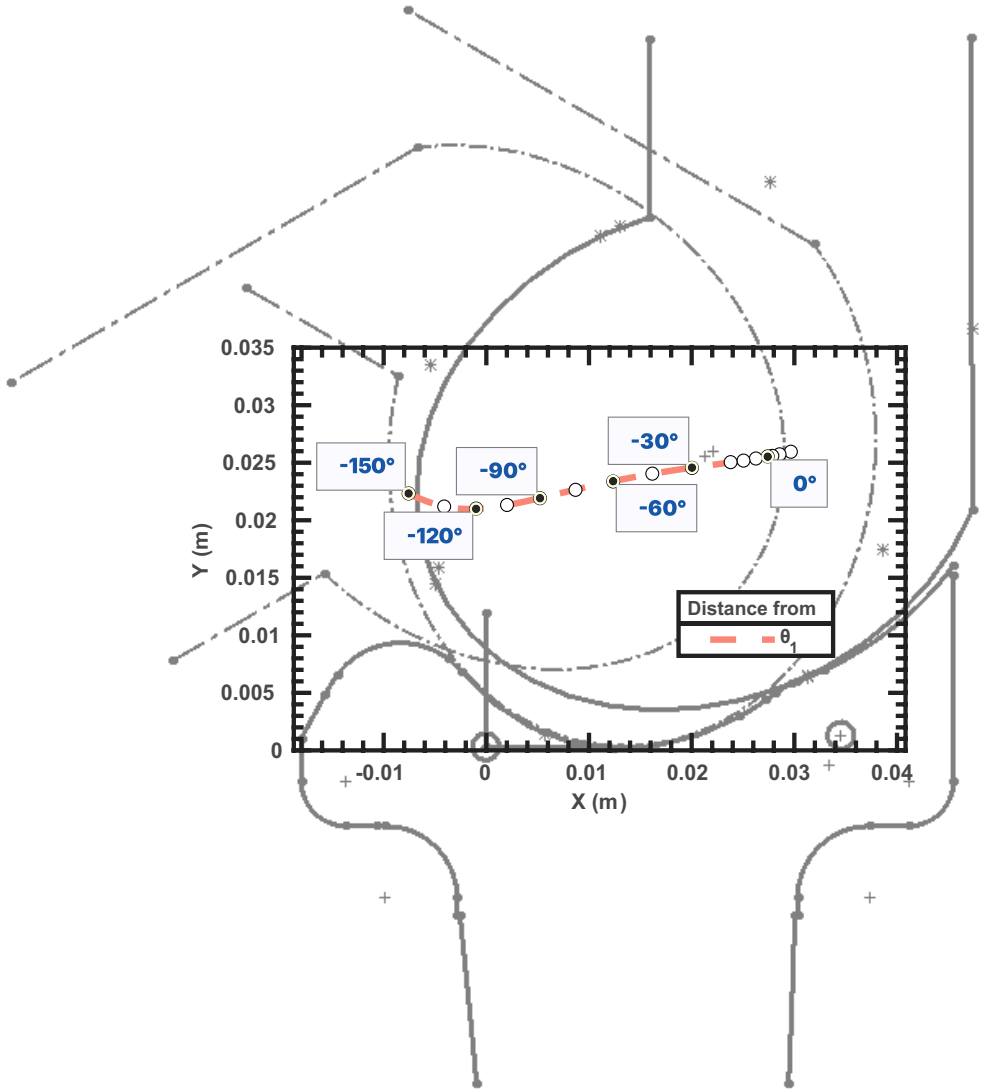


Figure 4.2: ICR for biomimetic robot knee as a function of knee angle. Measured from θ_1 on the tibia. Plot is overlayed on model of the knee at $\theta_k = 0^\circ$ (solid line), and -60° and -120° (dotted dash lines). White circles represent measured points in -15° increments from -150° to -15°, -5° increments from -15° to 0°, and then at 2°, 5°, and 10°.

The moment arm is therefore a scalar, and it is the same quantity whether it is computed from the position vector with Eq. (4.1) or measured directly with a tape measure. In this study it is calculated using the method specified by Young et al. [2019]. The scalar moment arm r_k about the joint axis $\hat{\mathbf{z}}_k$ (i.e., the Z axis in the knee frame) is calculated as

$$r_k = \mathbf{p}_{proj,xy} \cdot \frac{\mathbf{p}_f \times \hat{\mathbf{z}}_k}{\|\mathbf{p}_f \times \hat{\mathbf{z}}_k\|} \tag{4.2}$$

where $\mathbf{p}_{proj,xy}$ is the free muscle segment projected onto the XY–plane, which is normal to the joint axis, and $\mathbf{p}_f$ is the projected muscle segment vector. Moment arm is calculated in the tibial body frame.

Details of the existing model and the torque calculation methodology in the modeled robot are given by Bolen and Hunt [2019], Morrow et al. [2020].

4.2 Knee Torque Data Collection and Equipment

A test jig was built to collect knee torque measurements at different knee angles over its range of motion. The test stand frame is made predominantly out of 80/20 1515 series extruded aluminum components. The knee joint is free to rotate the tibia while the femur is fixed to the frame. A force sensor has one end connected to the tibia and the other is connected to a swing arm. The swing arm is tied to a winch with 3/16 inch Kevlar rope from Quality Nylon Rope routed over intermediate idler pulleys (see Fig. 4.1**B** and 4.1**C**). At each measured position of θ_k the winch would be locked and the BPA would be inflated and then deflated. This test procedure kept the measured results as isometric as possible. The knee joint is co-axial with gravity to eliminate gravitational torque (see Fig. 4.1**A** and

Fig. 4.1**C**).

Force data was collected using one of two different sensors. The first is a MODERN STEP 300 kg digital crane scale. The second force sensor is a CALT DYLY-103 100 kg S shaped load cell. The load cell is used in conjunction with a HX711 Load Cell Amplifier. Pressure is measured with a Freescale MPX5700 GP 5 V pressure sensor. Building air supply pressure is controlled with two pressure regulators in series. The first is a Parker model 20R113GC $0 - 120$ psi pressure regulator. The second is a Husky $3/8$ inch High Performance Air Regulator HDA72200. A Festo VTUG-10-MRCR-S1T-26V20-T516LA-UL-T532S-8K valve manifold was used to supply air to the actuator. Pressure and load cell amplifier data are sent to MATLAB via an Arduino Uno style Sparkfun BlackBoard C microcontroller.

The length measurements were made using a tape measure. When there was sufficient curvature in the BPA during knee extensor torque tests, a flexible piece of string was used to mark the axial length and then transferred to the tape measure. Angle measurements were taken with a Medigauge digital electronic goniometer or with MALENOO analog goniometers of 6, 8, or 12 inch lengths.

4.3 Hybrid Torque Calculation Method

A Hybrid Torque calculation method was developed to help uncover inconsistencies between the model and recorded torque values. This method measures the actual muscle length and angles of the robot to calculate force and joint torque. With this method, it is easier to determine which parts of the torque equation did not achieve their predicted

value and provide a correction to that part of the equation. The classical mechanics way to calculate the torque about a joint is to take the cross product of the position vector **d** and the force vector **f**, both defined in Section 4.1,

$$\mathbf{m} = \mathbf{d} \times \mathbf{f} \tag{4.3}$$

Knee torque is the scalar component of this moment about the joint axis, written M_z to make its axis explicit. Because the moment arm is the perpendicular distance from that axis to the line of action, the vector cross product of Eq. (4.3) reduces to an ordinary scalar product,

$$M_z = r\,F \tag{4.4}$$

where r is the scalar moment arm of Eq. (4.1), here measured directly with a tape measure as the perpendicular distance from the knee axis to the actuator's line of action, and F is the scalar force magnitude, calculated by measuring the BPA's P, l_0, l_{620} and l_{m} values and using the model fit equations determined in the following section. The hybrid method therefore replaces the model-predicted moment arm r_k of Eq. (4.2) and the model-predicted muscle length with their measured counterparts, leaving the force model as the only predicted quantity.

4.4 Improved Torque Model

The classical kinematic torque model assumed rigid links and inextensible muscle paths. Data collected during the first flexor experiments (Chapter 4) showed that this assumption does not hold for a pneumatic system: measured muscle lengths were consistently shorter than predicted, and measured torque deviated from prediction in a load-dependent manner. This section describes the corrections that were developed to account for a constant length offset, series compliance in the attachment hardware and artificial tendons, and the loss of usable muscle length when a BPA wraps around the joint.

These corrections are deliberately formulated to be identified from experiment rather than derived analytically, because the physical effects they represent are distributed and not separately measurable. The lumped compliance absorbs the cantilevered three-dimensional-printed attachment brackets, whose geometry is a printed CAD part rather than a prismatic beam with a tractable second moment of area, together with the bone fixtures, the cable winch of the test stand, and the artificial tendon. Predicting that aggregate stiffness from first principles would require finite-element models of each component with material properties (anisotropic printed polymer, braided sleeve, construction-dependent tendon stiffness) that are themselves unmeasured. The correction terms therefore parameterize only what the torque model needs: χ_0 absorbs the constant length discrepancy observed at zero load, χ_1 and χ_2 are effective stiffnesses in the bracket frame's principal directions, and χ_3 scales the wrapping-length loss. Each is identified by global optimization against the measured torque rather than solved in closed form, for two reasons. First, the terms enter the prediction through an implicit force balance, because the compliance

displacement depends on the force that it itself modifies, so the inverse problem has no analytic solution. Second, the parameters are only jointly identifiable: preliminary fits showed that widely different stiffness pairs can reproduce the training tests nearly equally well, so the adopted solution must be selected by cross-validated prediction on held-out tests rather than by residual minimization alone, which is what the multiobjective Pareto-front procedure of Section 4.5 provides.

Analysis of measured muscle length data shows that it was consistently shorter than the predicted muscle length at all joint angles. This shorter length results in less muscle force, a shifted RoM, and less torque in the experimental setup. That this shorter length occurs when $\varepsilon = 1$ and $F = 0$ indicates that a constant length offset should be used. Therefore, to produce an improved torque prediction model, l_{m} was recalculated as

$$l_{\mathrm{m}} = l_{LMT} - 2\, l_{fitting} - \chi_0 \tag{4.5}$$

where l_{LMT} is the path length of the musculotendon from origin to insertion point, $l_{fitting}$ is the length of the end caps, and χ_0 is a constant length offset.

Additionally, as the muscles produce more force, compliant components in the limb segments and attachment hardware deform. This deformation changes the effective braided pneumatic actuator (BPA) length, which alters contraction and therefore changes the force predicted by the BPA model. This effect is most apparent at high forces and low contractions. The purpose of the compliance model developed next is predictive, not structural: it exists so that the torque calculation can correct the actuator length for load-dependent deformation, and it makes no claim about the stiffness of any one component

in isolation. Accordingly, the stiffness values that populate the model, χ_1 and χ_2, are not measured on the hardware directly; they are identified by the optimization of Section 4.5 from the measured torque, and every stiffness value quoted later in this chapter is an optimizer output. The combined hardware compliance is approximated using Hooke's law in the bracket's principal directions.

A local bracket frame was established at the attachment bracket. The $\hat{\mathbf{x}}_{br}$ axis was aligned with the vector from the bracket reference point to the BPA attachment point and represents the stiffest direction of the bracket. The $\hat{\mathbf{y}}_{br}$ axis was chosen to point in the posterior direction with respect to the tibia frame, corresponding to the primary cantilever bending direction (see Fig. 4.4), while the $\hat{\mathbf{z}}_{br}$ axis completes the right-handed frame. In this frame the bracket is modeled as three independent linear springs along the principal axes, with stiffnesses k_x, k_y, and k_z in the x_{br}, y_{br}, and z_{br} directions.

For the two-rotation evaluator, the bracket frame was constructed from the vector joining the bracket origin to attachment point $\mathbf{p}_A$ or $\mathbf{p}_B$. The first rotation, through θ_z about the fixed space-frame axis $\hat{\mathbf{z}}_s$, points the intermediate axis $\hat{\mathbf{x}}_1$ toward the sagittal-plane projection $\mathbf{p}_{A/B}^{\text{sag}}$. The second rotation, through $-\theta_y$ about the current axis $\hat{\mathbf{y}}_1 = \hat{\mathbf{y}}_{br}$, brings the final bracket axis $\hat{\mathbf{x}}_{br}$ into alignment with the full three-dimensional point vector. Figure 4.3 shows these geometric targets. The resulting orientation and homogeneous transform are

$$\mathbf{R}_{sb} = \mathbf{R}_z(\theta_z)\mathbf{R}_y(\theta_y)^T, \qquad \mathbf{T}_{sb} = \begin{bmatrix} \mathbf{R}_{sb} & \mathbf{p}_{br} \\ \mathbf{0}^T & 1 \end{bmatrix}. \tag{4.6}$$

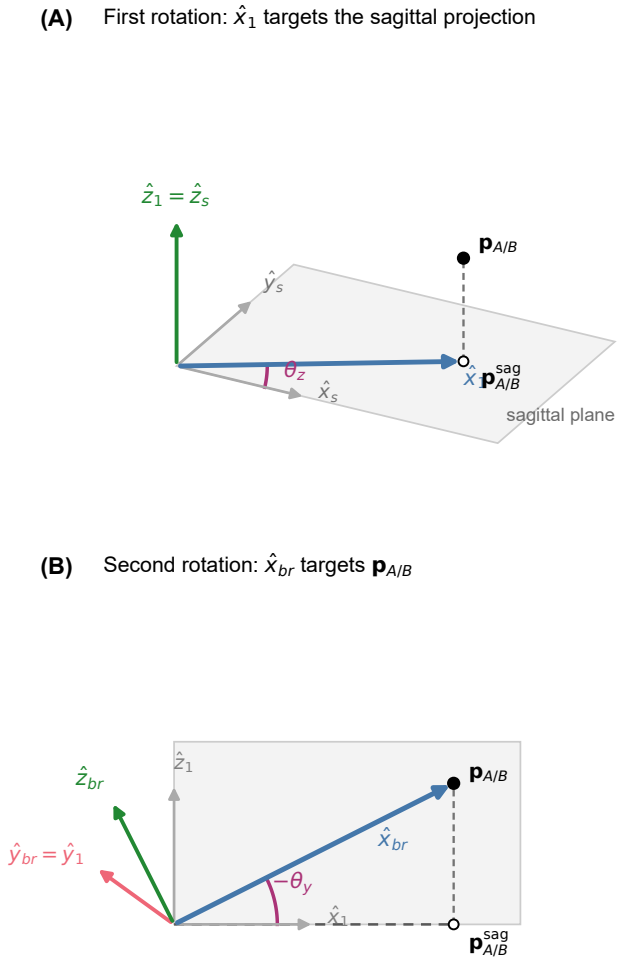


Figure 4.3: Two-rotation construction of the bracket frame from attachment point $\mathbf{p}_A$ or $\mathbf{p}_B$. **(A)** Rotation through θ_z about fixed $\hat{\mathbf{z}}_s$ aligns $\hat{\mathbf{x}}_1$ with the point's sagittal-plane projection. **(B)** Rotation through $-\theta_y$ about current $\hat{\mathbf{y}}_1 = \hat{\mathbf{y}}_{br}$ aligns $\hat{\mathbf{x}}_{br}$ with the full point vector.

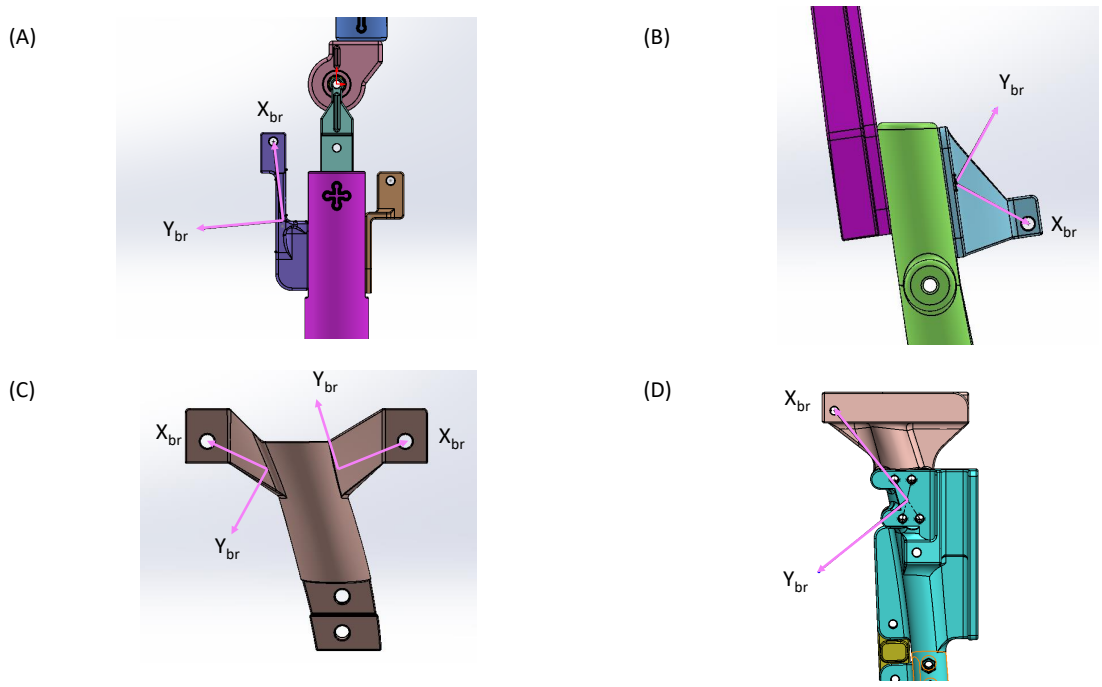


Figure 4.4: Bracket frame orientations for: **(A)** flexor configuration on the pinned knee, **(B)** extensor configuration on the pinned knee, **(C)** flexor and extensor configurations on the biomimetic knee with ∅10 mm BPAs, and **(D)** flexor configuration on the biomimetic knee with the ∅20 mm BPA.

$$\mathbf{K}_{br} = \mathrm{diag}\big(k_x,\ k_y,\ k_z\big) \tag{4.7}$$

These three principal stiffnesses are not independent in this model, and the reason follows from the frame construction itself. The first rotation of Eq. (4.6) is fixed by the sagittal-plane projection of the attachment point, but the orientation of the bracket frame about its own $\hat{\mathbf{y}}_{br}$ axis is a modeling choice: the pitch-only convention (no second rotation) and the two-rotation convention of Fig. 4.3 describe the same physical bracket and differ by exactly a rotation about $\hat{\mathbf{y}}_{br}$. A diagonal stiffness array is invariant under such a rotation, $\mathbf{R}_y(\theta)\,\mathbf{K}_{br}\,\mathbf{R}_y(\theta)^T = \mathbf{K}_{br}$ for all θ, if and only if the two stiffnesses in the plane of the rotation are equal, $k_x = k_z$. Adopting this y-symmetric form therefore makes the compliance prediction independent of the frame convention, and the redundant third stiffness is eliminated: with this array the two frame conventions produce identical torque predictions to numerical precision. Two stiffness values remain, χ_1 in the x_{br} and z_{br} directions and χ_2 in the cantilever bending direction y_{br}:

$$\mathbf{K}_{br} = \begin{bmatrix} \chi_1 & 0 & 0 \\ 0 & \chi_2 & 0 \\ 0 & 0 & \chi_1 \end{bmatrix} \tag{4.8}$$

The corresponding local compliance matrix is

$$\mathbf{C}_{br} = \mathbf{K}_{br}^{-1} \tag{4.9}$$

The force applied to the bracket was expressed in the bracket frame as $\mathbf{f}_{br}$. This force

vector can be written in terms of its magnitude and unit direction as

$$\mathbf{f}_{br} = F\,\hat{\mathbf{u}} \tag{4.10}$$

where $F = \|\mathbf{f}\|$ is the scalar force magnitude of Eq. (4.4) and $\hat{\mathbf{u}} = [u_x, u_y, u_z]^T$ is the BPA force direction expressed in the bracket frame. The bracket deflection vector, $\boldsymbol{\epsilon}_{br}$, was calculated from Hooke's law as

$$\mathbf{f}_{br} = \mathbf{K}_{br}\,\boldsymbol{\epsilon}_{br} \tag{4.11}$$

Solving Eq. 4.11 for the bracket deflection gives

$$\boldsymbol{\epsilon}_{br} = \mathbf{K}_{br}^{-1}\,\mathbf{f}_{br} = \mathbf{C}_{br}\,\mathbf{f}_{br} \tag{4.12}$$

Substituting Eq. 4.10 into Eq. 4.12 gives

$$\boldsymbol{\epsilon}_{br} = F\,\mathbf{K}_{br}^{-1}\,\hat{\mathbf{u}} \tag{4.13}$$

For the diagonal stiffness matrix in Eq. 4.8, Eq. 4.12 is equivalent to the component-wise expression

$$\boldsymbol{\epsilon}_{br} = \begin{bmatrix} F\,u_x/\chi_1 \\ F\,u_y/\chi_2 \\ F\,u_z/\chi_1 \end{bmatrix} \tag{4.14}$$

Although $\boldsymbol{\epsilon}_{br}$ is a three-dimensional deflection vector, the BPA force model depends

on the scalar change in effective BPA length (δ). Therefore, only the component of $\boldsymbol{\epsilon}_{br}$ along the BPA line of action directly changes the effective BPA length. This scalar bracket deflection is the projection of $\boldsymbol{\epsilon}_{br}$ onto $\hat{\mathbf{u}}$:

$$\delta_{br} = \hat{\mathbf{u}}^T \boldsymbol{\epsilon}_{br} \tag{4.15}$$

Substituting Eq. 4.12 into Eq. 4.15 gives

$$\delta_{br} = \hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \mathbf{f}_{br} \tag{4.16}$$

Using Eq. 4.10, Eq. 4.16 becomes

$$\delta_{br} = F \, \hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \hat{\mathbf{u}} \tag{4.17}$$

The scalar stiffness along the BPA line of action due to the bracket compliance is therefore

$$k_{\hat{u},br} = \frac{F}{\delta_{br}} = \left(\hat{\mathbf{u}}^T \mathbf{K}_{br}^{-1} \hat{\mathbf{u}} \right)^{-1} \tag{4.18}$$

This projected-compliance form is used because the force is applied along $\hat{\mathbf{u}}$, while the bracket is allowed to deflect independently along its local $\hat{\mathbf{x}}_{br}$, $\hat{\mathbf{y}}_{br}$, and $\hat{\mathbf{z}}_{br}$ directions. The reduced-order Hooke's law expression for the bracket deflection along the BPA line of action is therefore

$$F = k_{eq} \, \delta \tag{4.19}$$

Here k_{eq} is the series sum of every compliant element in the actuator path, and the bracket-only case is the special case with no tendon. When the BPA attaches directly to the bracket, k_{eq} excludes the tendon and equals $k_{\hat{u},br}$ of Eq. (4.18). When an artificial tendon was used, it was modeled as an axial spring with stiffness k_{ten}: bracket deflection and tendon stretch both shorten the effective BPA length along the actuator line of action, the two elements carry the same force, and their scalar compliances add in series,

$$k_{eq} = \left(\frac{1}{k_{\hat{u},br}} + \frac{1}{k_{ten}} \right)^{-1} \tag{4.20}$$

This series reduction is exact only under a specific set of conditions, each of which corresponds to a way the physical hardware could violate it. The bracket and the tendon must carry the same force, which holds for two massless elements coupled in series under quasi-static loading; a compliant coupling between them would break this. Their displacements must be kinematically additive along the line of action, with no displacement lost to off-axis geometry. The deflections must be small enough that the line of action $\hat{\mathbf{u}}$ does not rotate as the bracket deflects, which is why $\hat{\mathbf{u}}$ is evaluated once, at the undeflected configuration. The elements are assumed to carry no moment, so rotational compliance and force–moment coupling at the attachments are neglected. Both elements are linear, with stiffness independent of deflection, and the tendon is modeled without preload, so its tangent stiffness does not stiffen with load state. Finally, the sign convention and geometry are such that compliance deflection shortens the actuator, as Eq. (4.21) assumes; under a different geometry the same deflection would lengthen the effective actuator instead.

Since δ is the scalar compliance displacement along the BPA line of action, then the

compliance-corrected active BPA length used in the force balance is

$$l_m(\delta) = l_m(0) - \delta \tag{4.21}$$

where $l_m(0)$ is the initial muscle length calculated from (4.5).

For a specified set of correction parameters, the following calculation solves the attachment-compliance displacement at each knee angle; it is distinct from the optimization that identifies those parameters from experimental torque data (Section 4.5). The corresponding relative contraction is $\varepsilon^*(\delta)$. Since the BPA force depends on contraction, and contraction depends on the compliance displacement, δ is the root of the nonlinear force-balance equation

$$F_{\max} \cdot F_{\mathrm{BPA}}(\varepsilon^*(\delta), P^*) - k_{eq}\,\delta = 0 \tag{4.22}$$

The root is found with a bracketed scalar solver (`fzero` in MATLAB) on the physically admissible interval between zero deflection and the deflection that would bring the actuator to its maximum usable contraction. Termination is controlled by the bracket, not by a residual tolerance: the iteration closes in on the root and stops when the bracket has collapsed to adjacent floating-point values, so the returned δ is converged to machine precision. Two safeguards act as secondary stop conditions. If the actuator already sits at or beyond its maximum usable contraction at zero deflection, no root is sought and δ is set to zero; if the residual shows no sign change across the interval, the solver likewise returns $\delta = 0$ rather than a spurious root.

After solving Eq. 4.22, the corresponding force magnitude was used in Eq. 4.12 to

compute the vector bracket deflection. This deflection was transformed back into the limb reference frame and used to update the BPA attachment location. The routed BPA length, force direction, moment arm, and torque were then recalculated using the updated attachment location. In the present implementation, the optimized stiffness terms χ_1 and χ_2 were treated as effective compliance parameters for the attachment system rather than direct measurements of an isolated bracket. This allowed unmodeled compliance in the surrounding hardware to be represented by the reduced-order stiffness model.

Multiobjective optimization using `gamultiobj` in MATLAB was used to determine the length offset (χ_0) and the stiffness terms (χ_1 and χ_2) until the error between the model and the experimental data was minimized for the $l_0 = 48.5\,\text{cm}$ flexor BPA (Fig. 4.6**C**).

The extensor configuration required one further correction, for a loss with a different physical origin than the compliance terms. As the knee flexes, the extensor BPA bends over the front of the joint and makes contact with it, so part of the actuator follows the joint surface instead of a straight line. That wrapped section stops contributing usable contraction length: the pressurized wall on the inside of the bend carries less tension than the wall on the outside, which both unbalances the actuator's force and consumes part of the wrapped arc length that would otherwise shorten the path. A wrapping-loss term χ_3 was introduced to capture this loss, such that

$$\Delta l = \begin{cases} \chi_3 \; R \; |\theta_{\text{wrap}} - \theta_{\text{k}}| \; \zeta^2 & \theta_{\text{k}} \leqslant \theta_{\text{wrap}} \\ 0 & \theta_{\text{k}} \geqslant \theta_{\text{wrap}} \end{cases} \tag{4.23}$$

where Δl is the change in usable length of the BPA, R is the wrapping radius, $|\theta_{\text{wrap}} - \theta_{\text{k}}|$

is the angle through which the BPA is bent once contact begins, and θ_{wrap} is the knee flexion angle at which the BPA starts to bend over the knee joint, determined from solid modeling. ζ is the additive complement to relative strain, such that

$$\zeta = 1 - \varepsilon^* \tag{4.24}$$

so the wrapping loss grows quadratically as the actuator approaches its resting length and vanishes at full contraction. Combining (4.24) into (4.23), and subtracting Δl from (4.5) yields

$$l_{\text{m}} = l_{LMT} - 2\, l_{fitting} - \chi_0 - \Delta l \tag{4.25}$$

The optimization algorithm was again `gamultiobj` in MATLAB. This time, the muscle origins bracket was chosen. The bracket frame is shown in Fig. 4.4**B**. Since the bracket forms an arch shape and extends for 75 mm in the Z direction, for this configuration, we set χ_2 in the X_{br} direction, and χ_1 in the Y_{br} and Z_{br} directions. The optimization algorithm for this set solves for the best average goodness-of-fit (GoF) measures RMSE, FVU, and maximum absolute residual for two BPAs, while the third one was held out. Three independent rounds of optimization were performed, so that one of each of the BPAs could be held out as validation. Then we picked the solution with the shortest distance between the average normalized GoF and the validation data, while also requiring improved GoF values for all the individual BPAs, with the exception of maximum absolute residual on the $l_0 = 48.0$ cm BPA.

This initial identification was subsequently extended to the full archived test sets in

two stages. First, the flexor model was given a second compliant bracket at the origin-side attachment (a hip-frame cantilever carrying the BPA origin), so that the compliance of both ends of the actuator path is represented; the stiffness pair (χ_1, χ_2) is shared by the two brackets in fixed axis assignments (Appendix C, Table C6), leaving a three-term fit (χ_0, χ_1, χ_2). The flexor fit was cross-validated over the five pinned-knee flexor tests, holding out the encoder-corrected 47 cm test as validation. Second, the extensor refit locked (χ_1, χ_2) to the flexor front values, enforcing a single stiffness pair across the flexor and extensor configurations, and solved only the extensor-specific terms (χ_0, χ_3) on the six-test extensor validation pool, with the extensor length offset allowed to take negative values. The bounds and the adopted values of both stages are recorded in Appendix C (Tables C7 and C8); the fit outcomes are summarized in Section 4.7.

Figure 4.5 distinguishes the three-dimensional displacement of each bracket from the scalar displacement along the BPA force path. For bracket j, the displacement resolves into components $\eta_{1,j}$, $\eta_{2,j}$, and $\eta_{3,j}$ along the local bracket-frame basis, whereas only its projection $\delta_{br,j}$ along the corresponding force direction $\hat{\mathbf{u}}_j$ changes the actuator length. The two bracket projections act in series with the tendon elongation δ_{tendon}:

$$\begin{aligned} \boldsymbol{\eta}_{br,j} &= \sum_{i=1}^{3} \eta_{i,j} \hat{\mathbf{e}}_i^{(j)}, \\ \delta_{br,j} &= \hat{\mathbf{u}}_j^T \boldsymbol{\eta}_{br,j}, \qquad j \in \{1, 2\}, \\ \delta_{\text{series}} &= \delta_{br,1} + \delta_{\text{tendon}} + \delta_{br,2}. \end{aligned} \tag{4.26}$$

The goodness-of-fit metrics used to judge the optimization results are the root mean square error (RMSE), the fraction of variance unexplained (FVU), and the maximum

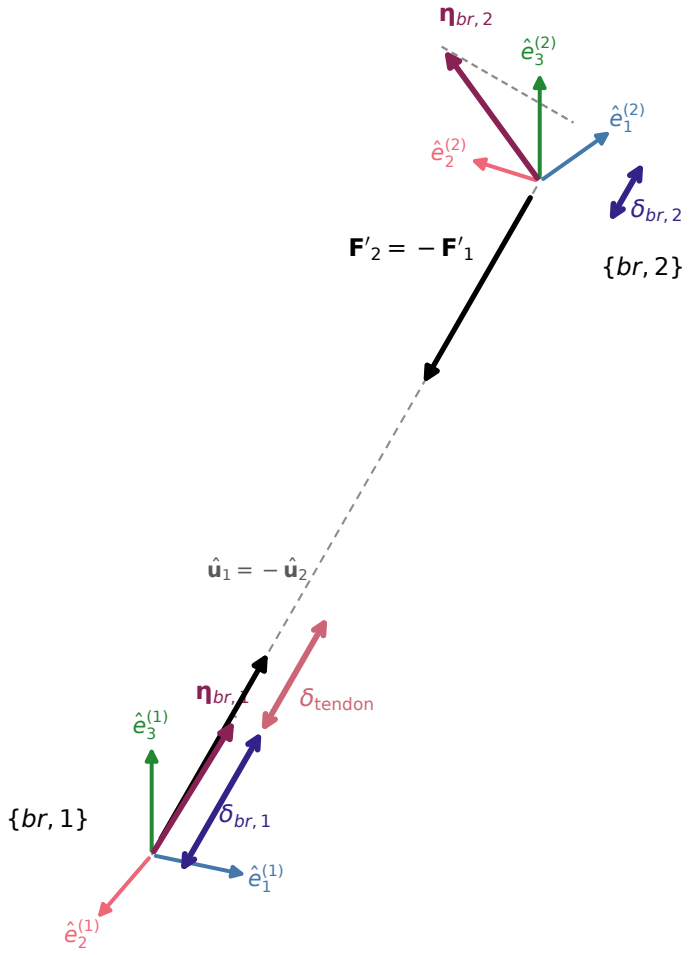


Figure 4.5: Two-bracket deflection and projection along the BPA force path. The force vectors at $\{br, 1\}$ and $\{br, 2\}$ point toward one another and satisfy $\mathbf{f}_2 = -\mathbf{f}_1$. The bracket projections $\delta_{br,1}$ and $\delta_{br,2}$ are offset from the corresponding coordinate frames for clarity, and tendon elongation δ_{tendon} continues collinearly from $\delta_{br,1}$. Colors follow a color-vision-deficiency-safe palette and are redundant with labels and line weights.

absolute residual, each computed between the measured and predicted isometric torque over the tested range of motion.

Each metric isolates a different aspect of disagreement between the two curves. RMSE reports the overall magnitude of the error and, when the curve shapes already agree, is dominated by the remaining level offset. FVU normalizes the error variance by the variance of the measured torque, making it scale-free: it asks whether the prediction reproduces the variation of the measurement rather than whether it is close in absolute units, is the complement of the Nash–Sutcliffe model efficiency [Nash and Sutcliffe, 1970], and takes values below one when the prediction outperforms the mean of the measurement itself. The maximum absolute residual is a worst-case local measure. A small shift of the predicted curve along the knee-angle axis produces large pointwise residuals where the torque gradient is steep and small residuals elsewhere, so this metric is sensitive to phase misalignment while responding only weakly to uniform level offsets.

This interpretation follows the standard decomposition of curve-comparison error into offset, amplitude, and pattern components [Taylor, 2001]: the mean square error between two curves separates into the squared mean offset, an amplitude mismatch, and a residual shape term. An equivalent three-term formulation, splitting waveform similarity into offset, amplitude, and shape, underlies the linear fit method developed for clinical gait waveform comparison [Iosa et al., 2014]. The metrics used here map onto this decomposition, with RMSE carrying the level disagreement, FVU the shape and amplitude agreement, and the maximum absolute residual serving as the phase-sensitive worst case. FVU is treated as the primary criterion because it is the only one of the three that is directly comparable across tests with different torque scales.

4.5 Optimization Algorithms and Cost Functions

Optimization enters this chapter in two places, and in both it replaces a calculation that cannot be carried out any other way. The correction terms of the improved torque model are quantities that no instrument measures directly: they are hidden inside an implicit force balance, and their best values are the ones that make the model's predicted torque match the measured torque. The route redesign is the mirror problem: the attachment locations are free design variables, and their best values are the ones that make the redesigned routes meet or exceed human torque capability. In both cases a search algorithm proposes candidate values, the physics model evaluates them, and the search proceeds until a stopping rule is met. This section states what is searched for, which solver was chosen and why, and when each search stops.

Two distinct optimization problems underpin the results of this dissertation and the design framework built on them. The first is the identification of the correction terms of Section 4.4 (χ_0, χ_1, χ_2, and χ_3) from the pinned-knee test data. The second is the redesign of the biomimetic knee route geometry, meaning the BPA origin $\mathbf{p}_1$, insertion $\mathbf{p}_{\text{end}}$, resting length, and tendon length (Appendix C), so that the flexor and extensor configurations meet or exceed human knee torque capability over the range of motion. Both are carried out in MATLAB [The MathWorks, Inc., 2025].

The correction-term identification is a genuine multiobjective problem: candidate parameter sets are judged simultaneously on RMSE, FVU, and maximum absolute residual (Section 4.4). It is solved with the genetic multiobjective solver `gamultiobj`, an implementation of the elitist nondominated-sorting genetic algorithm NSGA-II [Deb et al., 2002, The

MathWorks, Inc., 2025], which returns a Pareto front of parameter sets from which the adopted solution is selected by the holdout-validation procedure described above. The production runs use adaptive-feasible mutation and scattered crossover at a fraction of 0.8, with MATLAB's default Pareto fraction of 0.35; the population settings are stage-specific. The flexor two-bracket fit evolves a population of 150 for up to 600 generations and terminates early once the geometric average relative change in the spread of the Pareto front falls below 4×10^{-3}; the extensor refit uses a population of 50 for up to 150 generations with a spread tolerance of 1×10^{-3}. In both stages the parameters are searched in a normalized space, with length offsets in centimeters and stiffnesses on a base-ten logarithmic scale, so that the genetic search samples the stiffness bounds, which span an order of magnitude, uniformly by decade, and the initial population is seeded around the expected solution region. The search bounds and the adopted values are recorded in Appendix C.

The route redesign replaces the multiobjective formulation with a single scalar cost assembled from weighted penalty terms, because the route prediction (contact activation, wrapping, and compliance correction) is expensive to evaluate and only moderately smooth. The cost has the form

$$J = w_w J_{\text{worst}} + w_t J_{\text{torque}} + \textstyle\sum_a J_a + J_{\text{geom}} + J_{\text{len}}, \tag{4.27}$$

where J_{worst} is the largest fraction by which robot torque falls short of the human target at any angle across the range of motion, weighted heavily ($w_w = 10^5$); J_{torque} is the mean-squared shortfall fraction ($w_t = 10^3$), weighted two orders of magnitude below the worst-angle term so that no shortfall at a single angle can be traded away for a better

average; the J_a are configuration-specific auxiliary penalties described below; and J_{geom} and J_{len} are regularizers that keep the endpoints and actuator lengths near their seed values. Human torque targets are computed from the Gait2392 lower-extremity model [Delp et al., 2007, Seth et al., 2018]: the flexor is graded against the biceps femoris short head and the extensor against vastus medialis. Each target is a maximum-isometric torque curve: OpenSim moment arms, obtained as the partial derivative of muscle-tendon length with respect to the knee coordinate, $r(\theta_k) = \partial l_{\text{MT}} / \partial \theta_k$, are evaluated across the range of motion and multiplied by each muscle's maximum isometric force (provenance and exported tables in Appendix A). Hard constraints (bone clearance, wrap feasibility, actuator strain and contraction limits, tendon geometry) are enforced alongside the cost, and any candidate whose prediction fails is rejected with a prohibitive penalty.

The solution pipeline is common to both configurations. Each run begins with a full sanity evaluation of the seed design and a small smoke test of the solver setup, then executes `surrogateopt`, MATLAB's surrogate-assisted global solver for expensive black-box objectives, which builds a spatial interpolation surrogate over sampled candidate points and searches it adaptively [The MathWorks, Inc., 2025], in parallel under per-configuration evaluation budgets of 3,000 for the extensor route and 1,000 for the flexor route (the later cam variant of Section 4.7.1 uses 1,000 surrogate and 1,500 refinement evaluations), and finally refines the best feasible point with `patternsearch`, a derivative-free mesh-adaptive direct search method [Audet and Dennis, 2006, The MathWorks, Inc., 2025], for 15,000 further evaluations. The surrogate budgets sit well past the observed plateau of a few hundred evaluations, so they cap run time rather than define convergence. Each stage stops on its evaluation budget; the pattern search also stops early when its

mesh collapses below the mesh tolerance with the constraints satisfied, which is the usual exit once the design has converged. The entry scripts and their cost functions are archived with the code (`Opt_run.m`, `Opt_run_Ext.m`).

Although both configurations share this pipeline, their cost functions differ in ways that reflect the mechanics. The flexor BPA routes around a single moving wrap point, so clearance from the tibia and femoral condyles is enforced by continuous signed-distance constraints, and its cost adds an off-axis torque penalty (torques about the joint axes other than the knee axis, relative to the original single-BPA route) and a penalty if the wrap fails to release cleanly as the knee extends. The extensor BPA wraps over the front of the joint through as many as seven sequential contacts, so clearance is enforced instead by the contact-elimination rule that governs the route, and its cost penalizes torque overshoot above the human curve and excessive actuator contraction, which risks bottoming out at high flexion. The flexor target also carries a required torque margin above the human curve (1% in the initially reported redesigns, 5% in the adopted campaign), whereas the extensor target was the raw human curve in the initial redesigns and carries the same 5% required margin in the adopted campaign; both are set by a single `requiredTorqueMargin` knob in each configuration's context builder. With two BPAs per joint, the flexor route is mirrored about the mid-sagittal plane and the two actuator forces are summed, while the extensor model scales the force of a single shared route by the actuator count.

4.6 Isometric Knee Torque

With the actuator force model validated at the actuator level, the same model was used to predict isometric torque about the knee joint for the test stands described in Sections 4.1 and 4.3. Measured torque is compared against three predictions: the original rigid-body torque model, a prediction using the improved BPA force characterization alone, and the improved torque model of Section 4.4 after optimization. The hybrid torque method (Section 4.3) is used to isolate the sources of error.

4.6.1 Flexor on Pin Joint

Experimentally measured torque is compared with various prediction methods while using flexor BPAs on the pinned knee joint (Fig. 4.6). These results show that the original model and the improved BPA force model overpredict the measured torque at the knee for all joint angles, although the improved model performs slightly better (Fig. 4.6**A** and4.6**B**). However, when the measured parameters (l_m and r_k) are used in the hybrid model, then predictions closely match the experimental data. This indicates that the kinematics of the torque model are incorrectly predicting l_m and/or r_k.

The optimization yielded $\chi_0 = 11.9\,\mathrm{mm}$, $\chi_1 = 102\,\mathrm{kN\,m^{-1}}$, and $\chi_2 = 10.6\,\mathrm{kN\,m^{-1}}$ (Table 4.1). It can be seen that the model with these correction factors greatly improves the ability to predict the joint torque, even capturing the reduction in force at small joint angles that was completely missing from the previous model. The $l_0 = 45.7\,\mathrm{cm}$ BPA was used for validation (Fig. 4.6**D**). It can be seen that the improved model does a good job of predicting the torque with a different muscle length configured to operate over a different

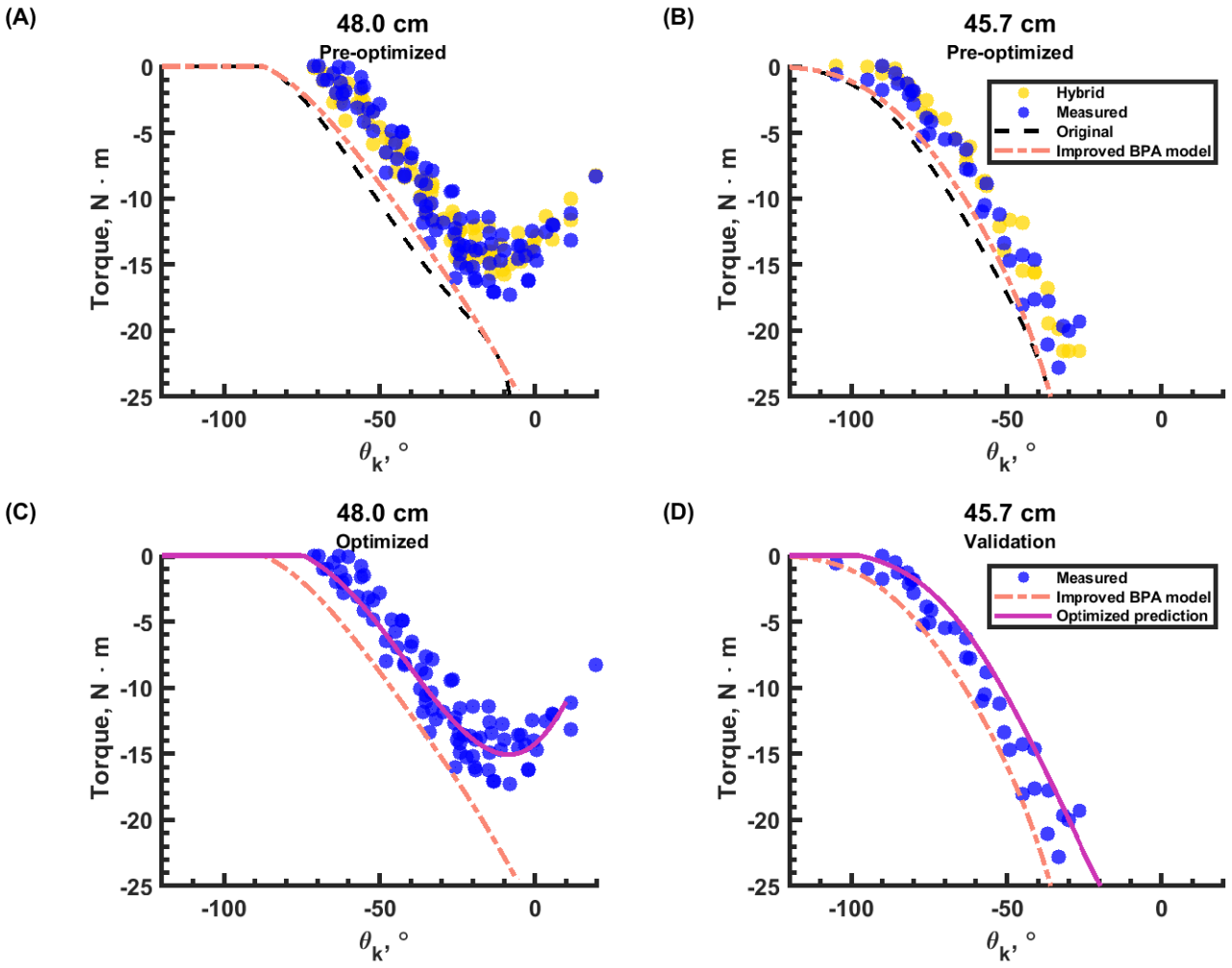


Figure 4.6: Results for the pinned knee joint using flexor BPAs. Dashed black line shows the torque predicted with our original model. Blue dots are the experimentally measured torque. Yellow dots represent the hybrid method of calculating torque. The magenta line represents the predicted torque. **(A)** Measured and pre-optimized torque prediction for the 48.5 cm BPA. **(B)** Measured and pre-optimized torque prediction for the $l_0 = 45.7$ cm BPA. **(C)** Theoretical and measured torque for the $l_0 = 48.5$ cm BPA using optimization results. **(D)** Predicted measured torque for the $l_0 = 45.7$ cm BPA using the optimization results

joint range of motion. The optimized torque prediction for the $l_0 = 48.5\,\text{cm}$ BPA has an RMSE $= 1.616$, FVU $= 0.0958$, and a maximum absolute residual of 3.2527; validation on the $l_0 = 45.7\,\text{cm}$ BPA gives RMSE $= 2.1214$, FVU $= 0.0878$, and a maximum absolute residual of 5.1690 (Table 4.2).

4.6.2 Extensor on Pin Joint

When the new stiffness and length offet model was used to predict torque for the knee extensor, measured torque was slightly higher than the original model calculates for the 48 cm resting length BPA, while it was slightly lower than expected in the 46 cm BPA, and it was much lower in the 42 cm. Only the K_1 and K_2 terms from the previous optimization results were used with the extensor configuration. Firstly, by inspection, the zero point for the measured torque was shifted to the right of the original model, indicating an offset that was at least opposite in sign from the χ_0 term in the previous section. Secondly, if we look at the BPA $l_0 = 46\,\text{cm}$ as an example, we see good correlation between the hybrid-method calculated torque and the originally predicted torque, but a divergence in the actual measured torque.

Results are shown in Fig. 4.7. The solution was found while holding the $l_0 = 41.5\,\text{cm}$ BPA for validation. The optimization resulted in $\chi_0 = -10.9\,\text{mm}$ and $\chi_3 = 0.123$. It should be noted here that the length offset χ_0 is almost equal in magnitude, but opposite sign, as the flexor configuration. The goodness of fit values for the torque prediction is found in table 4.2.

Table 4.1: Optimized coefficient values for flexor and extensor configurations. X_0 is a constant length offset (in meters), X_1 and X_2 are stiffness values (in N/m) in the bracket's X and Y directions, respectively, and X_3 is the unitless wrapping factor.

	$\mathbf{X_0}$	$\mathbf{X_1}$	$\mathbf{X_2}$	$\mathbf{X_3}$
Flexor	0.0119	1.02E+05	1.06E+04	–
Extensor	-0.0109	1.02E+05	1.06E+04	0.123

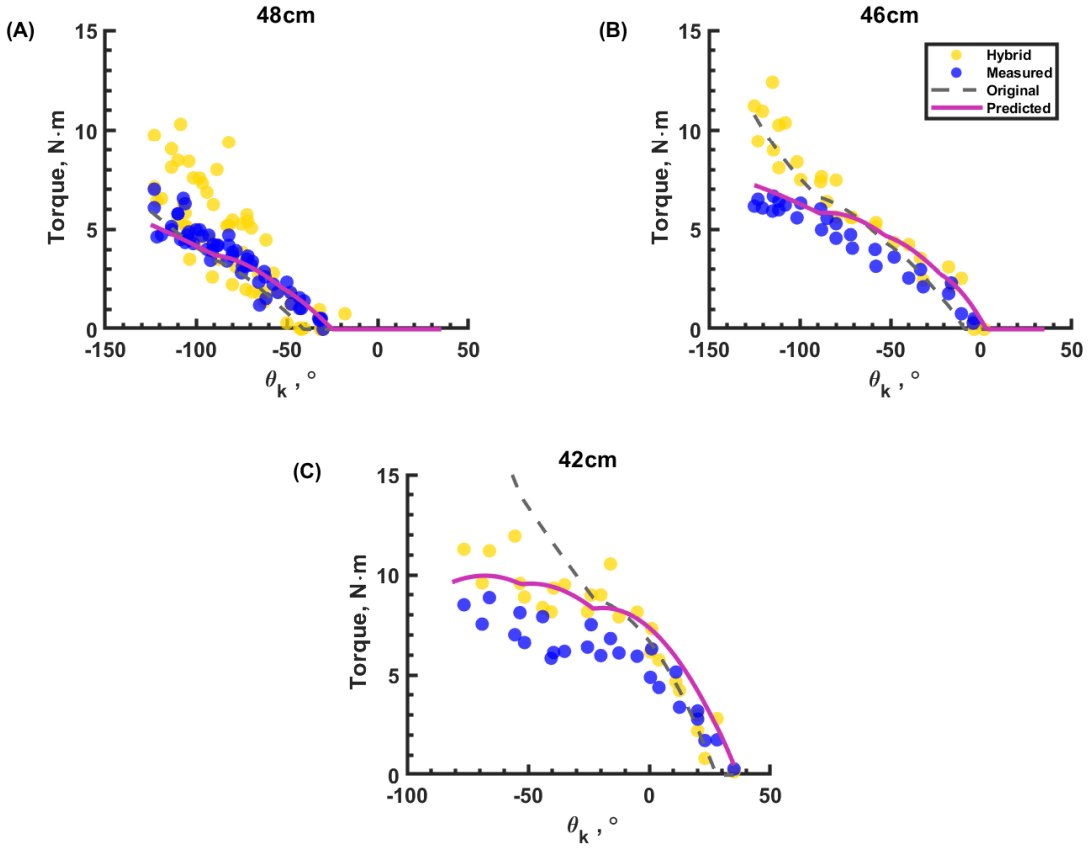


Figure 4.7: Pinned knee torque versus knee angle with the extensor BPAs. Blue dots are experimentally measured data, yellow dots are hybrid calculated torque, black dashed line is the original predicted values, and magenta line is the optimized predicted torque. For resting lengths, **(A)** 48.0 cm, **(B)** 45.7 cm, **(C)** and 41.5 cm.

4.6.3 Extensor on Biomimetic Knee

We obtained results using the biomimetic knee with a ∅10 mm BPA in the extensor configuration. The bracket frame origin is the centroid of the cantilevered segment that holds the muscle origin bracket to the proximal femur (see Fig. 4.4**C**, right side).

The torque results for a $l_0 = 51.8$ cm BPA are shown in Fig. 4.8. The torque values measured are significantly smaller than both the original prediction and back-calculated torque using the hybrid method. The results from the previous extensor configuration lead to a better torque prediction. GoF values can be found in table 4.2. Fig.4.9 shows the difficulty of modeling a BPA path that passes through the femoral condyles. Using the optimization results for extensor BPAs on the pinned knee joint, the $l_0 = 51.8$ cm BPA has an RMSE $= 0.8920$, FVU $= 0.3396$, and a maximum absolute residual of 1.4188.

4.6.4 Flexor on Biomimetic Knee

Fig. 4.10 shows the isometric torque values of the biomimetic knee from measurements, the Hunt model [Hunt et al., 2017a], and the predictions of the updated model. The values for χ_0, χ_1, and χ_2 derived from the flexors on the pinned knee were used. The bracket frame for the ∅10 mm BPA is shown on the left hand side of Fig. 4.4**C**, and for the ∅20 mm it is 4.4**D**. Fig. 4.10**A** represents a ∅10 mm $l_0 = 41.5$ cm BPA with a 21 mm artificial tendon made out of ϕ1.5 mm Shimano bicycle brake cable. The brake cable was wrapped four times. The BPA was pressurized to 604 kPa. The predicted values have an RMSE $= 0.3007$, FVU $= 0.0102$, and a maximum absolute residual of 0.4647 $N \cdot m$. Fig. 4.10**B** represents a ∅20 mm $l_0 = 42.0$ cm BPA with a 15 mm artificial tendon made out of Shimano bicycle

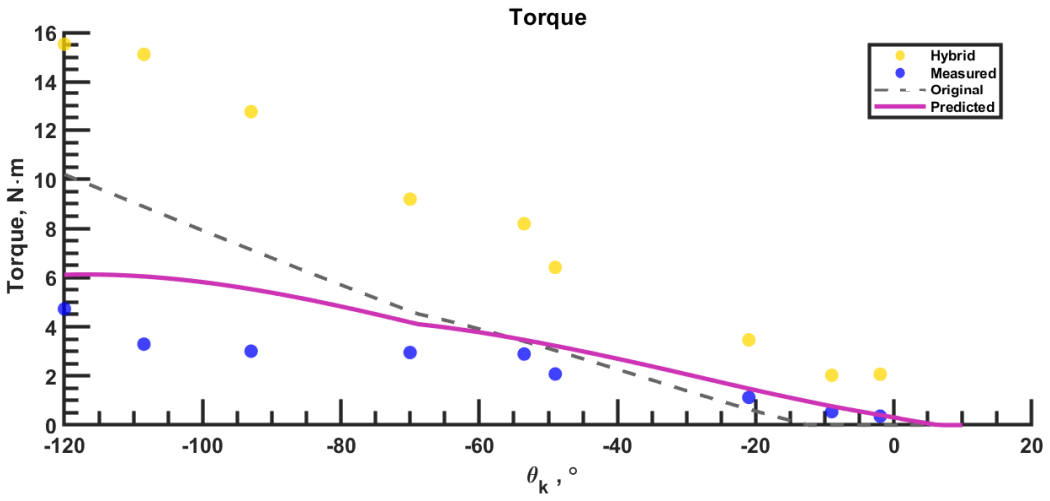


Figure 4.8: Measured versus expected isometric torque results using a ∅10 mm, $l_0 = 51.8$ cm extensor BPA on the biomimetic knee. Measured results are blue dots. Original prediction is a dashed black line. Optimized prediction is a solid magenta line. Hybrid calculation shown in yellow dots.

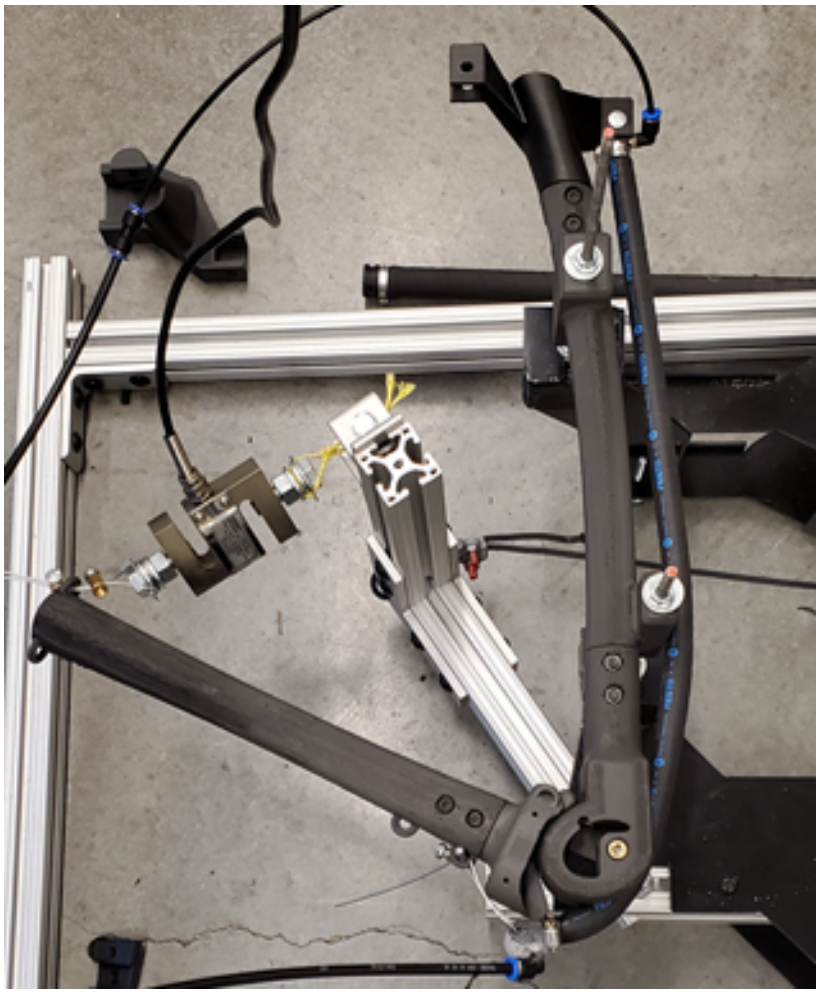


Figure 4.9: A configuration that is particularly hard to model. The BPA passes between the femoral condyles close to ICR.

brake cable (wrapped six times). Maximum human isometric torque values are shown as the black dotted line. The experiment was performed at 620 kPa and 325 kPa. At 620 kPa, the predicted values have an RMSE $= 1.8024$, FVU $= 0.0499$, and a maximum absolute residual of 3.5678 $N \cdot m$. The BPA torque does not exceed the magnitude of the human torque from ca. 60 – 100 degrees knee flexion. Allowing for a $\pm 20\%$ deviation from the human mean, the BPA does not produce human levels of torque from 78 – 103 degrees knee flexion.

Table 4.2: Goodness of fit for the improved torque model across test configurations. RMSE and maximum absolute residual are in N · m.

Configuration	**RMSE**	**FVU**	**Max. residual**
Flexor, pin joint, $l_0 = 48.5$ cm (fit)	1.616	0.0958	3.2527
Flexor, pin joint, $l_0 = 45.7$ cm (validation)	2.1214	0.0878	5.1690
Extensor, biomimetic knee, $l_0 = 51.8$ cm	0.8920	0.3396	1.4188
Flexor, biomimetic knee, ∅10 mm, $l_0 = 41.5$ cm	0.3007	0.0102	0.4647
Flexor, biomimetic knee, ∅20 mm, $l_0 = 41.5$ cm (620 kPa)	1.8024	0.0499	3.5678

4.7 Follow-Up Identification and Route Redesign

The identification work outlined at the close of the experimental chapters has since been completed, and its outcomes are summarized here; the coefficient values shown with the figures above (Table 4.1) belong to the initial single-bracket identification of those experiments.

The correction terms were re-identified over the full archived test sets with the two-stage procedure of Section 4.4: a two-bracket flexor fit cross-validated over the five pinned-flexor tests (the encoder-corrected 47 cm test held out), followed by an extensor refit

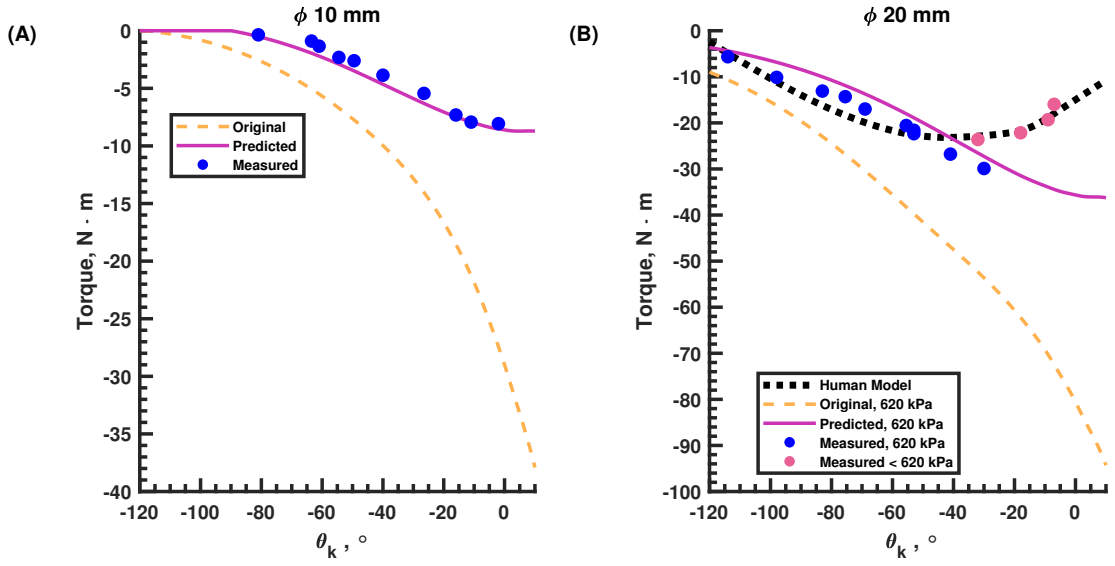


Figure 4.10: Biomimetic knee torque for **(A)** ∅10 mm $l_0 = 41.5$ cm flexor BPA with a 21 mm artificial tendon. Blue dots show the measured torque values. Magenta line is the predicted torque using the improved BPA characterization and optimization from the previous sections. The dashed light orange line shows the torque predicted using only the improved BPA characterization. **(B)** ∅20 mm $l_0 = 42.0$ cm flexor BPA with a 15 mm artificial tendon uses the same colors for 620 kPa results. Human values are given as a black dotted line. Due to the BPA exceeding the ultimate strength of the femur, torque measurements were taken at $P < P_{620}$ to show the desired RoM is achievable. Results for torque measurements at 560 kPa, 421 kPa, 325 kPa and 281 kPa (from left to right, respectively), are shown as pink dots.

with the stiffness pair locked to the flexor front. The adopted values are $\chi_0 = 3.94\,\mathrm{mm}$, $\chi_1 = 3.998 \times 10^4\,\mathrm{N\,m^{-1}}$, and $\chi_2 = 1.473 \times 10^4\,\mathrm{N\,m^{-1}}$ for the flexor (front row 77), and $\chi_0 = -6.38\,\mathrm{mm}$ with $\chi_3 = 0.158$ for the extensor on that same locked pair; full provenance is given in Appendix C, Table C8. This pair was adopted because it is the stiffness pair for which both redesign configurations meet the required 5 % torque-overshoot target simultaneously (flexor +5.00 %, extensor +5.043 % at -20.2° in the 2026-09-20 campaign). At these values, the corrected flexor model predicts the five pinned-flexor tests with per-test RMSE of 2.16/1.37/2.29/1.61/1.08 N m (48, 46, 47, 40-tendon, 41 cm; FVU at most 0.17), the 47 cm test being pure held-out validation. On the six-test extensor validation pool, per-test RMSE spans 0.49–1.52 N m against 0.95–3.62 N m for the uncorrected rigid model, an improvement on every pool test, with FVU below unity on every pool test except the 43 cm-tendon test (FVU 1.33 against a rigid-baseline FVU of 4.73). On the biomimetic extensor test ($l_0 = 51.8\,\mathrm{cm}$) the corrected prediction improves substantially on the baseline (RMSE 1.17 vs. 3.03 N m, FVU 0.67 vs. 4.49), so the shared-pair model now explains the bulk of that configuration's measured torque variance as well as its level. The solver question raised during the campaign was settled empirically: the genetic multiobjective solver was retained for the identification, where the Pareto front is required for holdout validation, and the surrogate-plus-pattern-search pipeline was reserved for the route redesign where it is already the method of record (Section 4.5).

The route redesign built on these values is likewise far advanced. The two-BPA flexor arrangement, motivated by the single-muscle shortfall at high flexion angles documented above, reached a feasible solution meeting the flexor torque target under the modeling assumption that both actuators share one attachment route rather than a physically

mirrored pair; making the mirrored pair explicit is the next design refinement. The flexor re-run under the newly adopted correction terms is complete: its design of record (`Bifemsh_20mm_Result.mat`, 2026-09-21) meets the 5 % target with a minimum margin of $+5.01$ % at -107°. The extensor route was redesigned directly under the adopted correction terms and meets or exceeds its human-scaled target with the required 5 % overshoot consumed as the thinnest margin at every angle of record: $+5.043$ % at -20.2° in the 2026-09-20 campaign, improved to $+5.02$ % by the 2026-09-25 re-run (`Vas_Pam_20mm_Result.mat`), later superseded as the extensor design of record by the cam redesign of Section 4.7.1, while respecting the wrap-release schedule and contraction limits of the route-elimination rule (Appendix C). These redesign solutions are the model definitions carried into the neuromechanical simulation work of Sections 7.2–7.3 and Chapter 8.

4.7.1 Redesigned-Route Results on the Gait2392 Robot-Body Model

The preceding paragraphs record the adopted correction terms and margin targets; this subsection records the quantitative outcomes of the redesigned routes on the Gait2392-derived robot-body model, whose muscle force and torque targets define the redesign objectives. All three result families were produced by the surrogate-plus-pattern-search pipeline of Section 4.5 under the adopted correction terms (Appendix C, Table C8) with one shared actuator plan: two 20 mm BPAs per route, a 620 kPa pressure ceiling, and a maximum usable contraction (KMAX) of 0.255 in the force model. Each family minimizes its own objective function (`objective_KneeExt20mm`, `objective_KneeFlexor20mm`, and the pulley variant `objective_KneeFlexorPulley` inside `Opt_run_pulley.m`): a unitless weighted sum

of normalized penalty terms, in which zero corresponds to no violated penalty, led by the worst-angle and mean-square torque shortfall against the required 5 %-overshoot human envelope, followed by overshoot, rest-length, strain, contraction, and mild design-change penalties. The three functions differ in penalty families and normalizations (the flexor and pulley objectives add off-axis-excess, wrap-release, and second-BPA anchor terms, and the pulley objective evaluates its torque requirement on the insertion-side torque after the tackle's force loss), so their numerical values quoted below are not commensurable with one another. The human-scaled requirements quoted below are the redesign contexts' `ctx.humanTorque` targets, the OpenSim isometric torque profiles of the reference human muscles over the same design-grid angles (`OpenSim_Vasti_Results.txt` for the extensor and `OpenSim_Bifem_Results.txt` for the flexor, loaded by `buildKneeExtContext20mm.m` and `buildKneeFlexorContext20mm.m`).

The extensor route, solved by `Opt_run_Ext.m`, reached the September design of record (`Vas_Pam_20mm_Result.mat`) at an objective of 0.1091, with the preceding 2026-09-20 campaign snapshot at 0.09770. Its minimum torque-margin fraction is 0.05025 at the worst-case point of the design grid (angle index 77 of the one-based MATLAB grid, the -20.2° point), so the required 5 % overshoot is consumed as the thinnest margin exactly as designed. Across the design range, the predicted two-BPA extensor torque spans $35.45\,\mathrm{N\cdot m}$ to $109.4\,\mathrm{N\cdot m}$ against the human-scaled requirement of $20.25\,\mathrm{N\cdot m}$ to $50.32\,\mathrm{N\cdot m}$: the route exceeds the human envelope in magnitude at every angle, with zero shortfall points.

The extensor route has since been redesigned once more by allowing the tibial-side route points to trace a cam profile, and this cam redesign is the extensor design of record (`Vas_Pam_20mm_Result_cam_20261014_0416.mat`, solved by `Opt_run_ExtCam`). The cam solu-

tion converts the razor-thin margin of the direct route into genuine design headroom. On the final geometry of record, which folds the optimizer winner together with a verified adjustment of the distal route point and tendon length, the minimum torque margin is $+30.83\,\%$ with respect to the required envelope, the human target plus the $5\,\%$ overshoot, and $+39.77\,\%$ with respect to the plain human target; the optimizer winner before that adjustment carried $+31.30\,\%$ and $+42.34\,\%$ at the same checks. The pressure required to meet the envelope spans 269 kPa–507 kPa across the range of motion, comfortably above the 150 kPa regulated floor of the VTUG valve manifold that supplies the hardware. The geometry confirms that the headroom is not bought with routing risk: the tibial wrap follows a bend radius of 96.9 mm, the guiding route point clears the femoral-condyle envelope by 43.1 mm at 10° of knee extension, and the moment arm spans 59.4 mm–102.8 mm (mean 72.4 mm) over the range of motion.

The flexor route, solved by `Opt_run.m` with the same two-BPA arrangement sharing one attachment route, improved its objective from a seeded starting evaluation of 2894 to a best value of 74.47 and meets the flexor torque target with a minimum margin fraction of 0.05007 against the required 0.05. Its predicted knee-axis torque (`TorqueZ` in the route model) spans $-81.23\,\mathrm{N\cdot m}$ to $-3.864\,\mathrm{N\cdot m}$ against the human-scaled requirement of $-23.16\,\mathrm{N\cdot m}$ to $-2.461\,\mathrm{N\cdot m}$ (peak magnitude $23.16\,\mathrm{N\cdot m}$), again exceeding the human envelope at every angle. The flexor results of record carry $\chi_0 = 3.937\,\mathrm{mm}$, $\chi_1 = 3.998 \times 10^4\,\mathrm{N\,m^{-1}}$, and $\chi_2 = 1.473 \times 10^4\,\mathrm{N\,m^{-1}}$, the adopted flexor terms, together with $\chi_3 = 0.1583$ loaded from the extensor refit and adopted as the flexor value of record; the preflight value of 0.6209 that appears once in the 2026-09-20 campaign log is superseded.

The Mono/Bi-pulley algorithms extend the route search space by allowing the artificial

tendon to be reeved over pulleys before insertion; Section 4.9 discusses what this buys. On the knee-flexor route of record, the pulley variant (`Bifemsh_20mm_Result_pulley_20260926_1652.mat`) converged to a collision-feasible layout with zero bound violations and a pattern-search exit flag of 1, at an objective of 0.08465 under its own objective function; because the direct-route and pulley objectives are different functions, this value is not commensurable with the direct route's 74.47. The best configuration on file drives the same two BPAs over a single tackle-line part with the pulley gain fixed at 1, the gain not being re-optimized. The redirect alone deepened the peak predicted flexion torque from 81.23 N · m on the direct route to 85.74 N · m on the same knee-axis measure `TorqueZ` (85.56 N · m on the insertion-side measure `TorqueInsZ`, which applies the force delivered at the insertion after the tackle's force loss), approximately 4.5 N · m more flexor torque, while holding the off-axis torque to at most 0.803 N · m.

The Bi-pulley batch applied the same reeving freedom to five muscle sites of the robot-body model's OpenSim leg, and all five runs failed the torque requirement (Table 4.3): every run closed with a nonpositive torque margin, and the medial-gastrocnemius and biceps-femoris-short-head runs terminated at the infeasibility sentinel. For the four runs that report a finite route-to-route clearance, the binding constraint was geometry rather than torque: their minimum pair clearances spanned 4.260 mm to 19.14 mm against the 25 mm separation required between adjacent routes, the sum of two 10 mm actuator radii and the 5 mm bundle margin. The medial-gastrocnemius run records no finite minimum clearance (`Inf` in the batch summary), so its failure rests on the infeasibility sentinel and its worst-constraint value of 10 alone, not on a clearance measurement. For the biceps femoris short head site, whose torque target is 40.2 N · m at the robot-body model's maximum

isometric force of 804 N, the pattern search exited with flag -2 and returned the sentinel objective of 101 from an initial objective of 1000.

Table 4.3: Bi-pulley batch on five muscle sites of the robot-body model's OpenSim leg (2026-09-26). Objective 101 is the batch's infeasibility sentinel; all five runs closed with a nonpositive torque margin. The soleus and tibialis-anterior runs converged feasibly (pattern-search exit flag 1), and their objectives are the normalized torque shortfalls; the remaining three runs exited with flag -2.

Muscle site	**Objective**	**Outcome**
soleus	0.9968	converged (flag 1), nonpos
tibialis anterior	0.9288	converged (flag 1), nonpos
rectus femoris	3.423 (worst constraint 0.2423)	nonpositive torque margin
medial gastrocnemius	101, infeasibility sentinel (worst constraint 10)	pattern-search exit flag -2
biceps femoris short head	101, infeasibility sentinel (worst constraint 10)	pattern-search exit flag -2

The robot-body model itself carries one provenance update from this campaign. The adductor magnus attachment point P2 was restored to its full-precision location $(-0.1357, -0.0929, 0.0591)$ m in the robot-body OpenSim model, displacing the rounded repository copy, which was disqualified for producing a zero-length via point that nullified the flexion torque; the regenerated mirrored model passed its validation with zero non-mirrored pairs. At the restored location, the muscle delivers 34.28 N · m of flexion torque, $1.12\times$ the human target of 30.61 N · m, and 27.62 N · m of adduction torque, $0.82\times$ the human target of 33.60 N · m. These redesigned routes remain optimizer-stage results: the robot-body OpenSim model contains no pulley entities, and no via points or markers were added to it for these redesigned routes (its eight pre-existing `MovingPathPoint` vias, the rectus-femoris and vastii patella-tracking points, one per muscle per leg, belong to the original muscle paths). The MuJoCo conversion tree of the simulation work (`mjc/gait2392_robot`, converted 2026-09-

24) predates both the 2026-09-26 pulley batch and the adductor-magnus restoration, so it carries one stale path point, the pre-restoration rounded `add_mag3` P2. The physical torque tests of the redesigned routes have begun (Section 4.9).

The same redesign pipeline was then extended from the knee to all 27 muscle-actuators of the robot-body model, and this whole-body campaign is reported here as an honest status rather than a completed result. The first pass reached 19 of the 27 actuators before the run stopped, and its chosen solutions showed that the penalty-based formulation itself was the binding limitation: all but one of the chosen routes violated the feasibility constraints, the sole exception being the biceps femoris long-head pair, whose worst-angle margin of $+0.03\,\%$ is indistinguishable from zero. The formulation was then restructured so that the task envelope, the joint-angle window each route must serve, acts as a hard feasible set rather than a penalty. In the restructured pass, twelve of the 27 actuators terminated on a concatenation error in the route builder (the three lumbar-wall actuators, tibialis posterior, the seven distal-foot actuators, and adductor magnus), leaving fifteen finalized routes. Eight of the fifteen were chosen from constraint-feasible configurations: semimembranosus, biceps femoris long head, tensor fasciae latae, soleus, lateral gastrocnemius, tibialis anterior, iliacus, and psoas. Of those eight, six also met the torque requirement at every angle, with worst-angle margins spanning $+0.25\,\%$ (biceps femoris long head) to $+250.7\,\%$ (tensor fasciae latae); the semimembranosus and soleus routes satisfy the geometry but remain short of their torque targets. The crash set and the two torque-short feasible routes define the remaining engineering: the route builder needs a fix for the concatenation failure, and the short routes need either larger actuators or relaxed corridors.

4.8 Sources of Error in Isometric Torque Generation

Our improved BPA characterization predicts torque more accurately than the original BPA model (see Fig. 4.6**A** and 4.6**B**). This only accounts for part of the error. There are certainly many factors that affect the torque results. Compliance in the testing system, positioning tolerances, flexibility of brackets, imperfect path modeling, and variation in individual BPAs are other factors.

Simplifying and testing the model revealed deficiencies in our previous analysis. The isometric system is not rigid. There is compliance in the winch cable, artificial tendons, and brackets that bend and stretch. We introduced a constant length offset term χ_0, axial stiffness χ_1, and bending stiffness χ_2. The bracket frame of reference was placed at the onset of the cantilevered section and rotated about Z. The optimization identified parameter values for a $\varnothing$10 mm BPA with $l_0 = 48.5$ cm in the simple pinned knee configuration. These results were validated with a second $\varnothing$10 mm BPA with $l_0 = 45.7$ cm.

The modeling of extensor BPA paths that wrap around the knee joint is an approximation with inherent error (Fig. 4.9). When the moment arm is short, small changes to it can have a large effect on the resultant torque [Young et al., 2019]. It was assumed that wrapping the BPA around the joint would be easier to account for than adding a patella, deterministically designing its location as a function of knee angle, and accounting for friction. At greater knee flexion angles, shorter l_0 BPAs were observed to stretch and compress (see Fig. 4.1**C**). The same loading sometimes caused the BPA to slip off the anterior head of the bolt that holds the femur to the knee. This produced a $\pm$ 20 mm displacement in the Z direction, changing both the force vector and the torque. The system designer, too, must

be mindful. Fig. 4.7 shows a deviation between measured and originally predicted results that widens as expected torque increases. This indicates system compliance. However, the hybrid-calculated torque matches or exceeds the predicted torque, except at greater flexion angles for the $l_0 = 41.5\,\text{cm}$ BPA, indicating a loss of functional l_m; this is why the χ_3 term was created in (4.23). Fig. 4.7 shows the improved fit from running the optimization that includes this term.

The biomimetic knee joint with an extensor BPA was used to validate the extensor optimization results. The corrected prediction more closely matches the measured torque values, although it remains higher than the measurements.

The nonlinear biomimetic robotic system presented here replaces rigid-body simplifications commonly used in other robotic and biomechanical systems. Instead, progressing from simple to complex systems elucidated compliance mechanisms that can lead to more accurate first-iteration designs. Technologies have been developed to predict system bending and axial compliance, a constant length offset, and a loss of usable length that occurs as artificial muscle bends around a joint.

In equation $m\ddot{\varepsilon}^* + c(\dot{\varepsilon}^*)\dot{\varepsilon}^* + k(\varepsilon^*)\varepsilon^* = F$ given $k(\varepsilon^*, P^*) = \left.\frac{\partial F}{\partial \varepsilon^*}\right|_{P^*} = -c_0 c_1 e^{-c_1\varepsilon^*} - 2c_2\varepsilon^* P^* e^{-c_2(\varepsilon^*)^2}$ then if we set mass m and external force F to zero we get $c(\dot{\varepsilon}^*) = (k(\varepsilon^*)\varepsilon^*)/\dot{\varepsilon}^*$

4.9 What Mono- and Bi-Pulley Routing Buys

The route redesign of Section 4.7 placed both 20 mm actuators on direct attachment routes. The Mono/Bi-pulley algorithms extend that search space by allowing the artificial tendon to be reeved over pulleys before insertion, and the flexor pulley run of 2026-09-26 quantifies

what the added routing freedom is worth on the knee-flexor route of record. The pulley variant converged to a collision-feasible layout with zero bound violations (pattern-search exit flag 1) at an objective of 0.08465 under its own objective function, which differs from the direct route's objective of 74.47, so the two values are not commensurable. The best configuration on file reeves the two actuators over a single tackle-line part at a gain fixed at 1:1, so the pulley acts as a redirect rather than a force multiplier; even so, the redirect alone deepened the peak predicted flexion torque from 81.23 N · m to 85.74 N · m, a gain of 4.51 N · m (5.6 %), while holding the off-axis torque to at most 0.803 N · m, under 1 % of the peak on-axis value; the predicted peaks of the direct and pulley routes exceed the human flexor torque scale of 23.16 N · m by factors of 3.51 and 3.70. The Bi-pulley batch establishes where that freedom runs out on anatomical routes. Across the five OpenSim muscle sites attempted (soleus, tibialis anterior, rectus femoris, medial gastrocnemius, and biceps femoris short head), every run closed with a nonpositive torque margin (objectives of 0.929 for tibialis anterior, 0.997 for soleus, and 3.42 for rectus femoris with a residual worst-constraint value of 0.242), and the medial-gastrocnemius and biceps-femoris-short-head runs terminated at the infeasibility sentinel (objective 101, worst-constraint value 10, exit flag -2). For the four runs that report a finite route-to-route clearance, the binding constraint was geometry rather than torque: minimum pair clearances spanned 4.26 mm to 19.14 mm against the 25 mm separation required between adjacent routes (two 10 mm actuator radii plus a 5 mm bundle margin); the medial-gastrocnemius run records no finite clearance, so its failure rests on the infeasibility sentinel alone. What the two algorithms buy is therefore reach and screening: the mono-pulley converts routing freedom into 4.51 N · m of additional predicted flexion torque without mechanical advantage and with

off-axis torque below 1 % of the peak, and the bi-pulley sweep screens candidate anatomical sites, showing that its reeved routes do not yet fit these corridors and redirecting the design effort toward clearance-relaxed attachment sites. These pulley solutions are optimizer-stage results; they have not yet been propagated into the OpenSim or MuJoCo model trees used for the simulation work of Sections 7.2–7.3, and the hardware torque tests of the updated compliant system have begun with the extensor route.

The first physical torque measurements of the redesigned extensor route are now on record. In the first test of the ∅20 mm extensor on the rebuilt route, three isometric points were held near full extension: measured knee torque of 1.163 N · m at a knee angle of 10°, 7.849 N · m at 11°, and 23.437 N · m at 10°, against hung loads of 3.7 N, 25.2 N and 76 N at gauge pressures of 388.5 kPa, 388 kPa and 592 kPa, respectively. The first and third points sit at essentially the same knee angle and span a twentyfold range of held torque, so they begin to map the route's response near extension as pressure and load vary. These three points are the start of the torque-versus-angle validation of the redesigned route against its required envelope; the full sweep, and the corresponding flexor measurements, are pending.

5 Inverted-Pendulum Balance Platform

5.1 Balance Platform and Inverted-Pendulum Testbeds

In parallel with the knee test stands, the author, together with McNeal and Williams, developed a family of hardware testbeds for the balance problem encountered by a humanoid: a robotic inverted pendulum actuated by BPAs at the ankle and standing on a motorized oscillating balance platform [Bolen et al., 2024b,a,c]. This chapter documents that family as engineering design work: one control configuration template runs across all three builds, and what each generation changed, and why, is recorded below. The balance and walking behavior that the finished hardware demonstrated is summarized afterward as preliminary work (Section 5.2). Three generations were built (Figure 5.1).

All three generations share one control template. A microcontroller reads body orientation, compares it against an upright set point, and commands the antagonist BPA pair with a correction proportional to the measured angle error: the command to one muscle of the pair is increased while the command to its antagonist is decreased by the same amount, about an equilibrium at which both valves sit half open, so that one muscle pressurizes as the other vents. Because the valves are on/off solenoid valves supplied from a manifold, the proportional command is realized by time-proportioning (pulse-width modulation of the valve duty cycle), with the switched pressure routed to the BPA chambers through a transistor-array driver stage, and with chamber pressures monitored by pneumatic pressure transducers. The pressure–force dynamics that this pulse-modulated scheme

excites are modeled in companion work [Elzein et al., 2025].

The first generation used a hinge ankle driven by an antagonist pair of 20 mm BPAs acting as plantarflexor and dorsiflexor. Its ankle angle could be measured by a shaft encoder mounted on the hinge axis, estimated from an inertial measurement unit (IMU) mounted on the tibia, or both, selectable as the feedback source for the proportional law; the IMU angle was obtained by integrating the gyroscope angular rate after a static bias correction, with the accelerometer available as a gravity reference. The distinctive feature of this generation is its transmission. The two BPAs operated in parallel on a four-wheel block-and-tackle (a reverse-pulley system) mounted on the foot, with the artificial tendon reeved through the pulleys and inserting on the tibia. Reeling the tendon through the pulley train multiplies the tendon travel delivered at the joint, at a proportional cost in transmitted force and an increased reaction force carried by the foot; the design consequences are examined in Section 4.8.

The second generation, IP 2.0, replaced the hinge with a ball-and-socket ankle, so that the foot could combine plantarflexion and dorsiflexion with inversion and eversion. Three muscles were attached: one soleus-type plantarflexor, and two dorsiflexors, one routed to act as an inverter and one as an everter, arranged so that the muscles can act antagonistically against one another as the demanded ankle orientation changes; in the controller, the two dorsiflexors shared a single complementary command against the plantarflexor. A ball-and-socket joint presents no shaft on which a rotary encoder can be carried, so angle feedback in this generation was IMU-only.

The final balance platform stands the inverted pendulum, its ankle driven by an antagonist BPA pair, on a motorized platform that oscillates the support surface beneath

the foot. An IMU on the tibia serves as a pseudo-vestibular sensor, feeding the same proportional law.

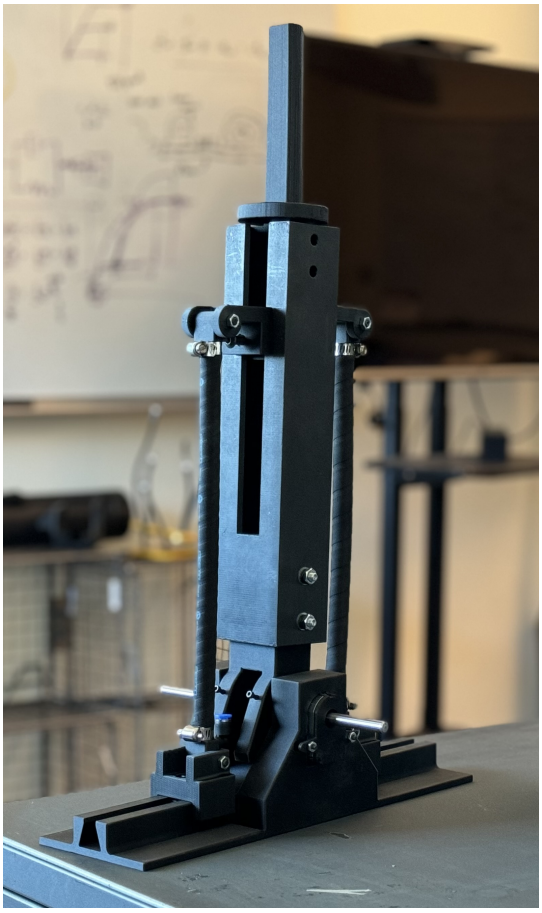


Figure 5.1: The BPA-actuated inverted-pendulum testbed. The pendulum shank rides on a hinge ankle above the foot assembly, and braided pneumatic actuators route down both sides of the shank to the foot, forming the antagonist ankle pair. Photograph from the AARL Inverted Pendulum project repository.

5.2 Preliminary Work: Balance and Walking Demonstrations

The testbeds of Section 5.1 were demonstrated by the author and presented in three conference posters [Bolen et al., 2024b,a,c]. They are preliminary work, and this section reports them as demonstrations.

The first-generation pendulum, with its hinge ankle, selectable encoder/IMU angle feedback, and reverse-pulley transmission, demonstrated self-balancing. Under proportional control of the antagonist 20 mm BPA pair with IMU pseudo-vestibular feedback, the leg held itself upright on flat ground, and balance was maintained when the same controller operated on the oscillating balance platform beneath the foot. The second generation, IP 2.0, with its ball-and-socket ankle, three-muscle plantarflexor/inverter/everter arrangement, and IMU-only feedback, both self-balanced and walked. Its walking had a narrow initialization window: the leg had to be started within about 5° of vertical, or it fell. Most testbed configurations held balance indefinitely. Both demonstrations used the same family of Festo BPAs characterized in Section 3.3.

Video recordings of the demonstrations exist as the author's records of his own experiments, and the test protocol matched the platform-oscillation balance studies of Peterka [2002]. The platform-oscillation timings and set angles can be recovered from these recordings, but they have not been extracted, and no such values are presented here. What the demonstrations do not constitute is a controlled measurement campaign. Chapter 8 defines the extension protocol that would measure disturbance rejection, sensor drift, and sustained operation under continuous platform oscillation before the same feedback architecture is compared on a full humanoid model.

6 The Sensory Afferent Database

6.1 A Structured, Model-Oriented Afferent Resource

The synthetic nervous system of Chapter 7 is assembled from physiological parts: Ia, Ib, and II afferents, the reflex pathways they drive, and the spinal circuits those pathways modulate. The properties of these parts are scattered across a century of experiments and dozens of animal preparations, and no single paper or review assembles them in a form a control engineer can use. The Sensory Afferent Database (SADb) was built within the AARL's NeuroNex Communication, Coordination, and Control in Nervous Systems (C3NS) effort to close that gap, and we presented it at Neuroscience 2023 [Bolen et al., 2023]. It is a database rather than a bibliography: each paper carries distilled mechanism notes, each sensory feedback pathway is a first-class record linked to the papers that established it, and each computational model of afferent control is linked to its source paper and to the pathway records it implements. The database lives in a five-table Airtable base (Papers, Feedback, Models, Review Papers, and Recorders) backed by the laboratory's Zotero libraries, with the DOI as the canonical identifier for every record.

The 2026 audit and curation campaign began from a three-way reconciliation, because the database's three homes did not agree. The author's personal Zotero collection held 538 items, the AARL group library's Biology subtree held 911 items across 27 collections, and the Airtable base held 99 papers. Matching record by record on DOI and normalized title showed that the personal collection was essentially a subset of the group library (all

483 personal DOIs present) and that the Airtable base was a small curated subset rather than a mirror: 95 of its 99 papers resolved to records in the group library's Biology subtree, while 443 personal items had never been entered at all. The gap was imported in verified batches, with every written record re-read and checked after writing, bringing the Papers table to 532 records.

Curation then proceeded under rules designed to make every note citable. A note is written from the paper's own text, abstract at minimum and full text when available, and never from memory; papers for which no text could be found were logged and left bare rather than guessed. Notes are one to three sentences of distilled mechanism, following the template of the laboratory's earliest records, and free-text fields are constrained by controlled vocabularies for animal preparations and afferent classes. Each feedback pathway is named by a fixed grammar of afferent, action, side, joint, and phase (for example, "Ib stance to swing"), so that pathway records compose into circuits the way the controller will use them. Ambiguous cases were flagged for a human ruling rather than decided silently.

The final curation wave added the structure the controller work needs: afferent class, animal-study potential, robot-and-simulation translation notes, and prune status fields, together with a corpus citation graph exposed as record-to-record links. Drafting was machine-assisted at scale, with 374 papers curated in fifteen batches; every batch passed automated vocabulary, abstract-paste, and duplicate gates before anything was written, and every record was verified against its grounding text after the write. Grounding remained the binding constraint throughout. A PDF-first rule recovered grounding for 47 of 55 flagged papers from the laboratory's own library, and a university-authenticated

browser session recovered abstracts for 24 of the 70 papers that had no findable text; the rest stayed bare by rule.

Pruning followed, in three stages, each approved before execution. First, 95 records left as off-topic or orphaned: 88 papers removed as satellite topics (machine-learning and aerodynamics methods, flow sensing without neurophysiology, and fish swimming biomechanics), as duplicates, or as isolated records stranded by an earlier strict flagging pass, plus seven orphaned pathway, model, and review records. Second, a cut list of 290 papers assembled from explicit rules for duplicates, monographs, and conference abstracts, and from a weighted rubric ranking the remainder by citation count, age, rating, network isolation, and record thinness, while 351 protected records (foundational authors, feedback-linked papers, highly cited work, and pre-1940 classics) were excluded from consideration. Third, a final round removed four papers and 47 review records whose links had been emptied by the earlier cuts. Every deletion was snapshotted so the corpus remains reconstructible.

6.2 Audited Corpus and Curation Coverage

The base held 957 records at the close of the campaign: 561 papers, 39 feedback-pathway records, 244 model records, 109 review records, and 4 recorder entries, 43 records below the laboratory's 1,000-record operating cap, with room to grow. Coverage of the retained papers is concentrated where the controller work needs it. Of the 561 retained papers, 523 carry curation notes, 426 carry animal-preparation tags, 118 are linked to named feedback pathways, 86 carry an afferent classification, and 258 carry notes translating the paper's

findings toward robot or simulation use; 550 carry DOIs, and 38 remain bare pending a findable text.

The citation graph was built over the full audited corpus of 943 papers, which includes the 382 records the prune later removed and which the packaged database retains under an archived flag. Within that corpus, 730 papers are connected by 11,130 directed in-corpus citation pairs, and a Louvain community partition of the graph resolves 18 named topic clusters, from the classics of afferent locomotor control and spinal interneuron circuit modeling through human balance and posture to bioinspired flow sensing and deep-learning methods. The partition earned its keep in the prune: the satellite clusters it exposed, the flow-and-flight sensing, fish swimming, and neural-network method clusters, match the topic groups the first prune stage removed. The corpus, its citation graph, and its cluster assignments are packaged as a self-contained laboratory tool: a single-file offline explorer offering searchable and pivotable tables, the citation-network map with focus-mode neighborhoods of any paper, and a generated knowledge base with an index page plus cluster, pathway, afferent, and animal pages, distributed with the flat corpus export as a zip that runs on any machine without a network connection.

6.3 AI-Assisted Literature Triage

Expanding a curated corpus is dominated by triage: deciding, paper by paper, what a record is about and whether it belongs. We evaluated AI tools for this task in a comparative study presented at Neuroscience 2024 [Bolen and Hunt, 2024]. The evaluation found the expected trade: the tools accelerated the population of the database, but their output still

required substantial human oversight for accuracy.

The 2026 campaign was built as the operational answer to that finding, a division of labor in which the machine drafts and the human rules. The machine contributed scale: batch drafting from grounded text, automated gates that rejected abstract pastes and duplicate records before writing, and per-record verification after it. The gates earned their place on caught failures, including a record that shipped another paper's abstract under its title until the grounding pass flagged it. The human held every decision that required judgment or carried risk: the curation vocabulary and its amendments, the 164 flagged judgment calls surfaced for ruling, and every single deletion in the prune, each stage of which waited for explicit approval. The gates and flags are not ceremony; they are the oversight the 2024 evaluation showed was necessary, made concrete.

6.4 Toward Robot and Simulation Use

The database exists to feed the sensorium of Chapter 7. Its pathway records enumerate the signal taxonomy the controller's sensory channels implement: Ia afferents carrying length and velocity, Ib afferents carrying force, II afferents carrying length, and contact receptors marking the stance–swing boundary (Section 7.2). The pathway naming grammar maps directly onto circuit edges, since a record's afferent class fixes the signal encoding, its phase fixes the gating, and its action fixes the sign, so translating a pathway record into a controller connection is a lookup rather than a literature search. The Models table carries the lineage the controller instantiates, from the two-level rhythm-generator and pattern-formation architecture [Rybak et al., 2006] through muscle-reflex walking [Geyer and Herr,

2010] to the two-layer neuromechanical walker the laboratory's controller descends from [Deng et al., 2019], and the 258 robot-and-simulation translation notes exist precisely to shorten the step from paper to wiring decision.

The corpus also reaches ahead to the sensorium extensions planned in Section 8.5. The human balance and posture cluster and the multisensory recalibration cluster are the entry points for the vestibular and ocular feedback work, already curated and already connected to the locomotor core through the citation graph. The knowledge base makes this reach practical for new students as well: the cluster and pathway pages compress a literature that took months to curate into an onboarding document measured in pages.

The honest accounting is that the resource is deeper than it is broad in places. The afferent classification covers 86 of 561 retained papers, concentrated in the pathway-rich core rather than spread across the corpus, and the flagged vocabulary gaps (vestibular and visual preparations among them) mark where the classification must grow before the extensions of Section 8.5 can be wired from the database alone. The growth policy follows from the prune: additions should be substantive records or clearly flagged stub twins, never bare rows, so that the corpus that the controller work and the next students inherit stays as searchable as the one this chapter documents.

7 Synthetic Nervous System Simulation

The preceding chapters validate the robot's mechanics: a force model for the braided pneumatic actuators (Chapter 3) and a corrected model of the joint torque those actuators deliver through the biomimetic knee (Chapter 4). This chapter builds the other half of the neuromechanical system, the controller, together with the simulation pipeline required to develop it. The controller is a synthetic nervous system (SNS): a network of conductance-based neuron and synapse models whose organization follows the mammalian spinal cord [Rybak et al., 2006, McCrea and Rybak, 2008, Deng et al., 2019]. The networks are not trained in the machine-learning sense; their parameters are set from physiological data and then tuned against explicit verification tests, which is what keeps the circuits interpretable: every neuron corresponds to an identifiable cell class, so the simulated system supports lesion, stimulation, and sensory-deletion experiments that cannot be performed on living subjects (Section 2.10).

Two definitions cover most of the neuroscience vocabulary used below. First, the locomotor controller is a two-level central pattern generator: a rhythm generator sets the timing of the step cycle, and a pattern-formation layer shapes that timing into phase-appropriate activation of antagonist muscle groups [Rybak et al., 2006]. Each leg's rhythm generator is a half-center pair, an extensor population and a flexor population that inhibit each other so that only one is strongly active at a time. Second, in the non-spiking (graded) formulation used for most of this work, each neuron is electrically a nonlinear

RC circuit: synaptic current charges a membrane capacitance through a leak conductance, and the resulting membrane voltage, tens of millivolts, is both the neuron's state and its output [Nourse et al., 2023]. Graded chemical synapses couple these units: a depolarized presynaptic cell opens a conductance in the postsynaptic membrane whose sign makes the connection excitatory or inhibitory. The sensory channels that close the loop are the muscle afferents, Ia signaling muscle length and velocity, Ib signaling muscle force, and group II signaling length, together with heel and toe contact sensors.

Within neuromechanical simulation, the controller built here belongs to the biologically grounded SNS line, one of the three lines compared in Section 2.9; the reflex-based and learned alternatives [Geyer and Herr, 2010, Song et al., 2021] are treated there. The ancestry is direct: the two-level organization implemented throughout this chapter is the rat-hindlimb CPG of Deng et al. [2019], built on the full-muscle hindlimb analysis of Young et al. [2019], and the research community and tooling around the approach, including SNS-Toolbox [Nourse et al., 2023], are surveyed in Section 2.8. Two-level CPG control has reached musculoskeletal macaques and muscle-driven robotic hind legs [Nishizaki et al., 2026, Fukuoka et al., 2021], though on non-humanoid platforms or with generic musculature. What this program adds is the combination the background identifies as unclaimed: an anatomically grounded two-level controller with Ia, Ib, group II, and contact feedback, driving a model whose actuators are the experimentally characterized BPAs of Chapters 3–4.

Building that controller required a simulation pipeline that did not exist: no physics engine carried the robot's custom muscle routing together with the BPA force model and an SNS. The pipeline assembled here has four parts. The robot's OpenSim musculoskeletal

model was converted to the MuJoCo physics engine and audited against its source [Delp et al., 2007, Todorov et al., 2012]. The quasi-static BPA force model was implemented in Python so that actuator forces inside the simulation come from the validated model of Chapter 3, not from a generic muscle approximation. The two-level spinal network was built in SNS-Toolbox and coupled to the converted model with proprioceptive and heel and toe contact feedback. Finally, because a result that holds in only one software implementation is difficult to trust, the same contact-driven rhythm core was re-implemented in Simulink and checked against its NumPy reference, and the archived AnimatLab models of the same lineage were re-run headless so that engine-level and network-level agreement could be separated; spiking-neuron mirrors of all three simulation families additionally test whether the conclusions depend on the non-spiking coding assumption. Section 7.2 documents these components as built, Section 7.3 and Section 7.4 report what they do, and Section 7.5 states what the spiking comparison establishes. The chapter begins with the prior AnimatLab walker whose limitations motivated this verification-first approach.

7.1 Prior Work: The AnimatLab Bipedal Walker

Before the simulation toolchain of this chapter existed, the author and Morrow constructed a neuromechanical model of a bipedal walker in AnimatLab, a graphical environment for building biomechanical systems actuated by synthetic neural systems [Cofer et al., 2010]. That model is the direct ancestor of the work reported here: its biomechanics and its two-layer controller architecture carry through every model in this chapter, and the limitations of its hand-tuned parameters are the reason the present toolchain emphasizes systematic verification.

The biomechanical model consisted of nine rigid bodies (pelvis, thigh, shank, foot, and toe segments) connected by sagittal-plane hinge joints with human-like rotational capabilities and actuated by antagonistic pairs of Hill-type muscles at the hip, knee, and ankle. The toe joint, the author's addition to the Li design, carried a passive spring instead of a muscle pair, simplifying the system. The mechanics followed the suspended bipedal walking model of Li et al. [2017], whose pattern generators were entrained by heel and toe mechanosensory receptors and hip-angle position sensors standing in for the group-I loading, cutaneous, and hip-joint afferents of the biological scheme; muscle afferents were not modeled. The Li model also required support against falling: a system with only planar joints cannot actively balance side to side.

The synthetic nervous network was built from non-spiking, leaky-integrator neurons and followed the two-layer CPG organization of Deng et al. [2019]. A rhythm-generator layer set the timing of the system; a pattern-formation layer reshaped the timing signals into phase-appropriate activation of antagonist muscle groups; and a sensorimotor layer containing Renshaw-cell-like recurrent inhibition (motor-neuron collaterals exciting inhibitory cells that feed back onto the same pool) drove the motor neurons. The sensorimotor layer received Ia (length and velocity) and Ib (tension) afferent feedback from the muscles, but the feedback gains were never systematically tuned, and no ground-contact signals were modeled. Every applied current, synaptic weight, and muscle-tension parameter was set by hand: each was adjusted by trial and error, one at a time, until the three layers oscillated in phase and the motor neurons produced stepping-like activation of the muscle pairs. That hand-tuning consumed the bulk of the project's effort, left no record of the sensitivity of the behavior to individual parameters, and did not generalize when new

feedback pathways were added; those limitations directly motivated the verification-first approach of the remainder of this chapter. Alternating, stepping-like activity and motion were demonstrated with the walker suspended in the air, and the results were presented at the 2021 NIH BRAIN Initiative Investigators Meeting [Bolen and Hunt, 2021].

7.2 A Neuromechanical Simulation Toolchain

This section documents the toolchain developed for the neuromechanical program. Its components are the AnimatLab walker environment in which the controller is prototyped, the OpenSim-to-MuJoCo conversion of the robot model with its BPA force interface, a Simulink block library of the SNS component models, and a cross-engine verification protocol that ties the implementations together. Each subsection presents one component as built and states what it does and does not establish; quantitative results follow in Section 7.3.

7.2.1 AnimatLab Body and Controller Organization

The walker representation and controller hierarchy are shown in Figures 7.1 and 7.3. The hierarchy separates rhythm generation, hip control, and knee–ankle pattern formation within left- and right-side neural subsystems. The current AnimatLab working model was instrumented for a new 10 s phase-1 run recording joint, neural, contact-sensory, and body-motion channels. This run was used to inspect recorded signals and simulation behavior, without treating it as evidence of stable walking or effective ground-contact gating. It is distinct from the earlier walker study of Section 7.1. The joint-motion recordings from this

run are reported in Section 7.3.

Since that run, the working model has been reorganized into a complete bilateral network. The right-side rhythm-generator, hip, and knee–ankle subsystems were mirrored from the left, inhibitory commissural pathways were added between the two rhythm generators, and each leg gained four biarticular muscle pathways (gastrocnemius, biceps femoris short head, semimembranosus, and rectus femoris) with human maximum isometric forces from the Gait2392 data set. The reorganized project opens in the AnimatLab environment without errors and exports a standalone simulation that runs headless. Its rhythm-generator half-centers hold a static latch under the tonic drive levels tested so far rather than oscillating, so conductance tuning toward robust oscillation is the active step; no locomotor claim is made for the reorganized model beyond its verified assembly.

Figure 7.2 illustrates the rhythm-generator, pattern-formation, and sensorimotor organization at the neuron and synapse level. These reference schematics complement the subsystem view by making the antagonist pathways and sensory connections visible.

7.2.2 MuJoCo Conversion and Synthetic BPA Knee

The robot's musculoskeletal model was converted from OpenSim to the MuJoCo physics engine so that a neural controller can drive it in closed loop [Delp et al., 2007, Todorov et al., 2012]. The MyoConverter program performs the base conversion [Caggiano et al., 2022, Wang et al., 2022]; because an automatic conversion of a model with custom muscle routing introduces specific defects, among them muscle paths that are not mirrored correctly between the two legs and joint rotation axes with reversed sign, the conversion is

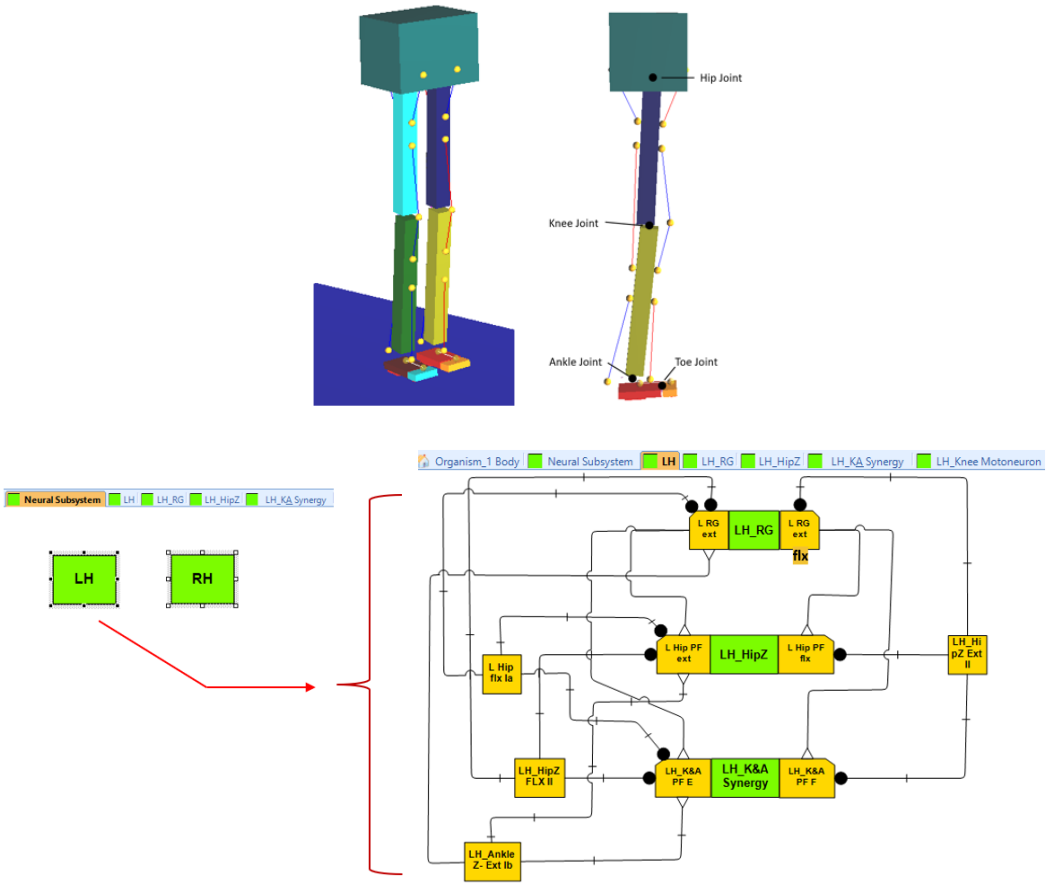


Figure 7.1: AnimatLab model and architecture illustrations. Top: the segmented walker and labeled hip, knee, ankle, and toe joints. Bottom: the left/right subsystem organization and an expanded view of the left-side rhythm-generator, hip, and knee–ankle modules. These architectural views describe model organization and are not snapshots of the phase-1 trajectory.

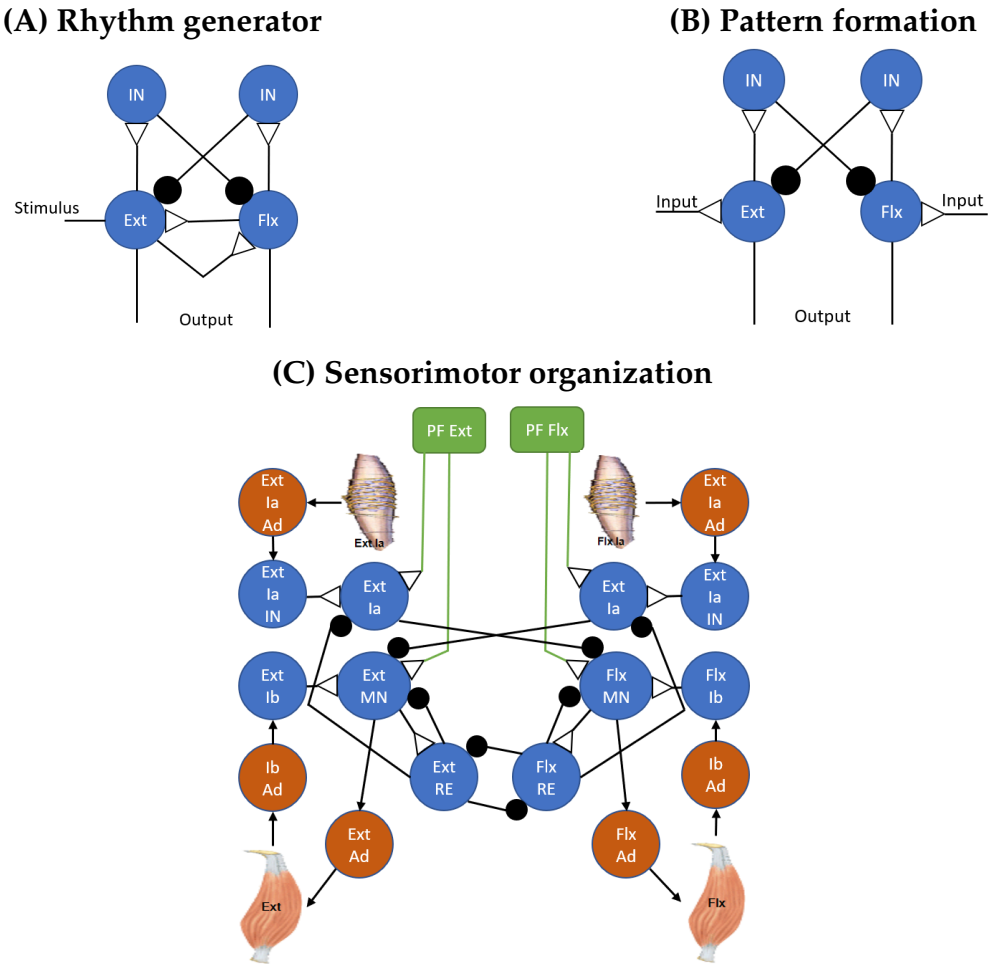


Figure 7.2: Reference schematics of **(A)** rhythm generation, **(B)** pattern formation, and **(C)** sensorimotor organization. Ext and Flx identify antagonist extensor and flexor pathways; IN denotes interneurons, MN motor neurons, and Ia and Ib muscle afferent pathways. Open triangles and filled circles distinguish excitatory and inhibitory synaptic connections. These schematics illustrate controller architecture and are not readouts from the phase-1 run.

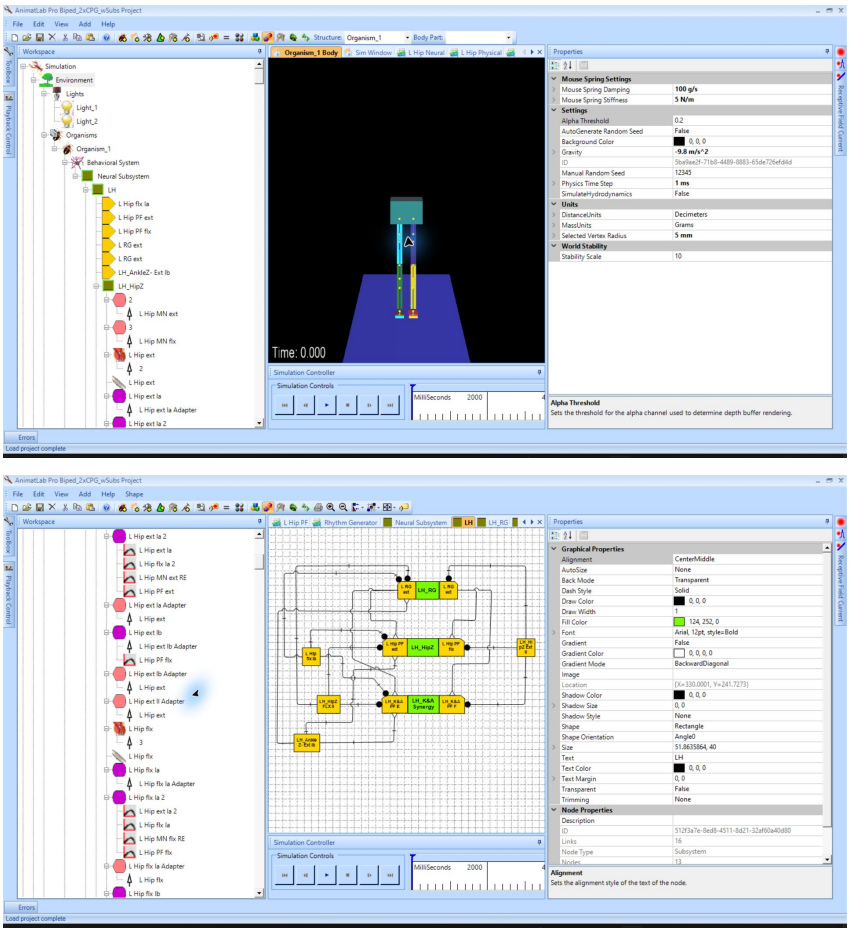


Figure 7.3: The saved AnimatLab working project opened in the application: the biomechanical body at initialization (top) and the expanded left-side controller subsystem (bottom). The workspace tree and property inspector expose the relationship between the body, sensory pathways, and neural modules. These interface captures were acquired without running a new simulation.

followed by repair scripts, short Python programs that correct each defect and re-audit the converted model against the OpenSim source. The conversion runs in a recorded software environment, with the engine and library versions fixed, so that it can be re-run from the source model and reproduced. Separately, the quasi-static BPA force model of Chapter 3 was implemented in Python so that it runs inside the simulation loop: at each time step it reads every actuator's length and extension rate from the MuJoCo tendon paths and returns the actuator force, including the correction-factor force balance of Chapter 4. A minimal SNS with reciprocal inhibition then drove a synthetic knee with antagonistic flexor and extensor BPAs. Figure 7.4 distinguishes the converted robot from this separate knee demonstration. The demonstration establishes the controller–actuator–joint interface; it does not establish SNS control of the converted robot or validated correction-factor torque predictions in MuJoCo.

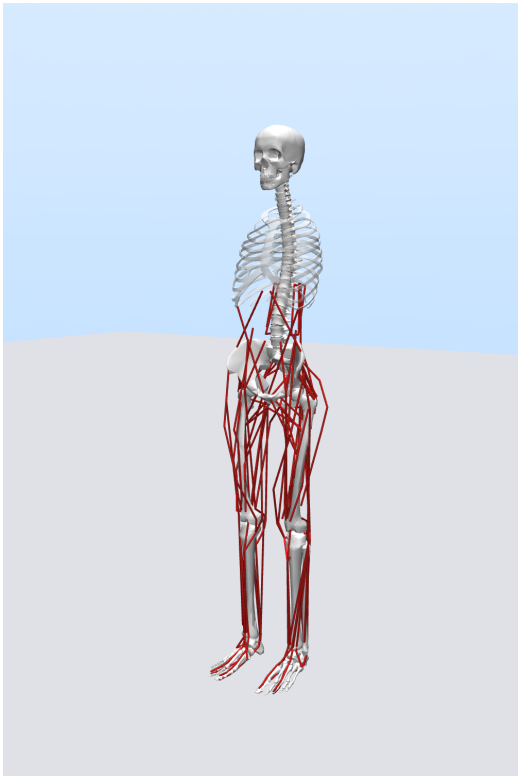
The conversion is complete and verified. The converted model carries all 92 muscle-tendon actuators, with attachment geometry audited against the OpenSim source, and the joint sign conventions of the robot's modified muscle paths were verified dynamically: each actuator was activated on its own in simulation, and the sign of the joint torque it produced identified its action at that joint. The check has to be dynamic because the attachment points of these paths move with joint configuration, so the routing cannot be read from a fixed table of moment arms. The Python implementation reproduces the MATLAB force calculation, including the correction-factor force balance. An SNS-Toolbox network couples to MuJoCo at a 2 ms neural step, matching the physics-engine timestep, with muscle force feedback entering as Ib afferent input. On this coupling, a 410-neuron two-level spinal network has been assembled for the converted model (per-leg rhythm-

generator half-centers, four phase-shifted pattern-formation groups per leg, motoneuron pools for all actuators, and Ia, Ib, and II afferent populations), and its rhythm generator has been verified in isolation to oscillate at an approximately 0.9 s cycle with the two legs in antiphase under tonic drive. Full-body closed-loop co-simulation is not yet running: leg-joint states develop non-finite values during longer simulations, with the foot-contact meshes the primary suspect under investigation.

7.2.3 Simulink SNS Library and Reflex Topology

A library of Simulink blocks was developed so that synthetic nervous systems can also be assembled and inspected in a standard controls-engineering environment. Its seven core component blocks are a non-spiking (graded) neuron, a graded chemical synapse, a spiking leaky integrate-and-fire neuron, Ia and Ib afferent encoders, muscle activation dynamics, and a generic BPA force block (Figure 7.5). The blocks implement the same neuron and synapse equations used by the SNS-Toolbox networks of the preceding subsection. The full library adds a synapse-sum junction for neurons receiving more than six synaptic inputs, three diameter-specific BPA force elements (10 mm, 20 mm and 40 mm) implementing the fitted force surfaces of Chapter 3, and a Hill-type biological muscle element, for twelve blocks before the spiking additions of Section 7.2.5. The core blocks were assembled into a reduced-order knee-reflex model with antagonist actuation (Figure 7.6): Ia stretch feedback excites the motor neuron of the stretched muscle and inhibits that of its antagonist (reciprocal inhibition), while Ib force feedback inhibits the motor neuron of the muscle that generated the force (autogenic inhibition). The library and the

(A) Converted robot



(B) Synthetic BPA knee

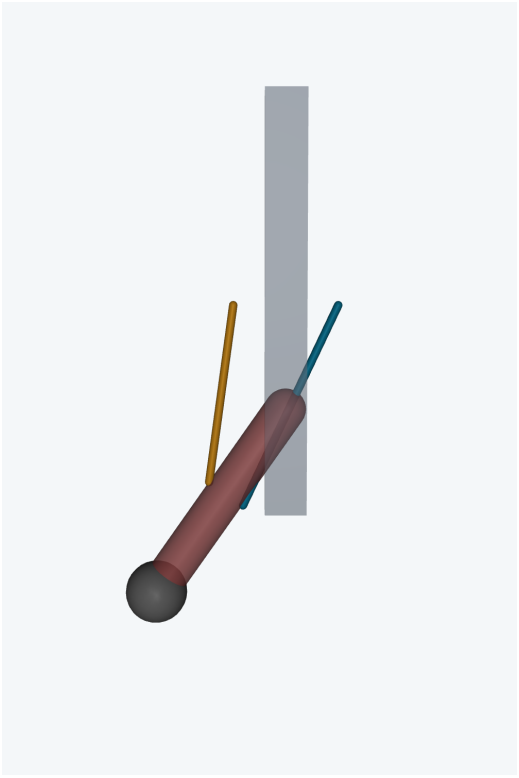


Figure 7.4: Native MuJoCo renders of **(A)** the converted Gait2392 robot at its saved reference pose, with muscle paths visible, and **(B)** the independent synthetic antagonist-BPA knee at an illustrative static angle of -35°. Cyan and gold identify its named flexor and extensor tendon routes; transparency exposes the routes through the view. The models are not shown to the same scale. These views document model construction, not SNS control of the converted robot or validated torque predictions.

demonstration run in base Simulink without additional toolboxes. The block icons derive their excitatory and inhibitory symbols automatically from the sign of each synapse, in a color-blind-safe palette, and the publication figures regenerate from scripts. To couple the controller to detailed mechanical CAD, the SolidWorks assembly is exported in URDF format and imported into Simscape Multibody; the import path has been verified on a trial knee assembly, and the tendon hardware is modeled in CAD but not yet inserted into the assembly. The model retains provisional mechanical parameters, the generic BPA force block remains a placeholder, and the saved runs are therefore not used to establish quantitative knee-regulation performance.

7.2.4 Cross-Platform Verification of the Contact-Driven Rhythm Core

Because the toolchain spans several simulation engines, a verification protocol was defined around the contact-driven rhythm core of the Walker-2-Layer (W2L) bipedal architecture: four rhythm-generator half-centers, one extensor/flexor pair per leg, plus graded pattern-formation and sensory cells, driven by heel-contact mechanosensory feedback. Each half-center is built from spiking neurons whose persistent sodium current, a slowly inactivating inward current, terminates each burst, so the mutually inhibiting extensor and flexor populations of a pair alternate [Deng et al., 2019]. The same core was implemented twice from a common connectivity specification, once as a NumPy reference integration through the SNS-Toolbox backend and once as native Simulink blocks from the library of Section 7.2, and both were driven with an identical protocol: a 12 s run at a 2 ms neural step, tonic drive with a brief kickoff pulse, and alternating 1 Hz heel-contact trains. Agreement was

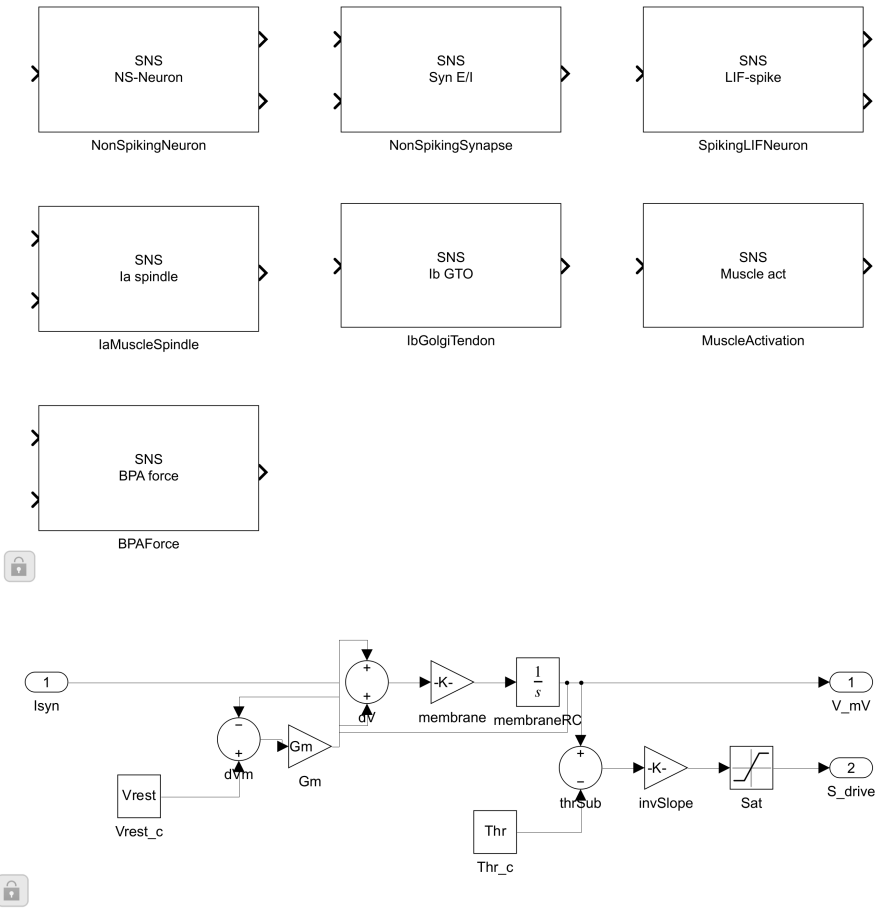


Figure 7.5: Native Simulink diagram exports showing the library's seven core masked blocks (top) and the internal implementation of the non-spiking neuron (bottom). The library blocks are arranged in a compact display grid for legibility; the full library also carries the synapse-sum junction, the diameter-specific BPA force elements, and the biological muscle element described in the text. The generic BPA force block remains a placeholder. These implementation views do not establish quantitative knee regulation or a completed CAD-coupled Simscape Multibody model.

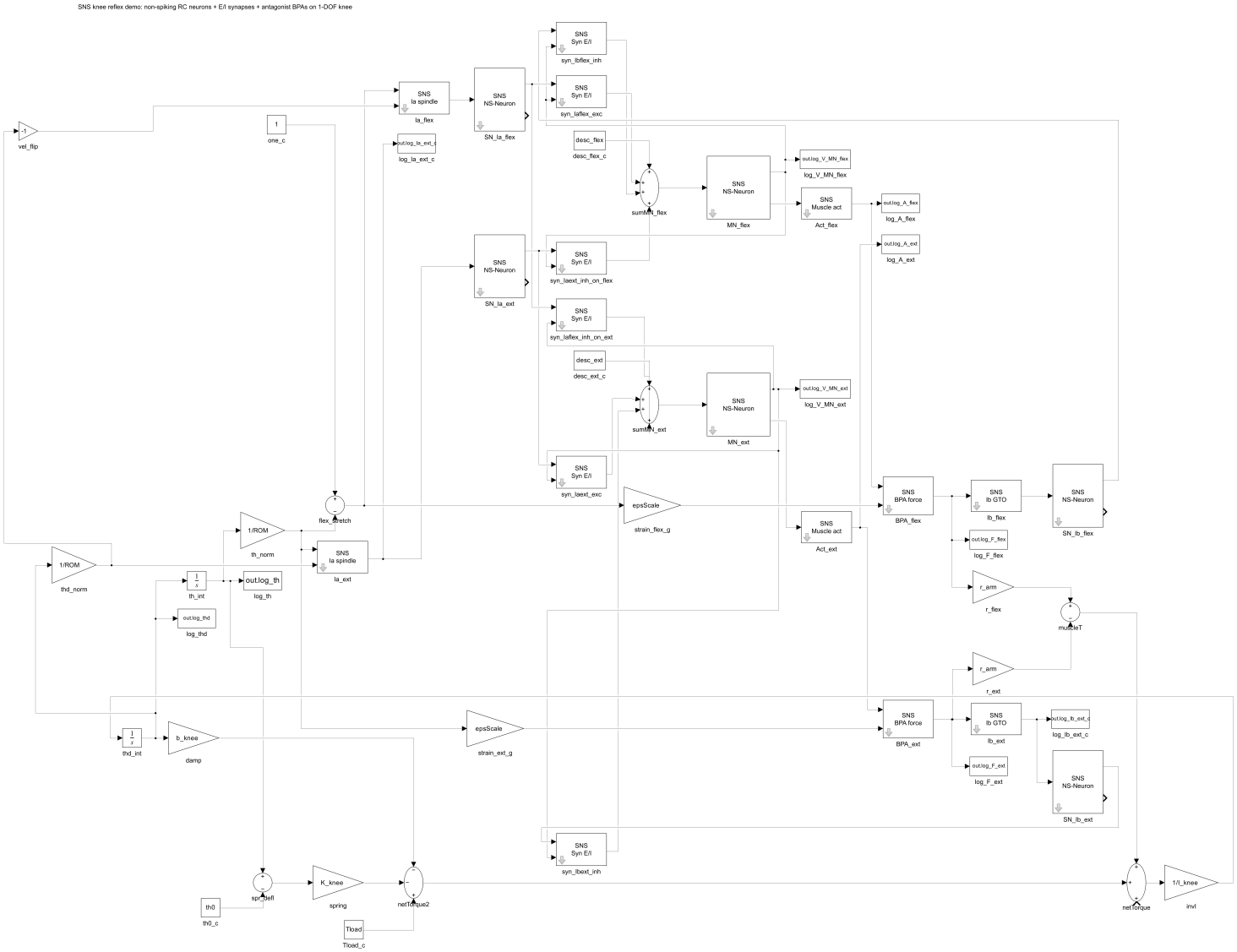


Figure 7.6: The complete reduced-order knee-reflex model exported from Simulink. The diagram connects Ia and Ib sensory pathways, excitatory and inhibitory synapses, antagonist motor neurons, activation and BPA force blocks, and the joint dynamics. Blocks and wires were arranged for display; source and destination ports were checked to confirm unchanged connectivity. Mechanical parameters and the BPA force block remain provisional.

scored on rhythm-generator burst period, left–right antiphase correlation, burst counts, and full-trace correlation against the NumPy reference. The archived AnimatLab reference models (the bilateral contact-driven model, the modern W2L model, and the Li walker) were additionally re-run headless from their standalone simulators with instrumented charts, so that engine-level agreement and network-level agreement could be separated. The protocol covers the rhythm core and the archived models only; full-network and whole-body behavior are reported as limitations alongside the results in Section 7.3.

7.2.5 Spiking-Neuron Mirrors of the Simulation Models

The controllers documented above all compute with graded (non-spiking) neurons, in which the membrane potential itself is the analog activity variable that drives the musculoskeletal model. To test that the conclusions drawn from those models are not artifacts of graded coding, spiking-neuron mirrors of all three simulation families were built, following one design rule: the motor-neuron layer stays non-spiking, because the musculoskeletal model consumes a continuous activation command, while the afferent encoders (the neurons that convert sensor signals into synaptic input), rhythm-generator half-centers, pattern-formation cells, interneurons, and commissural (cross-side) cells become leaky integrate-and-fire neurons operating at real membrane-potential ranges, so that the network carries spikes above the motor-neuron layer and analog voltages below it [Szczecinski et al., 2023, Nourse et al., 2023]. Event-based coding is expected to matter most in the afferent and rhythm layers, where mechanosensor onsets and phase transitions are discrete events, and least at the motor-neuron output, which the muscle activation

dynamics low-pass filter anyway.

The MuJoCo–SNS-Toolbox mirror (`build_network_spiking.py`) reproduces the production spinal network of Section 7.2 population for population and connection for connection, and is selected at build time by a single environment-variable switch so that the default build path is untouched; that the non-spiking pipeline is unchanged was proven by re-running its defaults gate and baseline evaluation to bit-exact agreement. Because SNS-Toolbox 1.5.2 ships no spiking bursting-neuron class, the rhythm-generator half-centers are adapting integrate-and-fire cells: burst termination is carried by spike-frequency adaptation of the firing threshold, whose time constant reuses the corresponding slow-inactivation value of the non-spiking build so that tuned adaptation values transfer between the two. Spiking cells rest at $-70\,\mathrm{mV}$, spike at $-50\,\mathrm{mV}$, and reset to $-60\,\mathrm{mV}$, with a tonic bias current parking their equilibrium a few millivolts below threshold, the analog cells' near-threshold operating point. The network sub-steps at 0.5 ms (four sub-steps inside each 2 ms physics step) so that spike timing is resolved without changing the physics integration, and non-spiking readout taps are appended for the runner's stance gates and neural logging so that all existing analysis code reads analog levels unchanged.

The mirror's synaptic weights are set by a calibration script rather than by hand. Spikes onto the non-spiking motor neurons are calibrated by mean conductance, so that a steady presynaptic train reproduces the non-spiking steady-state motor-neuron voltage (calibration residuals at or below $3 \times 10^{-4}\,\mathrm{mV}$); synapses between spiking cells are calibrated by firing-rate transfer (a 40 Hz source drives its target at 40 Hz, and a calibrated inhibitory synapse silences a 40 Hz-driven cell); the graded-to-spiking command pathway is scaled from the measured afferent current-to-rate map. The full calibration of record, with the

measured residual of every factor, is tabulated in Appendix B (Table B3).

In the Simulink family, three blocks were added additively to the SNS library of Section 7.2 (for fifteen blocks in total): a `SpikingNeuron` with the same internal synaptic summation as the non-spiking block, an event-wired `SpikingSynapse` whose conductance jumps by a fixed increment per presynaptic spike and saturates at its maximum, and a `HybridSpikingSynapse` that carries the same conductance dynamics onto a non-spiking postsynaptic cell. The blocks were verified against the analytic single-spike conductance decay (maximum deviation 6.2×10^{-7} µS), against a verbatim re-implementation of the SNS-Toolbox update equations on a hybrid two-neuron circuit (identical spike count and maximum spike-time deviation 11.75 µs), and against the SNS-Toolbox numpy backend itself (mean interspike-interval difference 0.19 µs). Spiking twins of the knee-reflex and beverage-holding demonstrations were then built by copy-and-patch: only the sensory interneuron layer and its synapses are replaced by spiking cells, while the motor neurons, muscles, joints, and measurements are identical to the committed demonstrations, and the twin is run at an operating point matched to the baseline gains.

In the AnimatLab family, spiking copies of the runnable standalone exports were produced by a conversion script that rewrites copies of the model files, leaving every original file untouched: neuron thresholds are moved from the graded regime (unreachably positive) into the spiking regime, every graded chemical synapse type is re-emitted as a per-spike synapse with the same identity and amplitude (interpreted as a per-spike conductance increment with 10 ms decay), and the per-connection class tag that AnimatLab stores independently of the synapse type is flipped to match. A calibrated variant sets each cell class's threshold just below its measured graded peak, and an ablation that converts

the thresholds but keeps the graded synapses isolates which change does what. Each copy passes a structural validator (neuron, connection, synapse-type, chart, and stimulus sets identical to its original, and every class tag consistent) and runs headless to its configured end. The 2023 spiking walker model of the same lineage, which was tuned as a rate code from birth, serves as the spiking reference baseline for this family.

Each mirror is verified against its non-spiking twin by a fixed gate set rather than by inspection: a topology gate requires the connection multiset to be identical between the two builds under three gain configurations; a rhythm gate requires self-sustained alternating rhythm under constant drive with the period within 20 % of the analog build; a basin gate requires the rhythm to persist under 1 % multiplicative jitter on every scalar parameter; an air-regime smoke gate compares cadence, duty, and joint ranges of the deafferented suspended-air preparation; and a regression gate requires every committed non-spiking artifact to reproduce its recorded values exactly with the spiking blocks present in the library. The mirrors carry mapped parameters only, with no new tuning, so their results (Section 7.4) establish what transfers structurally and what requires re-tuning, and any fresh tuning campaign is future work.

7.3 Neuromechanical Simulation Results

7.3.1 AnimatLab Phase-1 Recordings

The AnimatLab phase-1 run described in Section 7.2 produced synchronized joint, neural, contact-sensory, and body-motion recordings over 10 s. Basic checks confirmed finite values and a uniform 0.2 ms sampling interval in both exported charts. Each chart contained

50,001 samples from 0 through 10 s; nine trailing zero-filled rows beyond the configured chart end were excluded. Figure 7.7 shows bilateral hip, knee, and ankle angles. Neural and contact-sensory recordings remain archived with the run; the contact-sensory channels record membrane voltage rather than contact force. These recordings demonstrate signal acquisition and alternating activity in this run, rather than validated locomotor performance. The model included an externally applied horizontal force during the first second, and neither stable walking nor effective ground-contact gating was established.

7.3.2 Cross-Engine Verification of the Contact-Driven Rhythm Core

The verification protocol of Section 7.2.4 was applied to the W2L rhythm core. Between the NumPy reference and the native Simulink reimplementation, both legs' rhythm-generator burst periods matched at 1.000 s versus 1.000 s (0.0% error), the left–right antiphase correlation reproduced the reference exactly ($r = -0.525$), and all ten expected bursts occurred in the scored window. The full voltage traces cross-correlate at $r = 0.99996$–0.99998 at a -2 ms lag with an RMSE of 0.099–0.111 mV (Figure 7.8); the residual lag is consistent with the one-step delay of the discrete Simulink solver. The same protocol surfaced an architectural fact rather than a defect: with the heel-contact trains removed, the full network holds a static latch under tonic drive with kickoff alone, in every implementation — so phasic contact feedback is part of the drive architecture of the W2L design, not an optional perturbation.

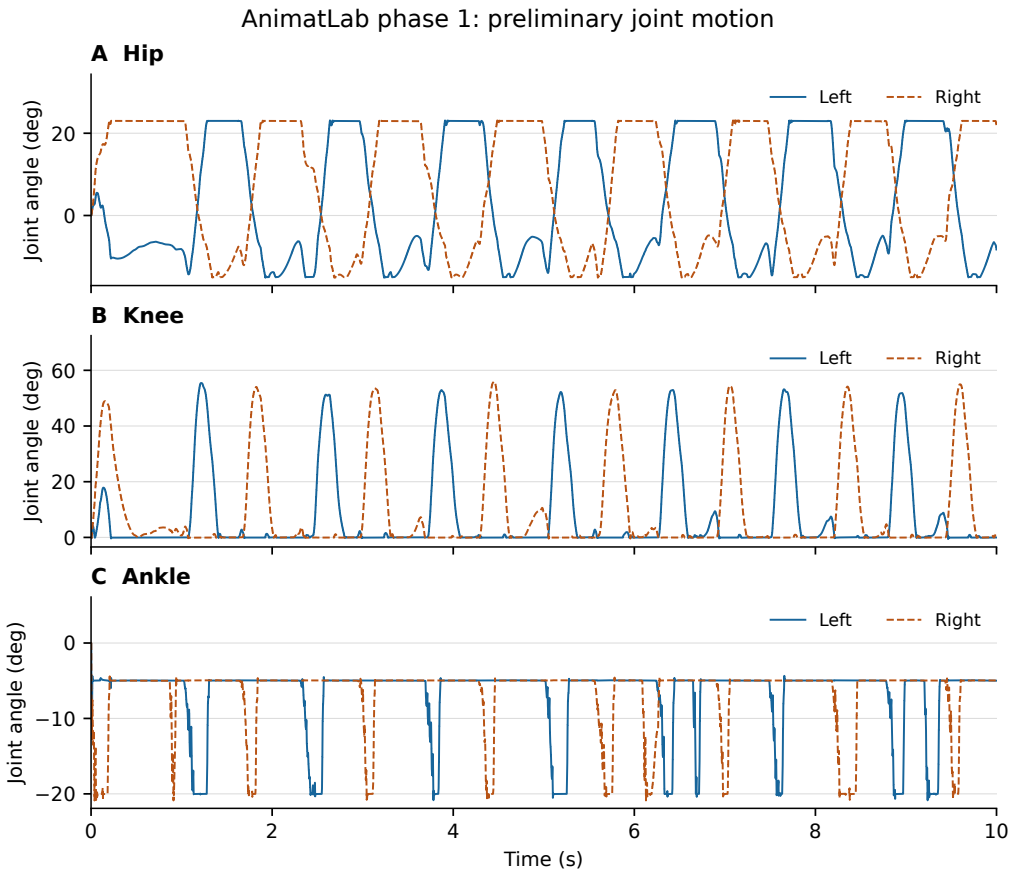


Figure 7.7: AnimatLab phase-1 joint motion: **(A)** hip, **(B)** knee, and **(C)** ankle angles in the model's recorded joint conventions. Blue solid and orange dashed traces denote left and right sides. All 50,001 samples within the configured 0–10 s chart window are retained without filtering. The model includes an external horizontal force during the first second. These traces document joint motion in an instrumented run; they do not establish stable walking or effective ground-contact gating.

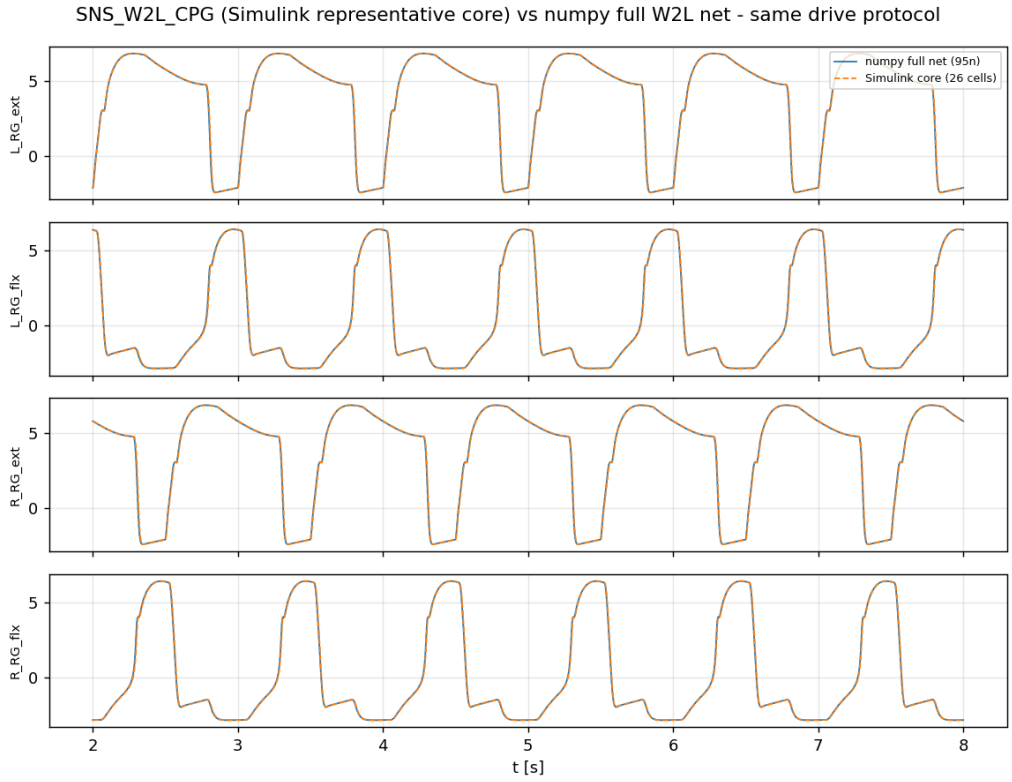


Figure 7.8: Left and right rhythm-generator extensor half-center voltages of the W2L contact-driven rhythm core under the shared verification protocol, from the NumPy reference (SNS-Toolbox backend) and the native Simulink reimplementation. Scored metrics — burst period, left–right antiphase correlation, and burst counts — agree exactly, and the full traces cross-correlate at $r \geqslant 0.99996$ at a -2 ms lag.

7.3.3 Headless Validation of the AnimatLab Reference Models

The archived AnimatLab reference models were re-run headless from their standalone simulators with instrumented charts, and eleven targetable rhythm and gait metrics were extracted automatically; all eleven pass their acceptance bands. In the bilateral contact-driven model, the rhythm generators oscillate at periods of 0.450 s (left) and 0.467 s (right) with a left–right lag of 0.529 cycle (cross-correlation $r = 0.976$), confirming alternating gait timing. In the modern W2L model, the joints oscillate at 2.25 Hz; its flexor half-centers peak at -56.4 mV and never cross the -55 mV threshold, so the rhythm is carried by clean extensor/flexor antiphase ($r = -0.996$) through the graded pathway rather than by flexor spiking. The Li-model rerun [Li et al., 2017] walks at $0.637\,\mathrm{m\,s^{-1}}$ with right and left step periods of $1.299(24)$ s and $1.333(80)$ s and duty factors of 0.443 and 0.555, reproducing the published behavior of the original model in its original engine.

7.3.4 BPA-Actuated Walker in MuJoCo

The walker geometry of Section 7.1 was exported to MuJoCo and driven by antagonistic BPA actuator pairs under the split contact-driven network, with actuator forces computed by the BPA model of Chapter 3. In the air-stepping gate — the harness carrying 70% of the 411 N body weight — the walker produced clean knee-flexion excursions at a 1.542 s period with a knee flexion range of 63.5°, peak BPA forces of 880 N (left) and 1372 N (right) at a peak pressure of 372 kPa — 60% of the 620 kPa ceiling — and 7 and 5 heel-contact episodes on the left and right. This run demonstrates the complete chain from the characterized BPA force model to a multi-joint walker stepping under synthetic neural control, with

actuator sizing operating inside the characterized pressure range.

The same campaign reported its failures with equal weight. On the ground, the W2L gate model stands (full-support stance) but does not walk: heel duty is zero and forward displacement is $+0.001\,\mathrm{m}$ over the run. The MuJoCo re-implementation of the Li model produced no stance episodes in any of seven drop-protocol variants. Together with the latch result of Section 7.3.2, these negatives localize the open problem precisely: the contact-driven rhythm architecture itself transfers across engines intact, but translating it to unsupported ground locomotion requires the balance and weight-bearing layers defined in Chapter 8, and those layers are not present in these models.

7.3.5 Gait-Validation Sweep against the Locomotion Reference Library

The four production walking configurations of the converted robot-body model were evaluated against a library of 43 human locomotion reference cycles: 32 running trials from eight subjects at four speeds, ten walking references (seven at prescribed speeds from $0.5\,\mathrm{m\,s^{-1}}$ to $2.0\,\mathrm{m\,s^{-1}}$, plus three self-selected), and one further reference cycle, `Case_40_motion`, the predicted-walking cycle of the Falisse predictive-simulation dataset (period 1.11 s, duty factor 0.57). The four configurations span the two controller lineages of this program. The human-derived lineage contributes the production walking configuration (`s3k`), whose two-level network was tuned against the human walking reference, its pruned derivative (`s3kpruned`), and the six-synergy variant (`syn6`), whose pattern-formation layers drive the motor pools through a six-synergy basis extracted from human gait data. The robot-walker lineage contributes the W2L-layout variant (`w2lvar`), which restructures the rhythm

and pattern-formation layers after the contact-driven W2L circuit of Section 7.2.4. Each configuration was scored with the production pipeline's cycle-shape kinematic score, in which 0 denotes perfect agreement and -25 is the fixed no-cycle fallback, both on its tuning-reference evaluation and against every library cycle (Table 7.1).

Table 7.1: Gait-validation sweep of the four production walking configurations against the 43-cycle locomotion reference library. The score is the production pipeline's cycle-shape kinematic score (0 denotes perfect agreement); the tuning-reference score is each configuration's evaluation on the tuning pipeline's walking reference, and the library mean averages the same score over all 43 references (among genuine gait cycles, higher scores indicate closer agreement).

	Tuning-ref. score	**Library mean score (43 refs)**	**Most negative reference (score)**
`s3k`	-186.4	-8776	`Case_40_motion` (-8841)
`s3kpruned`	-176.1	-6924	`Case_40_motion` (-6986)
`syn6`	-200.4	-9961	`Case_40_motion` (-10040)
`w2lvar`	-144.5	-5808	`subject08_Run_20002` (-5896)

The ranking is identical in the two evaluations: the W2L-layout variant agrees most closely with both its tuning reference and the library on average (`w2lvar` at -144.5 and -5808), the six-synergy variant least closely (`syn6` at -200.4 and -9961), and the two production configurations fall between, with pruning moving `s3k` toward closer agreement (-186.4 and -8776 for `s3k`, against -176.1 and -6924 for `s3kpruned`). Every configuration's library scores are one to two orders of magnitude more negative than its tuning-reference score, which quantifies how far a network tuned to a single walking reference transfers to the library's diverse reference cycles; the gap is not an artifact of the 32 running trials, because the production configuration's mean over the ten walking references alone (-8816) is slightly more negative than its running-only mean (-8761) and essentially its overall

library mean (-8776). The most negative single score for three of the four configurations falls on the same library reference, `Case_40_motion` (-8841 for `s3k`, -6986 for `s3kpruned`, and -10040 for `syn6`), while the W2L-layout variant's most negative reference is a running trial, `subject08_Run_20002` (-5896). All four configurations produce right-leg-only or asymmetric gait in these evaluations. At the family level, the sweep's reading is that the robot-walker-lineage configuration, not the human-derived ones, agrees most closely with the library, while the six-synergy configuration, the most strongly human-derived in its pattern formation, agrees least closely.

7.3.6 Comparison of the Production Walker Variants

The same campaign removed individual pathway classes from each configuration and re-evaluated the tuning-reference score, identifying which pathways each lineage depends on (Table 7.2). In the production configuration, removing the Ia afferent pathway freezes the model outright: the evaluation detects no gait cycles and returns the fixed -25 fallback. Removing the Renshaw cells, the group-II afferents, or all afferent classes at once degrades the score to -267.3, -255.2, and -267.8, respectively, against the full model's -186.4, while removing the Ib, contact, or interleg commissural channels individually shifts the score by smaller amounts toward zero (-180.3, -183.9, and -174.9). The six-synergy variant behaves differently: with all afferents removed it scores -190.5 against its full model's -200.4 and still walks bilaterally, with ten cycles on each leg, a knee-flexion minimum of -71.2°, and a 0.684 s period, so its afferent complement is not what sustains its gait. The W2L-layout variant depends on both the Ia pathway (removed: -273.5) and the weak

rhythm-generator mutual excitation (removed: -279.9) against its full model's -144.5 with four right-leg and six left-leg cycles, while removing its Renshaw cells leaves a bilateral gait at -142.4.

Table 7.2: Pathway-removal sweep on the tuning-reference evaluation. The score is the production pipeline's cycle-shape kinematic score (0 denotes perfect agreement; -25 is the fixed no-cycle fallback, a failed evaluation that scores above every genuine gait in this table). The full-model rows repeat each configuration's unmodified production evaluation.

Configuration	**Removed pathway**	**Score**
`s3k` (full model)	none	-186.4
	Ia afferents	-25 (no cycles detected)
	all afferent classes	-267.8
	Renshaw cells	-267.3
	group-II afferents	-255.2
	Ib afferents	-180.3
	contact channels	-183.9
	interleg commissural pathways	-174.9
`syn6` (full model)	none	-200.4
	all afferent classes	-190.5
`w2lvar` (full model)	none	-144.5
	Ia afferents	-273.5
	weak rhythm-generator mutual excitation	-279.9
	Renshaw cells	-142.4

Drive-perturbation and push testing splits the field along family lines. The production configuration never fell in 15 stand and push evaluations (maximum push sway 0.123 m), and its walk survives drive perturbations of $+0.01$, -0.01, and -0.02 (scores of -201.9, -177.0, and -171.2), freezing only at $+0.02$; its pruned derivative survives the unperturbed evaluation (-176.1) and the -0.02 perturbation (-168.2). The six-synergy and W2L-layout variants fall under every ± 0.01 and ± 0.02 drive perturbation tested, with pelvis

tilt reaching 39° to 40°. The sweep therefore identifies the production configuration and its pruned derivative as the robust pair of the family, and locates the fragility of the six-synergy and W2L-layout variants under drive perturbation.

7.4 Spiking-Neuron Mirror Results

The spiking-neuron mirrors of Section 7.2.5 were evaluated against their non-spiking twins in the same order in which the families were introduced: the MuJoCo spinal network, the Simulink demonstrations, and the AnimatLab walker family. All numbers in this section come from the gated verification campaign whose artifacts and reproduction commands are catalogued in Appendix B.

7.4.1 MuJoCo Spinal-Network Mirror

The non-spiking production baselines were first re-established on the machine of record: all three current production configurations reproduce their documented evaluation scores exactly, including the default build's kinematic score of -161.56754173676563, and after the mirror's selector was added the default build gate ($(410, 376, 1186)$ populations, inputs, and connections) and that same baseline score reproduce digit for digit, proving the non-spiking pipeline untouched.

The topology gate passes exactly: under the default, fully tuned, and joint-layer pattern-formation gain configurations, the spiking mirror's directed connection multiset is identical to the non-spiking build's (1,186, 7,908, and 7,762 connections respectively), with identical input lists, identical motor-neuron maps, and exactly the expected number of readout taps

(12, 12, and 16). The mirror is structurally the same network.

Under constant drive (network alone, 20 s), the rhythm gate splits. The spiking mirror self-sustains clean bilateral antiphase — rhythm-generator extensor and flexor cells firing at 4.9 Hz and 3.2 Hz per leg, with the swing readout sustained in every window — but its envelope period is 0.621 s against the analog build's 0.399 s, a 55.6 % lengthening that misses the gate's 20 % tolerance (Figure 7.9). The basin gate passes in full: the rhythm persists under 1 % multiplicative jitter on every scalar parameter in all 12 seeded trials, with a final-window swing minimum of 4.58 mV against the 5.13 mV baseline (margin 0.89×) and no divergent trials. In the deafferented suspended-air preparation, the mirror air-steps rhythmically with full-amplitude joint swings (knee range 106.0° against the twin's 117.9°), but its cadence is roughly ten times faster than the non-spiking twin's 0.47 Hz air-stepping regime in this preparation, so the air-regime gate fails on the cadence clause alone.

One untuned ground-walking evaluation was run as a stretch attempt: the spiking mirror at the non-spiking winner's mapped parameters, with no re-tuning. It scores worse overall (kinematic score −236.35 against −161.57), as expected for an untuned mapped starting point, but it produces a genuine bilateral gait — eight right-leg and nine left-leg cycles, a cycle-window knee flexion minimum of −79.8°, no non-finite states, and the model stays up — whereas the non-spiking winner's left leg is frozen (zero cycles) in every recorded run of its study, including this session's baseline (Figure 7.10, Table 7.3). This is a single run with no statistics, reported as a lead for the fresh tuning campaign that the mirror's plan calls for, not as an established gait result.

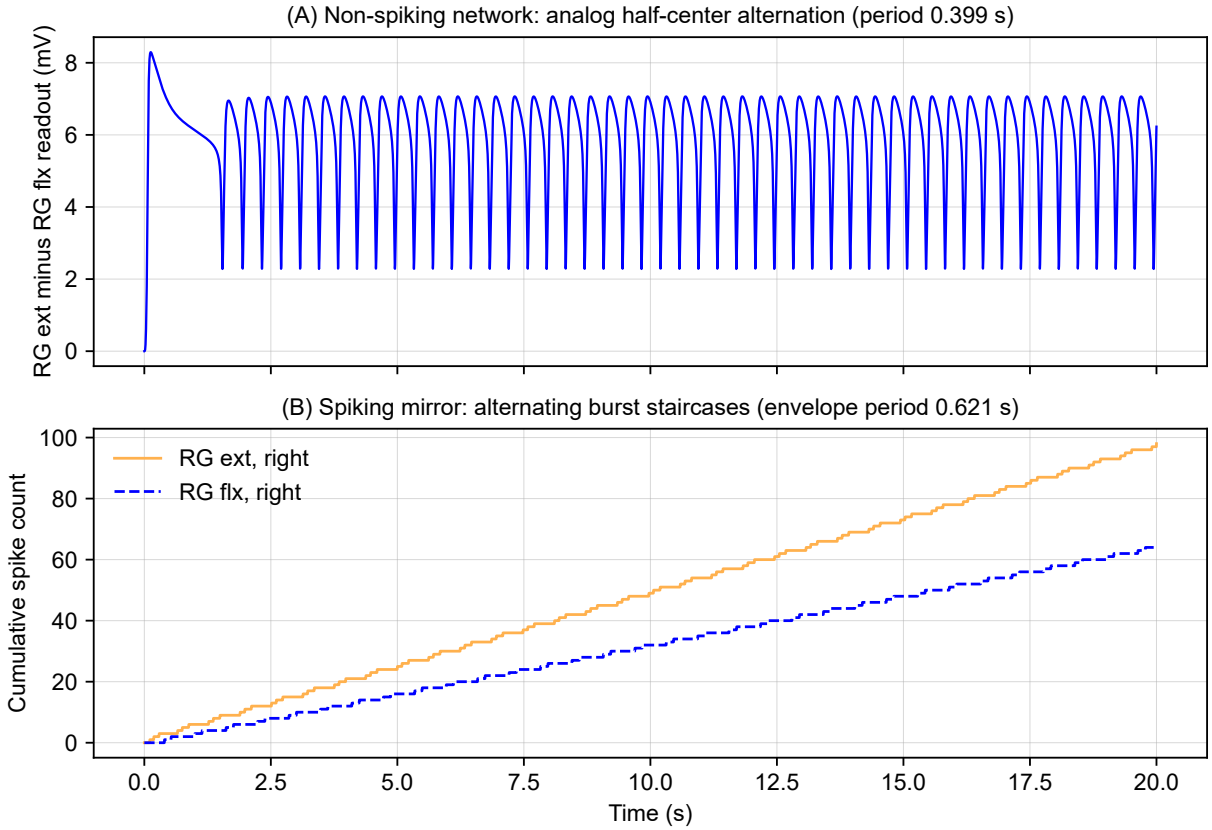


Figure 7.9: Constant-drive rhythm of the right rhythm-generator under network-only simulation. **(A)** The non-spiking network's extensor-minus-flexor readout voltage alternates irregularly about a 4.8 mV swing (larger only in the startup transient), with an average period of 0.399 s. **(B)** The spiking mirror's raw extensor (solid orange) and flexor (dashed indigo) cells fire in alternating bursts whose envelope period is 0.621 s, 55.6 % longer than the analog build.

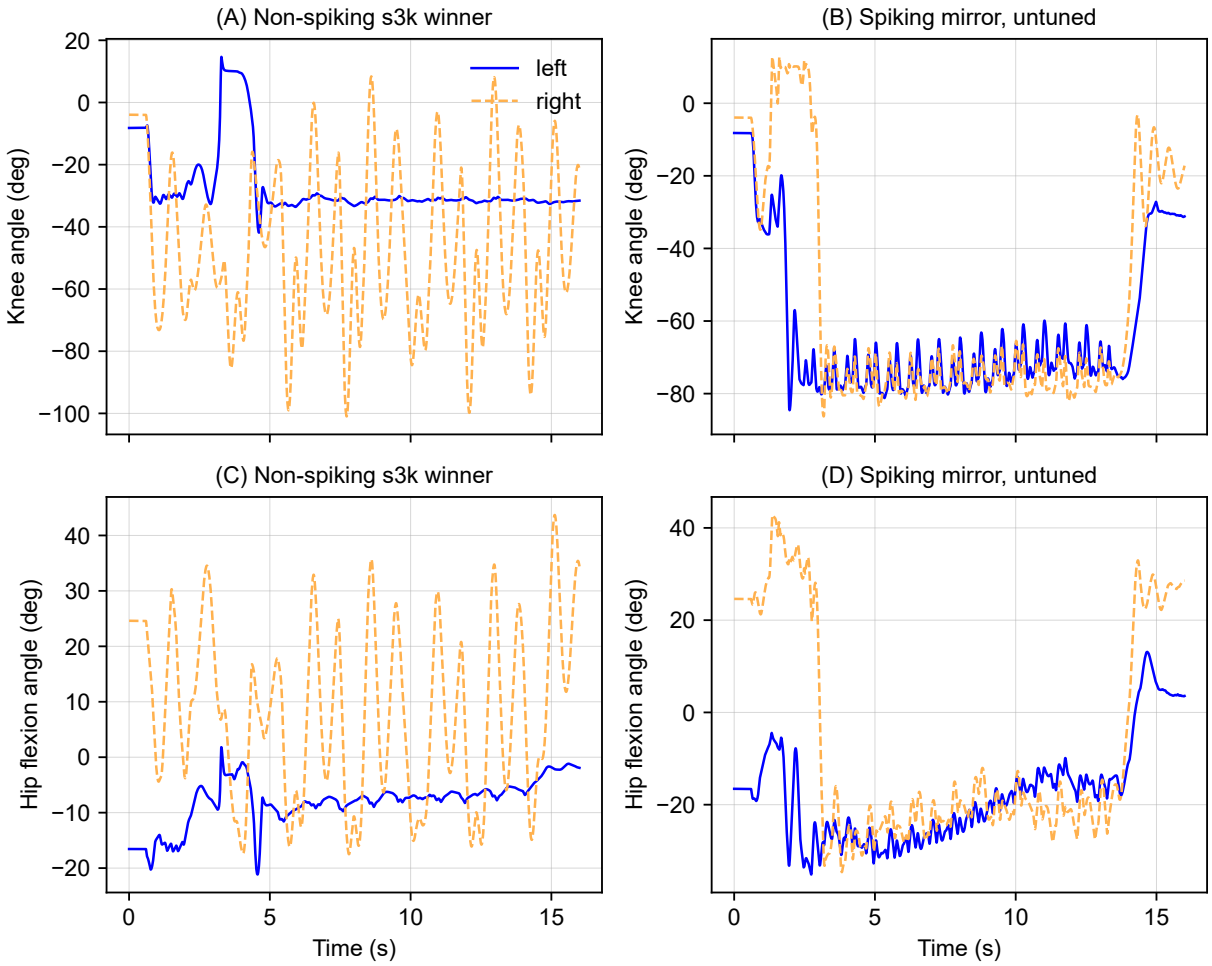


Figure 7.10: Ground-walking joint angles of the converted MuJoCo robot at the non-spiking winner's parameters. **(A, C)** The non-spiking winner steps with the right leg only; the left knee moves through a single slow excursion without cycling. **(B, D)** The untuned spiking mirror cycles both legs with deep knee flexion. Solid indigo traces denote the left leg, and dashed orange traces denote the right leg.

Table 7.3: Untuned spiking mirror versus its non-spiking twin at the same winner parameters, ground evaluation (16 s). The cycle knee minimum and stance fractions are computed over detected gait cycles; the kinematic score is the cycle-shape error metric of the production pipeline (0 denotes perfect agreement; higher scores indicate closer agreement). The spiking mirror carries mapped parameters only, with zero re-tuning.

Metric	**Non-spiking winner**	**Spiking mirror (untuned)**
Kinematic score	-161.57	-236.35
Gait cycles, right / left	9 / 0 (left frozen)	8 / 9 (bilateral)
Cycle knee minimum	-64.3°	-79.8°
Contact duty, right / left	0.637 / 0.985	0.799 / 0.856
Maximum pelvis tilt	22.1°	19.2°
Non-finite states / fell	no / up	no / up

7.4.2 Simulink Spiking Demonstration Twins

The three new library blocks were verified at the unit level before any demonstration was rebuilt. The single-spike conductance transient matches the exact exponential to 6.2×10^{-7} µS, saturating pulse-train accumulation matches to 4.6×10^{-5} µS, and a hybrid two-neuron circuit matches a verbatim re-implementation of the SNS-Toolbox update equations spike for spike (62 of 62 spikes, maximum spike-time deviation 11.75 µs, maximum voltage deviation 1.2×10^{-2} mV). The same circuit run through the SNS-Toolbox numpy backend agrees with the Simulink copy to 0.19 µs in mean interspike interval and 0.81 µV in final voltage. With the spiking blocks present in the library, every committed demonstration and units test reproduces its recorded value exactly, including the committed two-neuron units test at 6.07×10^{-4} mV.

At the matched operating point, the spiking twin of the knee-reflex demonstration is indistinguishable from the baseline in its transient and within one degree in its settle

behavior (Table 7.4, Figure 7.11): both read 15.79° at 0.05 s, and the twin settles 0.9° higher with a slightly hotter, faster antagonist alternation, because the rate-coded pathway adds the synaptic filter time constant that graded synapses do not have. The spiking twin of the beverage-holding demonstration likewise holds the cup at essentially the same level, with the reflex-off sag identical to the baseline and the reflex modulation within 0.02° (Table 7.5).

Table 7.4: Knee-reflex demonstration, committed baseline versus spiking twin at the matched operating point (5 s runs; settle statistics over the last 2 s).

Metric	**Baseline**	**Spiking twin**
Knee angle at 0.05 s	15.79°	15.79°
Settle mean (last 2 s)	43.5°	44.4°
Settle range	41.6°–44.9°	41.9°–46.7°
Extensor / flexor activation (window means)	0.53 / 0.53	0.58 / 0.56
Antagonist alternation	0.5 Hz–2 Hz	7.4 Hz (dominant peak)
Ia interneuron firing at settle	graded, subthreshold	95 Hz; Ib 0 Hz

Table 7.5: Beverage-holding demonstration, committed baseline versus spiking twin (10 s pour; sag measured after 2 s).

Metric	**Baseline**	**Spiking twin**
Maximum sag, reflex on	2.00°	2.02°
Maximum sag, reflex off	7.85°	7.85°
Reflex modulation (off minus on)	5.85°	5.83°
Biceps activation over the pour	0.392 → 0.417	0.392 → 0.447
Triceps activation over the pour	0.400 → 0.316	0.400 → 0.254
Biceps Ia firing rate (reflex on)	graded	166.6 Hz (mean)

A representative rhythm-generator-to-motor-pool subnetwork demonstrates the hybrid motif end to end: spiking half-centers drive, through a chain of inhibitory interneurons,

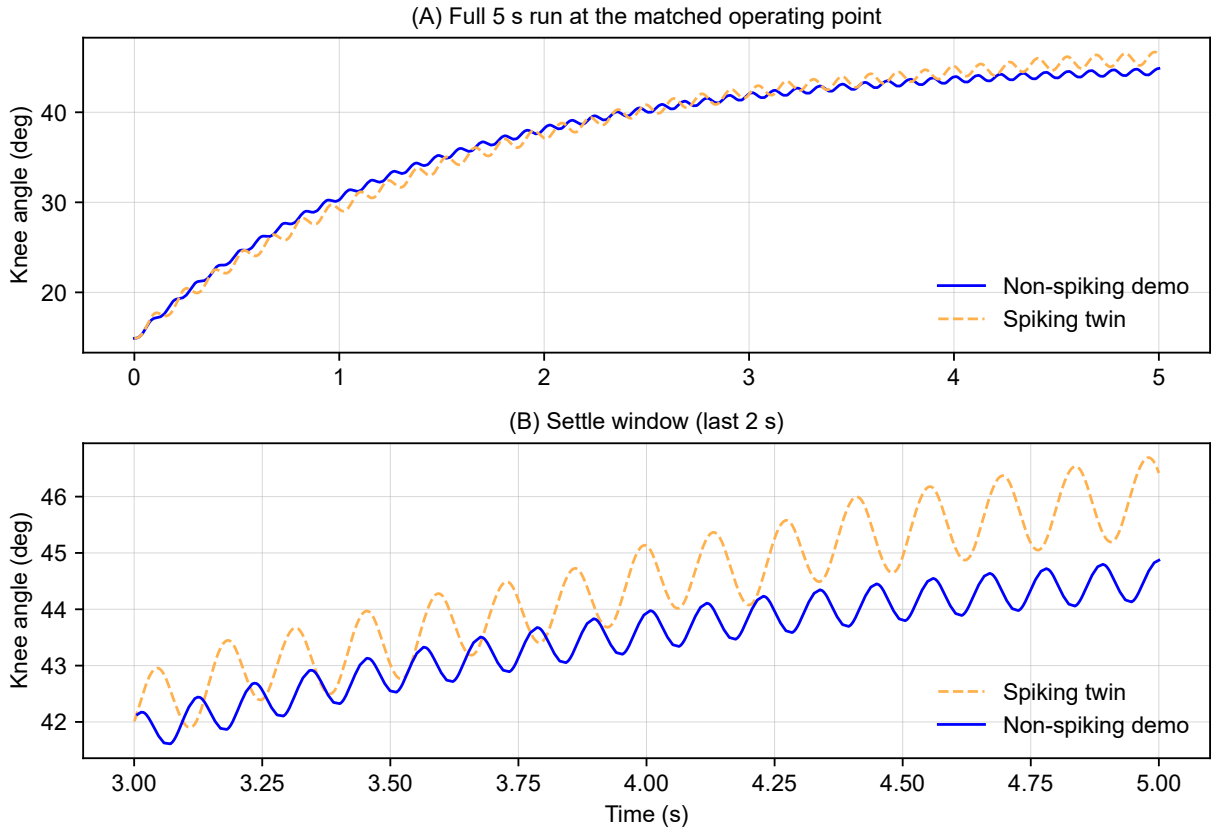


Figure 7.11: Knee angle of the Simulink knee-reflex demonstration: committed non-spiking baseline (solid indigo) versus spiking twin (dashed orange) at the matched operating point. **(A)** Full 5 s run. **(B)** Settle window (last 2 s); the spiking twin's alternation band is slightly wider.

a non-spiking three-motor-neuron pool whose voltage stays graded over a 5 mV depth (-52.0 to -46.9 mV) and tracks its rhythm generator's rate envelope. Burst alternation in that subnetwork is partial: the extensor half-center bursts four times in 12 s while the flexor half-center keeps a low-rate floor through the extensor bursts. This incompleteness is reported as is, and the full 410-population Simulink spinal-network conversion is scoped as future work for the same reason.

7.4.3 AnimatLab Spiking Conversions

The AnimatLab family produces the campaign's clearest negative result, and it is reported as such. The graded baselines were re-established first: the two-layer walker's rhythm generator oscillates at 0.443 s (2.26 Hz), the bilateral variant at 0.462 s with the two legs folded near antiphase (lag -0.42 cycle, $r = 0.869$), and the already-spiking 2023-lineage model — the family's spiking reference — bursts at 1.305 s with folded antiphase of $+0.494$ cycle ($r = 0.703$), reproducing its period of record. The current reorganized project's standalone export remains latched under the tonic drives examined, consistent with Section 7.2.

No spiking copy of the three graded models preserves the rhythm (Table 7.6). The two-layer conversions die at every threshold and decay tried (two or zero rhythm-generator spikes in 5 s); the bilateral variant latches with one side's half-centers saturated at 442 Hz–450 Hz; and the reorganized project's copy converts its latched baseline into strictly synchronous 11.8 Hz co-bursting with no alternation — the left extensor and flexor half-centers emit identical 58-spike trains whose spike-count envelopes correlate at $r = 1.000$ at zero lag.

The thresholds-only ablation, which keeps the graded synapses, preserves near-baseline joint oscillations, isolating the synapse conversion as the change that breaks the rhythm (Figure B1): per-spike conductances at the graded amplitude either starve the reciprocal loop of inhibitory charge or saturate it, whereas the graded synapses read the membrane potential continuously and keep oscillating through subthreshold voltages. A spiking version of this family therefore needs its synaptic amplitudes and decays re-tuned as a rate code, which is how the 2023 spiking reference was built in the first place, rather than copied parameter by parameter.

Table 7.6: AnimatLab spiking conversions versus their graded baselines (5.1 s headless runs; joint ranges in degrees where the rhythm survives). The thresholds-only ablation converts the neurons but keeps the graded synapses.

Variant	**Baseline (graded)**	**Spiking copy**
Two-layer walker	0.443 s rhythm; hip -15 to $+23$, knee -1 to $+60$	dead (2 spikes; hip pinned near $+6$)
Bilateral variant	0.462 s; antiphase -0.42 cycle	latched; one side at 442 Hz–450 Hz
Reorganized project (standalone export)	latched (tonic drives)	synchronous 11.8 Hz co-bursting; same-leg envelopes $r = 1.000$ at lag 0
Thresholds only (graded synapses)	hip -15 to $+23$, knee -1 to $+60$	oscillates: hip -15.1 to $+23.2$, knee -0.8 to $+47.9$

The spiking conversions of the two-layer family, the calibrated-threshold variants, and the decay probe are catalogued variant by variant, with neuron, synapse-type, and connection counts and the structural-validation outcome of each, in Appendix B (Table B5), together with the full gate table of the MuJoCo mirror (Table B4).

7.5 What Spiking Changes and Does Not

The spiking-mirror campaign (Sections 7.2.5 and 7.4) allows the dissertation's neural-coding assumption to be examined directly, and its outcome divides cleanly into what the coding regime changes and what it leaves alone.

What spiking does not change is the behavior of the controlled model, provided the design rule and the calibration hold. Because the motor-neuron layer stays non-spiking and the spikes onto it are calibrated by mean conductance, the interface that the actuator and torque models of this dissertation attach to, a continuous activation command per muscle, is unchanged by construction. The Simulink twins make the point quantitatively: with only the sensory interneuron layer converted, the knee-reflex transient is identical, the settle band moves by 0.9°, and the beverage-holding reflex modulation moves by 0.02°. To the musculoskeletal model, a calibrated 40 Hz train and the graded voltage it replaces are the same command; the activation dynamics of the muscle model filter the spike train the same way the biological membrane would. The actuator-level and joint-level conclusions of Chapters 3–4 therefore do not depend on the neural coding chosen above the motor-neuron layer.

What spiking changes first is timing. The mirror's self-sustained rhythm runs 55.6 % slower than the analog build at the same constant drive, and its air-stepping cadence is roughly ten times faster than the non-spiking twin's in that preparation — deviations in opposite directions in the two tests, which is itself informative, since a single scaling error would move both the same way. The campaign traced the constant-drive period to two identifiable knobs, the threshold-increment magnitude and the adaptation time constant,

and demonstrated alternation/period trade-offs along them; the untuned starting point, not a structural failure, is the honest reading. Any period-matched spiking controller is therefore a short tuning campaign away, and the mirror's mapped parameters are the correct starting point for it.

What spiking changes second is robustness, in the mirror's favor. The analog network's constant-drive rhythm sits close to a bifurcation, a parameter region where an exact re-implementation in a different integration stack can settle to the tonic fixed point instead of oscillating, so marginal winners risk not surviving a solver change. The spiking half-centers passed the perturbation basin gate in full, with the rhythm persisting under 1 % jitter on every scalar parameter including the spiking-specific ones, at $0.89\times$ of the baseline swing. For the hardware-transfer program of Chapter 8, where the network will be re-implemented on embedded hardware with finite-precision arithmetic, a rhythm whose basin is wide is the safer substrate, and the sub-stepping cost of resolving spikes (0.5 ms network steps inside the unchanged 2 ms physics step) is modest.

What spiking does not tolerate is a naive parameter copy at the synapse level. The AnimatLab conversions show that converting graded synaptic amplitudes to per-spike increments of the same magnitude breaks the reciprocal half-center loop outright — the loop is either starved of inhibitory charge or saturated — while converting only the thresholds changes nothing, and the 2023 spiking walker of the same lineage walks because it was tuned as a rate code from birth. Synaptic strength in a spiking network is a rate-code quantity that must be derived from the presynaptic firing rate and the conductance decay, exactly as the MuJoCo mirror's calibration script derives it, not copied from the graded value. This is the one place where the campaign's negative results should temper

expectations: spiking conversion is an exercise in re-derivation, and the calibration-first, gate-everything procedure of Section 7.2.5 is the transferable method.

Finally, the one place where the untuned spiking mirror outperforms its tuned non-spiking twin, producing a bilateral ground gait where the winner's left leg is frozen, is a single run with no statistics, and it is claimed only as a lead. But it points at the mechanism the design rule was after: the mirror's afferents are event-coded, its adaptation is spike-history-based, and its inhibitory charge arrives in discrete quanta, so the phase relationships that the analog build locks in around a frozen leg are not the only stable operating regime of the same topology. Establishing whether that regime survives tuning, and whether event-coded contact afferents are what unlock it, is precisely the closed-loop CPG comparison program defined in Chapter 8, and the mirror is now the instrument for running it.

8 The Work as a Whole and Future Work

8.1 Completed Foundation Studies

Four studies completed during this dissertation establish the foundation for the experimental chapters and for the controller program defined in the remainder of this chapter. First, the placement optimization of Morrow, Bolen, and Hunt found artificial muscle attachment locations that reproduce human torque profiles and ranges of motion about the trunk and leg joints [Morrow et al., 2020]; it converted the earlier finding that transplanted human attachment locations do not transfer directly to a BPA-actuated robot [Bolen, 2019, Bolen and Hunt, 2019] into a usable design tool. The route redesigns of Section 4.7 rerun that tool under the corrected torque model. Second, the Sensory Afferent Database (Chapter 6) organized sensory-afferent literature into a structured, model-oriented resource [Bolen et al., 2023], and a follow-up study evaluated AI-assisted literature triage for expanding it [Bolen and Hunt, 2024]; those data inform the sensory channels of the neural controller. The database has since been audited and expanded into a curated laboratory tool: of the 943 papers in the audited corpus, 833 carry structured curation notes, 95 carry an afferent classification, and 363 carry notes translating each pathway toward robot or simulation use, packaged with the corpus citation graph (18 topic clusters, 11,130 directed citation pairs) as a self-contained explorer for lab use. Third, an inverted-pendulum balance platform with BPA-driven plantarflexion and dorsiflexion (Chapter 5) demonstrated pseudo-vestibular feedback using inertial measurement units [Bolen et al., 2024b,a,c]. Fourth, companion

work by Elzein, Lutz, Bolen, and Hunt produced a state-space, optimization-based model of pressure and force dynamics in pulse-modulated BPAs [Elzein et al., 2025], supplying dynamic actuator behavior beyond the static force characterization reported in Chapter 3.

8.2 Synthesis: From Actuator Characterization to Neuromechanical Control

Taken together, the chapters of this dissertation form a coherent design chain for biomimetic legged robots, in which each stage feeds the next. At the actuator level, the maximum isometric force of Festo BPAs was shown to depend strongly on resting length, correcting a systematic over-prediction by the manufacturer's tool at short resting lengths, and a nondimensional force surface was derived that predicts isometric force from relative strain and relative pressure with only three diameter-dependent coefficients (Chapter 3). At the joint level, the improved actuator model was combined with a compliance-corrected musculotendon kinematics model to predict isometric knee torque across the full range of motion for pinned and biomimetic knee joints, and the biomimetic knee driven by optimized BPAs was shown to meet or exceed human isometric torque over most of the range of motion (Chapter 4). The remaining links connect the mechanical system to its controller: the balance-platform studies demonstrated in hardware that a BPA-actuated limb can hold itself upright under pseudo-vestibular feedback (Chapter 5), the Sensory Afferent Database supplies the afferent organization that the controller's sensory channels mirror (Chapter 6), and the simulation framework joins the converted robot model to a synthetic nervous system so that closed-loop experiments can run before hardware exists (Chapter 7).

The chain was assembled deliberately. The force model supplies the actuator terms for the torque model; the torque model supplies the biomechanical model for designing joint controllers; and the validated biomimetic knee demonstrates that the actuator architecture can reproduce human joint-level force capabilities. The neural layer is no longer only a proposal: the OpenSim-to-MuJoCo conversion, BPA coupling interface, SNS-Toolbox spinal architecture, sensory channels, and staged validation workflow described in Sections 7.2–7.3 provide a working framework for neuromechanical experiments. The AARL's prior AnimatLab walker (Section 7.1) explains why that framework emphasizes systematic verification: its hand-tuned two-layer CPG produced gait-like suspended stepping, but its afferent feedback did not generalize to weight-bearing motion.

The cross-platform verification campaign (Sections 7.2.4–7.3.4) adds a design constraint that runs through this chain: a synthetic nervous system is a mathematical object, and it must survive translation between the tools in which it is built, verified, and deployed. Reproducing the contact-driven rhythm core with exact period agreement and trace correlations above 0.9999 between independent NumPy and Simulink implementations, with the archived AnimatLab models reproducing their published rhythms headless in the same protocol, establishes that the SNS formalism, rather than any particular simulator, carries the dynamics. The persistent-sodium half-center with a fixed inactivation time constant, the graded synapse, and contact-driven rhythm are therefore portable primitives: they can be tuned in whichever engine is fastest and trusted in whichever engine is most physical. The campaign's negative results are equally structural. The feedforward CPG of the prior walker and the contact-driven core of the W2L architecture sit at deliberate, opposite points of the sensory-dependence spectrum. The former oscillates without feedback but could

not bear weight; the latter will not oscillate at all without phasic heel feedback, a load dependence that echoes the decerebrate-cat locomotion of Section 2.5. The missing piece for unsupported walking is balance, a network layer rather than an engine feature, and the BPA-actuated walker's peak forces at 60% of the pressure ceiling indicate that, once that layer exists, the actuation margin is already there.

With the mechanical characterization and the simulation framework in place, the dissertation's remaining task is the controller: a synthetic nervous system that schedules these actuators the way the nervous system schedules muscles, and, eventually, the physical robot that the controller runs on. The stages that follow are experiments the completed work already makes possible: each draws directly on the validated actuator model, the corrected torque model, the afferent database, and the simulation framework as delivered. The stages are a neuromechanical simulation pipeline brought from assembly to closed-loop tuning (Section 8.3), a two-layer CPG with full proprioceptive and contact feedback (Section 8.4), extension of the sensorium with vestibular, ocular, and cerebellar contributions (Section 8.5), physical realization on a treadmill robot with real-time neural control (Section 8.6), and, beyond steady walking, dynamic whole-body maneuvers (Section 8.7). Closed-loop quantification, ground-robustness comparison, and hardware transfer each stand as independent research programs enabled by the completed framework. Some of this work is planned for the remainder of the dissertation project; the later stages extend naturally into postdoctoral research.

8.3 A Neuromechanical Simulation Pipeline

The first stage converts the validated biomechanical design into a physics simulation that a neural controller can drive. The pipeline has three parts: the musculoskeletal model converted from OpenSim to MuJoCo, the coupling that connects a synthetic nervous system to that model, and the controller itself, which is the subject of Section 8.4. Much of this stage is now in place (Section 7.2): the OpenSim-to-MuJoCo conversion is complete and audited, the SNS-Toolbox coupling runs at a 0.1 ms neural step with force feedback, and the full two-layer spinal network has been assembled with its rhythm generator verified in isolation. The remaining blocker is non-finite leg-joint states in full-body co-simulation, with the foot-contact meshes the primary suspect; once it is resolved, the plan shifts from assembly to tuning and validation.

8.3.1 From OpenSim to MuJoCo

The starting point is the author's modified OpenSim model: the Gait2392-derived robot variant in which the actuator set has been reduced and adapted to the BPA hardware and the muscle origin, insertion, and wrapping points have been updated to the optimized robot routing [Bolen and Hunt, 2019, Morrow et al., 2020]. The model is kept in OpenSim 4.x format and converted to MuJoCo [Todorov et al., 2012] using MyoConverter [Caggiano et al., 2022, Wang et al., 2022], which converts OpenSim musculoskeletal models to MuJoCo models with optimized muscle kinematics and kinetics; the completed conversion and its audit are documented in Section 7.2. The conversion is checked by inspecting the generated model, verifying joint naming and ordering for later sensor and actuator

mapping, confirming that the expected number of muscle-tendon actuators exists with plausible attachment geometry, and fixing any conversion gaps by revising the OpenSim model or accepting manual edits in the MuJoCo model.

The MuJoCo model represents actuation with muscle-like actuators that carry an internal activation state, so that SNS outputs can drive them through activation dynamics. At each timestep, MuJoCo updates the kinematics and the muscle variables (e.g., muscle-tendon lengths and velocities) needed to compute proprioceptive signals.

8.3.2 Coupling the Synthetic Nervous System

The neural controller is implemented with SNS-Toolbox [Nourse et al., 2023], which supports large networks of spiking and non-spiking neurons and couples directly to MuJoCo. At each timestep, SNS receives Ia, Ib, and II proprioceptive channels plus binary heel and toe contact signals from the MuJoCo model, processes them through the neural core described below, and returns per-muscle activation commands that are fed into the MuJoCo muscle activation states.

The coupling itself is in place; its staged validation remains. Before any gains are tuned against human gait, the mechanics-only loop will be verified with open-loop activation patterns, confirming the sign and magnitude behavior of torques and forces. The network will then be exercised with a minimal sensor set, contact signals plus a single proprioceptive channel, and only later with the full Ia/Ib/II set. Once stable, the activation strengths and neuron-to-muscle mapping parameters will be tuned across a few walking speeds. A Simscape-based mechanical model could substitute for MuJoCo for validation

of specific mechanical subsystems, but Simscape does not natively host the spiking and non-spiking neuron models of an SNS, so MuJoCo remains the primary platform. The AARL's quadruped has already traveled part of this road: a MuJoCo model of the BPA quadruped was coupled to SNS-Toolbox, produced closed-loop coordinated stepping, and was transferred to hardware with potentiometer and pressure feedback on a treadmill [Lutz, 2025]; the present pipeline extends that approach from the quadruped to a leg built on human anatomy. Comparable neuromechanical simulation frameworks, such as FARMS, are emerging in parallel [Arreguit et al., 2023].

8.4 A Two-Layer CPG with Sensory Feedback

The prior AnimatLab walker (Section 7.1) demonstrated both the promise and the limitation of this architecture, and its outcome belongs on record here because it defines the requirements for the work that follows. With the rhythm-generator, pattern-formation, and sensorimotor layers oscillating in phase, the suspended model produced alternating stepping-like motion at approximately 4 Hz, faster than the roughly 1 Hz of typical human gait, but adjustable through the sodium-channel activation parameters and, once ground contact exists, subject to further slowing from ground-reaction forces and load-dependent feedback. Preliminary results from that model were presented at the 2021 NIH BRAIN Initiative Investigators Meeting [Bolen and Hunt, 2021].

When the earlier demonstrated configuration was lowered onto a simulated ground plane, the same patterns could not support the body or generate forward motion: afferent pathways had been wired but not tuned, and functional ground-contact gating of stance

and swing had not been implemented. The current instrumented phase-1 model records left and right foot and toe contact channels, but their use for closed-loop phase control has not yet been validated. In the suspended walker, the CPG acts as a feedforward controller: rhythm generation alone, without functional ground-contact feedback, produced gait-like stepping, exactly as the deafferented-CPG physiology of Section 2.5 predicts. What feedforward stepping cannot do is support the body against the ground, so the critical path from stepping-like motion to weight-bearing walking is the proprioceptive and contact feedback channels described below, and the verification tests that tune them. That premise is well supported: biomechanics and sensory feedback have been shown to regularize the behavior of different locomotor CPG architectures [Deng et al., 2022], so the present work contributes an application of this principle to the suspended walker. The AnimatLab walker model, a direct descendant of that earlier work, continues to serve as the prototyping environment for the network while the MuJoCo pipeline is brought up.

8.4.1 Architecture

The neural core follows the two-level organization of the mammalian locomotor CPG established by the Rybak group [Rybak et al., 2006, McCrea and Rybak, 2008, Rybak et al., 2015]: each leg has a rhythm generator (RG) circuit, which produces the basic locomotor rhythm and controls its period, and a pattern formation (PF) circuit, which shapes the rhythm into phase-appropriate activation of the flexor and extensor motoneuron populations. A left–right rhythm generator circuit coordinates the two legs through commissural inhibition and excitation, following models of left–right coordination [Molkov

et al., 2015, Danner et al., 2016]. The per-leg PF output drives the motoneuron pools, here the BPA activation commands for the hip, knee, and ankle muscle groups of each leg.

8.4.2 Sensory Feedback Channels

Three classes of proprioceptive feedback, together with discrete contact feedback, will close the loop between the MuJoCo model and the CPG, mirroring the biological afferents organized in the Sensory Afferent Database [Bolen et al., 2023]:

- **Ia afferents** (muscle spindle primary endings) signal muscle length and velocity, providing stretch reflex pathways and phase-dependent modulation of the PF network.
- **Ib afferents** (Golgi tendon organs) signal muscle force, providing load-dependent regulation of stance-phase extensor activity.
- **II afferents** (muscle spindle secondary endings) signal length, contributing to position sensing and interjoint coordination.
- **Contact receptors** on the heel and toe provide discrete stance and swing boundary signals, driving phase transitions and weight-support switching.

8.4.3 Verification Tests

Before attempting locomotion, the network will be verified with simple simulation tests in which a tonic stimulus is applied to the muscle motoneurons, to the pattern-formation layer, and to the rhythm-generator layer, and the resulting outputs are compared with the

corresponding experimental literature. The first of these tests is complete at the rhythm-generator level: under tonic drive, the isolated SNS-Toolbox rhythm generators of the assembled spinal network oscillate with a period of approximately 0.9 s with the two legs in antiphase (Section 7.2); in the AnimatLab instance of the network, conductance tuning toward robust rhythm-generator oscillation is still under way. The closest biological analogue is the tonic-stimulation air-stepping protocol of Selionov et al. [2009], in which tonic central and sensory stimuli facilitate involuntary stepping in humans; analogous deletion and stimulation phenomena are reproduced by the two-level CPG model itself [Rybak et al., 2006] and by neuromechanical implementations [Markin et al., 2010, Shevtsova et al., 2016]. Comparing the simulated tonic-stimulation responses with these studies will validate the sign, gain, and phase structure of the implemented circuits before any walking-speed tuning is attempted.

8.5 Extending the Sensorium: Vestibular, Ocular, and Cerebellar Contributions

Once stepping is stable, three supraspinal contributions will be added. First, vestibular feedback will be simulated with IMU-like sensing of body orientation and angular velocity, extending the AARL's balance-platform work, in which an inverted pendulum with pseudo-vestibular IMU feedback maintained balance on an oscillating platform [Bolen et al., 2024b,a,c] and analytic neural networks reproduced human balance dynamics [Hilts et al., 2019]. Second, ocular feedback will be modeled to investigate how visual stabilization cues can help reset IMU drift, a well-known practical problem for inertial sensing on robots, by treating visual measurements of the horizon and surroundings

as a low-frequency correction to the vestibular estimate. Third, a cerebellar layer will be explored as an adaptive gateway between higher-level locomotion commands and the CPG: a structure that can gate between different gaits or movement patterns (e.g., walking, running, and turning) and tune the phase relationships among them, in line with the cerebellum's established role in adaptation and coordination. Multilevel spinal-plus-cerebellar synthetic nervous systems have already been demonstrated for insect-like legs [Grummer et al., 2024], and hierarchical cortico-cerebellar-spinal models have been proposed for adaptive step-length control in bipedal locomotion [Chen et al., 2025]. The proprioceptive–vestibular interaction literature [Akay and Murray, 2021] will guide how these signals are weighted.

8.6 Physical Realization: A Treadmill Robot with Real-Time Neural Control

The final stage returns to hardware. The robot will walk on a treadmill with real-time feedback and control implemented on an embedded computing stack: Teensy microcontrollers for the low-latency per-joint loops (valve control, pressure sensing, and contact sensing) and an NVIDIA Jetson-class companion computer for the SNS network evaluation and activation computation. A serial connection to a host computer will be used only for data logging and high-level commands (e.g., run, walk, or stimulate a named neural pathway), keeping the control loop fully embedded. Walking control of a physical musculoskeletal biped with McKibben muscles has recently been demonstrated with spring-loaded inverted pendulum controllers [Li et al., 2024], supporting the feasibility of embedded control on anthropomorphic pneumatic legs. An actuation-level design lever also remains open:

the BPA's limited contraction can force long musculotendon routings, a burden visible in the extensor redesign bounds of Appendix C and encountered in the author's master's routing work [Bolen and Hunt, 2019]. A block-and-tackle "reverse pulley" arrangement, demonstrated on the AARL's self-balancing inverted-pendulum platform, addresses this by trading transmitted force for travel: paired BPAs pulled through compound pulleys on a tendon coupled to the limb, increasing range of motion at the cost of reduced transmitted force and increased reaction force on the structure [Bolen et al., 2024b,a,c]. Because joint torque is computed in the body frame, configurations that turn that reaction force to advantage may exist, making the arrangement a candidate for joints whose optimized routes demand impractically long actuators. This division of labor mirrors the SNS-Toolbox backend philosophy, which targets embedded computing platforms for real-time neural simulation [Nourse et al., 2023].

Because the static force and torque models of Chapters 3 and 4 are computationally inexpensive lookup-table candidates, they will serve as the feedforward model inside the real-time controller, while the dynamic pressure–force behavior under pulse-modulated control will draw on the state-space model of Elzein et al. [2025]. Embedded execution of the complete CPG-plus-reflex network on this stack is the intended demonstration of the dissertation project: a robot that walks with human-like biomechanics under a biologically grounded neural controller, and whose nervous system, unlike a living animal's, can be instrumented, stimulated, and lesioned at will.

8.7 Toward Dynamic Whole-Body Maneuvers

Beyond steady walking, the long-term goal of this line of research is dynamic, whole-body maneuvering: transitions between gaits, reactive balancing, and ultimately ballistic maneuvers that require coordinating the full actuator set through brief, precisely timed activation bursts. The author's aspiration for this program is a robot that can perform a kickflip, a maneuver that, like its skateboard namesake, demands explosive coordinated extension, precise timing, aerial body orientation, and controlled landing. It is exactly the kind of behavior that exercises every layer of the proposed architecture, from contact receptors and spinal pattern generation to vestibular orientation sensing and cerebellar coordination, and it would demonstrate that biomimetic actuation and neuromechanical control scale from steady locomotion to dynamic whole-body movement.

9 Conclusion

This dissertation moved the biomimetic humanoid robot project from an unvalidated design concept to a quantitatively characterized actuator set and a validated knee-level biomechanical design, and it defined the path from those experimental foundations to neuromechanical control.

At the actuator level, the analysis in Chapters 3 and 4 produced novel equations for calculating isometric force in Festo 10 mm and 20 mm-diameter BPAs. The work elucidated the relationship between maximum BPA force at 620 kPa and resting length, $F_{620}(l_0)$, a previously unreported characteristic of these actuators that the manufacturer's tool predicts with the opposite trend, and it produced a nondimensional force surface, $F^*(\epsilon^*, P^*)$, with only three diameter-dependent coefficients. Taken together, these fits predict isometric BPA force across resting lengths from a small number of measurements, correct the large over-predictions of the manufacturer's tool at short resting lengths, and outperform published single-length models. In a design workflow, these equations make it possible to select actuator diameter and resting length and to evaluate whether a proposed muscle routing can meet force and range-of-motion requirements, before committing to a prototype.

At the joint level, the improved actuator model was combined with a compliance- and wrapping-corrected musculotendon kinematics model and validated on two knee designs. The hybrid torque calculation method isolated the sources of error between prediction and measurement, and the resulting correction terms (a constant length offset, an equivalent-

stiffness representation of series compliance, and a wrapping-loss term) substantially improved torque prediction on the pinned knee and transferred to the biomimetic four-bar knee. The biomimetic knee driven by optimized BPAs was shown to meet or exceed human isometric knee torque over most of the range of motion, confirming that the optimized muscle routing of the author's earlier work is realizable in hardware and identifying the high-flexion regime where additional flexor capacity is needed.

Beyond the specific models, the dissertation's methodological contribution is the actuator-to-joint design chain itself: characterize the actuator, correct for the compliance and geometry of the transmission, validate against human biomechanics, and only then close the loop with a controller. The remaining gap, the neural layer, is scoped in Chapter 8: a MuJoCo model converted from the AARL's OpenSim models via MyoConverter (the conversion, the SNS-Toolbox coupling, and the assembled two-layer spinal network with its verified antiphase rhythm generator are already in place), wrapped in an SNS-Toolbox synthetic nervous system implementing a two-layer CPG with per-leg rhythm generation and pattern formation, left–right coordination, and Ia, Ib, II, and contact feedback, verified first by tonic-stimulation tests against the literature, then extended with vestibular, ocular, and cerebellar layers, and finally deployed on embedded hardware controlling a treadmill-walking robot. The portability of these neural primitives was verified directly: the contact-driven rhythm core reproduces with exact period agreement and trace correlations above 0.9999 between independent NumPy and Simulink implementations, the archived AnimatLab reference models reproduce their published rhythms headless, and a BPA-actuated walker in MuJoCo air-steps under synthetic neural control with peak actuator forces at 60% of the pressure ceiling, while the honest negatives of the same campaign

localize the remaining gap in the balance layer, not in the rhythm, the actuators, or the engines. The biomimetic platform that results will permit neuromechanical experiments (neural lesions, sensory deletions, and pathway stimulation) that are impossible in living animals, closing the cycle between better robots and better biological understanding, and opening the way toward dynamic whole-body maneuvers.

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APPENDICES

Appendix A: Muscle Force and Joint Torque Calculations

This appendix documents the reference implementation of the force, torque, and compliance equations used in Chapters 3 and 4. When present in MATLAB code, file names are given so that any number in this dissertation can be traced to the code that produced it.

A.1 Human muscle force calculations

A.1.1 OpenSim Thelen2003 and Millard2012 Formulations

OpenSim's Thelen2003Muscle comprises a contractile element (first-order activation dynamics with distinct rise and fall time constants; active force–length curve; Hill-type force–velocity curve) in series with a compliant tendon, plus a parallel passive fiber element, with fiber pennation. Unlike the MuJoCo actuator, the fiber length is *not* the musculotendon length: at every instant the model solves the force equilibrium $F_{\text{tendon}}(L_{\text{mt}} - L_{\text{fiber}} \cos\alpha) = F_{\text{fiber}}(L_{\text{fiber}}, \dot{L}_{\text{fiber}}, a)$, so tendon compliance filters and delays force transmission and stores energy. The tendon is a quadratic-then-linear "strip" curve anchored at the slack length L_s (force zero until L_{mt} stretches the tendon beyond L_s). Millard2012EquilibriumModel, OpenSim's newer default, retains the same equilibrium structure but replaces the curve shapes (an asymmetric Gaussian active-force–length curve with a plateau, an exponentially stiffening tendon) and adds fiber damping; it is numerically better behaved at extreme lengths. Both OpenSim models therefore differ from the

converted MuJoCo model in the same single respect: series-elastic compliance.

A.1.2 OpenSim Moment Arms and Human Torque Targets

The human torque targets of Section 4.5 are OpenSim tendon-excursion moment arms: for each muscle, the moment arm about the knee is the partial derivative of muscle-tendon length with respect to the knee coordinate, $r(\theta_k) = \partial l_{\mathrm{MT}} / \partial \theta_k$, evaluated across the range of motion. The exported analysis tables archived with the test data (`OpenSim_Vasti_MomentArm.txt`, `OpenSim_Vasti_Results.txt`, `OpenSim_Bifem_MomentArm.txt`, `OpenSim_Bifem_Results.txt`) carry the per-muscle moment arms and the corresponding maximum-isometric torque curves (moment arm multiplied by each muscle's maximum isometric force) for vastus intermedius, lateralis, and medialis, and for the biceps femoris short head; the optimization loads the vastus medialis and biceps femoris short-head columns as its targets (`buildKneeExtContext20mm.m` lines 119–140; `Knee_Flexor_Data_20mm.m` line 546). This derivative definition and the geometric perpendicular-foot construction of Equation (A4) below are equivalent for a straight muscle path crossing a planar revolute joint; the derivative form remains correct for the wrapping and moving-path-point geometry of the biological knee, which is why the human targets are computed in OpenSim rather than by the robot-side construction.

A.2 Muscle Model Formulations and Cross-Platform Force Comparison

A.2.1 MuJoCo Muscle Actuator (as converted)

The conversion from OpenSim to MuJoCo re-expresses each Thelen2003 muscle as a MuJoCo `general` actuator with the built-in muscle dynamics. Activation is a first-order process with distinct rise and fall time constants (0.01 s and 0.04 s in the converted model). Force is computed from the actuator's normalized length $\bar{L} = (L - L_{\min})/(L_{\max} - L_{\min})$, where the length range is taken from the converted geometry, as

$$F = F_{\max} \left[a\, f_L(\bar{L})\, f_V(\dot{L}) - f_P(\bar{L}) \right], \tag{A1}$$

with $a \in [0, 1]$ the activation, f_L a force–length curve that peaks inside the normalized range, f_V a force–velocity curve, and f_P a passive element. The peak force scale $F_{\max}$ (actuator parameter `gainprm[2]`) is set by the converter such that the peak force over the model's range of motion matches the OpenSim `max_isometric_force`; audited across all 84 active muscles, the deviation is $\leqslant 15\%$ (Sec. A.2.2). Two properties matter for interpretation: (i) the tendon is *rigid*, so there is no series-elastic element and no tendon slack length, and force at the joint equals the contractile-element force immediately; and (ii) all curve parameters are shared defaults of the converted model rather than per-muscle fits.

A.2.2 Cross-Platform Force Comparison

Both models were driven with identical kinematics and identical activations along the subject01 walking trajectory (the same IK frames used for the static-optimization back-solve).

The OpenSim side is the StaticOptimization solution's own muscle forces (full Thelen dynamics with compliant tendon); the MuJoCo side replays the identical joint trajectory with those activations. Table A1 reports, per muscle, the coefficient of determination R^2 between the two force time series, the shape correlation, and the best-aligning lag [Provenance: `Code/MuJoCo_SNS/spinal/_muscle_force_compare.py`; per-muscle values for all twelve analyzed muscles are archived in `muscle_force_compare.csv` beside Figure A1].

Table A1: MuJoCo vs. OpenSim muscle force along the subject01 walk at matched activations. Lag in analysis frames ($\approx$ 17 ms each).

Muscle	R^2	shape corr^2	lag [frames]
soleus	0.742	0.800	1
med. gastrocnemius	0.888	0.894	2
lat. gastrocnemius	0.884	0.887	2
tibialis anterior	0.945	0.945	0
vastus lateralis	0.941	0.945	1
rectus femoris	-1.15	0.384	0
semimembranosus	0.565	0.621	0
biceps femoris short	-0.26	0.868	0
psoas	0.927	0.934	1
gluteus maximus (2)	0.926	0.943	1
tensor fasciae latae	0.933	0.935	0
peroneus brevis	0.977	0.989	0

For eight of the twelve representative muscles the two formulations agree with $R^2 =$ 0.87–0.98 and mean-force ratios within 10%, with *no* systematic phase shift (best-aligning lag 0–2 frames, $\leqslant$ 34 ms); soleus and semimembranosus agree moderately ($R^2 = 0.74$ and 0.57, respectively). The absence of a phase lead in the rigid-tendon model indicates that, along physiological trajectories, missing tendon slack compliance does not desynchronize force transmission; its signature is instead visible in waveform *shape*: for the soleus (the

largest series-elastic energy store in human gait), the OpenSim compliant tendon carries substantial force through mid-stance while the MuJoCo rigid-tendon force follows the activation dips. This is the expected smoothing of the series-elastic element, visible in the soleus panel of Figure A1. The two remaining outliers are anatomically explained: rectus femoris was rerouted in the conversion over the vastii's patella-tracking via point, and its moment-arm polyfit deviates from OpenSim's path (the same coupler artifact noted for the vastii); biceps femoris short head agrees in shape ($r^2 = 0.87$) but carries a scale offset. We conclude that the converted model reproduces OpenSim muscle force generation along physiological trajectories to a degree sufficient for neural-control studies, and that if tendon-elasticity phenomena (energy storage, force smoothing) become material to a claim, the appropriate upgrade is a compliant-tendon formulation rather than parameter retuning of the rigid-tendon model.

A.3 BPA Force Model Implementation

A.3.1 Goodness-of-Fit Metrics

All fitted quantities in this dissertation are scored by the shared routine `Functions/Go_OfF.m`: the root-mean-square error RMSE $= \sqrt{\frac{1}{n}\sum_i (y_i - \hat{y}_i)^2}$, the adjusted fraction of variance unexplained FVU $= \frac{n-1}{n}\ \sum_i (y_i - \hat{y}_i)^2 \ /\ \sum_i (y_i - \bar{y})^2$ (so that $R^2 = 1 - \text{FVU}$), and the maximum absolute residual $\max_i |y_i - \hat{y}_i|$ (`Go_OfF.m` lines 45–52). Lower is better for all three, which makes the triplet directly usable inside optimization objectives.

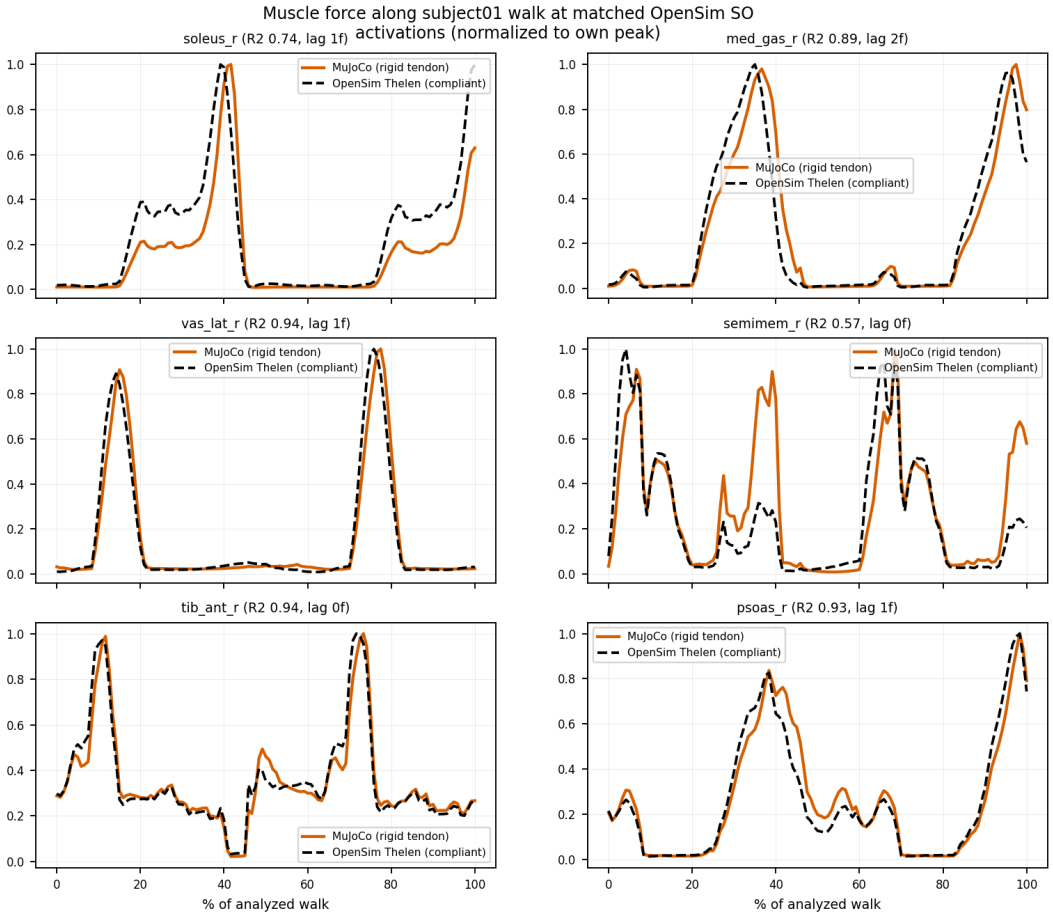


Figure A1: Normalized muscle force along the analyzed walk at matched OpenSim static-optimization activations (MuJoCo: rigid tendon; OpenSim: Thelen compliant tendon). Soleus illustrates the signature of the missing series compliance: OpenSim force is carried through mid-stance by tendon stretch while MuJoCo force follows the activation.

A.3.2 Strain Measures

The musculotendon path length l_{LMT} is the sum of straight segments between route via points, transformed into a common frame at each frame crossing. The contractile strain used by the force model subtracts the artificial tendon and both end-cap fittings from the path length before normalization by the resting length,

$$\varepsilon \;=\; \frac{l_0 - \left(l_{LMT} - l_{ten} - 2\,l_{fitting} - \chi_0\right)}{l_0}, \tag{A2}$$

where χ_0 is the constant length offset of Section 4.4 (zero in the uncorrected model). [Provenance: `MonoPamDataExplicit.m` get.Contraction, lines 210–217; `minimizeFlxPin.m` lines 415–426. End-cap fitting length $l_{fitting} = 25.4$ mm is a constructor default, `MonoPamDataExplicit.m` line 69.]

The maximum free-load contraction at 620 kPa is $\varepsilon_{620} = (l_0 - l_{620})/l_0$, where l_{620} is the BPA length at 620 kPa with no external load (code variable `KMAX`; `MonoPamDataExplicit.m` line 278). The relative strain is $\varepsilon^* = \varepsilon/\varepsilon_{620}$ (line 281).

A.3.3 Normalized Force Surface

The normalized isometric force surface of Equation (3.4) is implemented in closed form in `Functions/f_festo.m` (lines 1–26),

$$F^* = c_0\left(e^{-c_1\varepsilon^*} - 1\right) + P^*\,e^{-c_2(\varepsilon^*)^2}, \qquad F^* \leqslant 0 \;\Rightarrow\; 0, \tag{A3}$$

with $P^* = P/620\,\text{kPa}$. The zero-clamp $F^* \leqslant 0 \Rightarrow 0$ is applied by the callers (e.g. `Functions/festo4.m` line 35), not inside `f_festo.m` itself. The coefficients were fitted with `fittype` nonlinear least squares and least-absolute-residual robustness (`Testing_Data/bpaFits.m`, lines 45–60 and 115–140, lower bounds $[0, 0, 0]$) and are reported in Table 3.2. The code additionally holds a 40 mm fit not used in this dissertation ($c_0 = 0.1224$, $c_1 = 10.47$, $c_2 = 2.023$; `f_festo.m` lines 17–20).

At runtime the torque model may alternatively evaluate a spline interpolation of the manufacturer's datasheet curves (`Functions/festo4.m`, backed by `FestoLookup.mat`), with force clamped to zero for $\varepsilon^* > 1$ or $F^* < 0$. Equation (A3) replaced this lookup in the corrected model.

A.3.4 Maximum Force

The resting-length-dependent maximum force of Equation (3.5) is implemented in `Functions/maxBPAforc` (10 mm: line 46; 20 mm: line 57; duplicate for 20 mm in `Functions/Fmax20.m`) with the fixed 7.5 mm end-contact offset. The fitted 20 mm coefficients carry confidence intervals $a_1 \in [1.4745, 1.5011]\,\text{N kPa}^{-1}$ and $a_2 \in [0.0238, 0.0258]\,\text{kPa}^{-1}\,\text{m}^{-1}$ (`Fmax20.m` lines 26–27). Total force is the product $F = F^*(\varepsilon^*, P^*)\, F_{620}(l_0)$, resolved into a vector along the force direction (`MonoPamDataExplicit.m` lines 283–287).

A.4 Muscle Path Length and Moment Arm

Path length accumulates the Euclidean norm of each segment between via points, applying the body-frame transformation whenever the path crosses between the femur and tibia

frames (`MonoPamDataExplicit.m` lines 78–95, total at lines 122–133).

The vector moment arm is implemented as the perpendicular foot of the joint origin onto the line of action,

$$\mathbf{r} \;=\; \mathbf{p} \;-\; \hat{\mathbf{u}}\,(\hat{\mathbf{u}}\cdot\mathbf{p})\,, \tag{A4}$$

where $\mathbf{p}$ is the vector from the joint origin to the muscle-path crossing point and $\hat{\mathbf{u}}$ is the unit force direction (`MonoPamDataExplicit.m` lines 156–168; identical implementations in `MonoMuscleData.m` lines 111–123, the `Mom` function of `minimizeFlxPin.m` at line 539, and `MonoPam_mult.m` lines 186–201). This is the same quantity as the cross-product projection of Equation (4.2); the planar scalar moment arm about the joint axis is the magnitude of the in-plane components, computed as `hypot(mA(:,1), mA(:,2))` (`predictKneeFlexor20mm.m` line 217). Torque about the joint z-axis is $\mathbf{m} = \mathbf{r} \times \mathbf{f}$, evaluated only where the BPA is in tension (values are discarded for $\varepsilon < -0.02$, the excessive stretching region; `MonoPamDataExplicit.m` line 308).

A.5 Compliance Model Implementation

The compliance-corrected muscle length entering Equation (4.5) is computed as $l_m = l_{LMT} - \chi_0 - l_{ten} - 2\,l_{fitting}$ (`minimizeFlxPin.m` lines 415–426; identically `minimizeExtX3.m` line 345). In the two-bracket fit of record, the insertion-bracket stiffness matrix and the projected scalar stiffness along the force direction are

$$\mathbf{K}_{br} = \mathrm{diag}(\chi_1,\, \chi_2,\, \chi_1), \qquad k_{\hat{u},br} = \left(\hat{u}^{\,T}\mathbf{K}_{br}^{-1}\hat{u}\right)^{-1}, \tag{A5}$$

matching Equation (4.8), and the origin-side second bracket contributes an additional scalar compliance c_{b2} in series (Appendix C, Table C6). As executed by the evaluator of record, both bracket frames are assembled pitch-only, at `minimizeFlxPin.m` line 289 (insertion) and line 306 (origin), with the two-rotation assemblies carried as commented alternatives; the insertion array is $[X_1, X_2, X_1]$ (line 387) and the origin array $[X_1, X_1, X_2]$ (line 400). The two-rotation-capable twin of the same block is `minimizeFlxPin2brk.m` lines 387–418. An earlier arm-1 array $[X_1, X_2, X_2]$ is superseded and survives only in the provenance comment at line 387, and the legacy single-bracket evaluator uses $[X_1, X_2, X_1]$ (`minimizeFlxPinX3.m` line 267).

The artificial tendon is modeled as an axial spring with stiffness

$$k_{ten} \;=\; \frac{n\,A_c\,E}{L_{ten}}, \tag{A6}$$

with $n = 2$ parallel cable strands ($\times 6$ for the 20 mm assembly), $A_c = 1.51 \times 10^{-6}\,\mathrm{m}^2$ (19-strand cable), and $E = 193 \times 10^{9}$ Pa (`MonoPam_mult.m`, function `Spr()`, lines 627–637; the evaluators obtain the spring rate through `Spr(bpa_i)`, `minimizeFlxPin.m` line 156). Series combination with the projected bracket compliance gives $k_{eq} = \left(k_{\hat{u},br}^{-1} + k_{ten}^{-1}\right)^{-1}$ (`minimizeFlxPin.m` lines 394–409).

The scalar compliance displacement δ of Equation (4.22) is found numerically with `fzero` on the bracket $[0,\, l_m - l_{620}]$,

$$F_{620}(l_0)\,F^*(\varepsilon^*(\delta), P^*) - k_{eq}\,\delta = 0, \tag{A7}$$

with $\delta \equiv 0$ when the uncorrected relative strain already exceeds unity (`minimizeFlxPin.m`

lines 434–441). The vector bracket deflection is then $\boldsymbol{\epsilon}_{br} = \mathbf{K}_{br}^{-1}\,(\|\mathbf{f}\|\,\hat{\mathbf{u}})$ (line 461), the cable stretch is $\|\mathbf{f}\|/k_{ten}$ (line 478), and the attachment point, path, moment arm, and torque are recomputed. For the two-BPA flexor grouping, the coupled balance shares one common force between both muscles and is solved analogously on $[0, F_{620}]$ (`MonoPam_mult.m` lines 556–612).

A.6 Wrapping-Loss Implementation

For the pinned-knee extensor, Equation (4.23) is implemented segment-wise over the wrap arc with radii $R_1 = 12\,\text{mm}$ and $R_2 = 40\,\text{mm}$ and segment boundaries $\theta_1 = 27^\circ$ and $\theta_2 = -23^\circ$ (`minimizeExtX3.m`, nested `Contraction`, lines 327–370). For the biomimetic-knee 20 mm cell, the arc-length factor is the routed bend measure $b(\theta_k)$ (arc length swept by the wrap, built with the `wRap` bend radius in `buildKneeFlexorRoute20mm.m` lines 265–268), giving $\Delta l = \chi_3\, b(\theta_k)\,(1-\varepsilon^*)^2$ (`MonoPam_mult.m` lines 398–441), with a hardcoded-geometry fallback for degenerate routes (lines 428–434). The loss enters only the force-producing strain; the kinematic (prediction) strain excludes it (`minimizeExtX3.m` lines 236–240).

Appendix B: Neuron Model Equations

This appendix documents the neuron and synapse models used by the synthetic nervous systems in this project, both in the AnimatLab walker model and in the SNS-Toolbox implementation of the MuJoCo pipeline (Chapter 7). The authoritative model state is the working AnimatLab project file (`Biped_2xCPG_wSubs/Biped_2xCPG_wSubs.aproj`); after the network reorganization described in Section 7.2, a fresh standalone export (`Biped_2xCPG_wSubs_Standalone.asim`) runs headless and carries the rhythm-generator charts, and the earlier phase-1 instrumented export (`walk new new tester added 2 axis_phase1.asim`) records the run reported in Section 7.3.

B.1 Spiking Neuron Model

Each spiking neuron is a conductance-based leaky integrator with threshold and adaptation,

$$C_m \dot{V} \;=\; -\,g_{\text{leak}}\,(V - E_{\text{rest}}) \;+\; I_{\text{syn}} \;+\; I_{\text{app}}, \tag{B1}$$

where V is the membrane potential, C_m the membrane capacitance ($g_{\text{leak}} = C_m/\tau_m$ with membrane time constant τ_m), E_{rest} the resting potential, I_{syn} the summed synaptic current, and I_{app} any applied (tonic stimulation) current. A spike is emitted when V exceeds the firing threshold, after which V resets and an after-hyperpolarization (AHP) of amplitude A_{AHP} with time constant τ_{AHP} is applied; the threshold itself accommodates with the firing history (relative accommodation r_{acc}, time constant τ_{acc}). Optional calcium channels are

defined but disabled in the current model ($G_{\text{max,Ca}} = 0$).

The parameter values shared by the neurons in the walker model are given in Table B1.

Table B1: Spiking neuron parameters in the AnimatLab bipedal walker model (`Biped_2xCPG_wSubs`). All neurons share these values except the motor neuron threshold noted below.

Parameter	**Symbol**	**Value**
Resting potential	E_{rest}	$-60\,\text{mV}$
Membrane time constant	τ_m	$5\,\text{ms}$
Initial firing threshold	V_{th}	$-55\,\text{mV}$ (motor neurons: $-59\,\text{mV}$)
Relative accommodation	r_{acc}	0.3
Accommodation time const	τ_{acc}	$10\,\text{ms}$
AHP amplitude	A_{AHP}	$1\,\text{mV}$
AHP time constant	τ_{AHP}	$3\,\text{ms}$

B.2 Non-Spiking (Graded) Chemical Synapses

Pattern-formation and motoneuron-drive connections in the walker use graded chemical synapses, whose conductance is a clipped linear ramp of the presynaptic potential,

$$g_{\text{syn}}(V_{\text{pre}}) \;=\; g_{\max}\;\text{clamp}\!\left(\frac{V_{\text{pre}} - V_{\text{th}}}{V_{\text{sat}} - V_{\text{th}}},\,0,\,1\right), \qquad I_{\text{syn}} = g_{\text{syn}}\left(V_{\text{post}} - E_{\text{syn}}\right), \tag{B2}$$

where V_{th} and V_{sat} are the synaptic threshold and saturation potentials and E_{syn} the reversal potential. The walker model instantiates three non-spiking synapse types:

- **Nicotinic ACh (excitatory drive):** $E_{\text{syn}} = -10\,\text{mV}$, $g_{\max} = 0.5\,\mu\text{S}$, $V_{\text{th}} \to V_{\text{sat}} = -65\,\text{mV} \to -20\,\text{mV}$.

- **Hyperpolarizing IPSP (inhibition):** $E_{\text{syn}} = -70\,\text{mV}$, $V_{\text{th}} \rightarrow V_{\text{sat}} = -55\,\text{mV} \rightarrow -40\,\text{mV}$.
- **Depolarizing IPSP:** $E_{\text{syn}} = -55\,\text{mV}$, $V_{\text{th}} \rightarrow V_{\text{sat}} = -60\,\text{mV} \rightarrow -40\,\text{mV}$.

Spiking chemical synapses and electrical synapses (rectifying coupling weight 0.1, non-rectifying 0.2) are also defined in the model and are available for the sensory afferent pathways of the MuJoCo/SNS implementation.

B.3 Relation to the SNS-Toolbox Implementation

These are the standard AnimatLab firing-rate/conductance models. SNS-Toolbox [Nourse et al., 2023] implements corresponding classes of non-spiking and spiking neurons with conductance-based synapses, and the AnimatLab parameters above served as the physically motivated initialization for the first SNS-Toolbox instantiation of the spinal network: a 406-neuron network with per-leg rhythm-generator half-centers, phase-shifted pattern-formation groups, motoneuron pools for all 92 muscle-tendon actuators of the converted model, and Ia, Ib, and II afferent populations. In that implementation the non-spiking membrane and graded-synapse equations take the same forms as Equations (B1) and (B2) with the spiking threshold and accommodation terms removed, and the afferent resting-tone offsets are zeroed so that Ia and Ib signals are pure activity measures; the rhythm-generator layer has been verified to oscillate at a human-like period with the two legs in antiphase under tonic drive (Section 8.4). An explicit parameter-mapping and validation campaign between the two implementations remains future work; the tonic stimulation tests of Section 8.4 correspond to constant I_{app} injections into Equation (B1) at the motoneuron,

pattern-formation, and rhythm-generator layers respectively.

B.4 Spiking-Mirror Implementation, Calibration, and Verification

This appendix documents the numeric details behind the spiking-neuron mirrors of Section 7.2.5, in the same file-traceable form as Appendix A: the cell and synapse equations as implemented, the calibration factors of record, the full verification-gate table of the MuJoCo mirror, and the variant-by-variant AnimatLab conversion table. The campaign's artifacts live under `Code/MuJoCo_SNS/spinal/reports_spiking_20261002/` (reports, gate scripts, logs, and metric files) and `Code/MuJoCo_SNS/spinal/` (the mirror builder `build_network_spiking.py` and its calibration script `calibrate_spiking.py`); the Simulink blocks are built by `Code/Matlab/SNS_Sims` and tested by `sns_units_test_spiking.m` in the same folder.

B.4.1 Adapting Integrate-and-Fire Cells and Hybrid Synapses

Between spikes, every spiking cell of the MuJoCo mirror obeys the membrane equation

$$C_m \dot{V} \;=\; -\,g_m\,(V - V_{\text{rest}}) \;+\; I_{\text{syn}} \;+\; I_{\text{bias}}, \tag{B3}$$

with a spike emitted when V reaches the firing threshold θ, after which V resets to V_{reset} and θ increases by the increment $\Delta\theta$; between spikes θ decays back toward its initial value with time constant τ_θ, bounded by the adaptation floor. The tonic bias I_{bias} parks the equilibrium a few millivolts below threshold, reproducing the analog cells' near-threshold operating point. Table B2 lists the per-class values as coded (SNS-Toolbox's `SpikingNeuron`

class; the toolbox's membrane update is dimensionally consistent only at $g_m = 1\,\mu\text{S}$, which the mirror therefore uses throughout). The rhythm-generator adaptation time constant reuses the analog network's persistent-sodium inactivation value, so that tuned adaptation values transfer between builds; at the rhythm gate's operating point it is 0.35 s.

Table B2: Spiking cell parameters of the MuJoCo mirror (real millivolts; `build_network_spiking.py`, `SPIKE` table, with the rhythm-generator increment superseded by the calibration of record). All cells rest at $-70\,\text{mV}$ with membrane conductance 1 µS and the tonic bias of Table B3.

Cell class	θ **(mV)**	V_{reset} **(mV)**	$\Delta\theta$ **(mV/spike)**	τ_θ **(s)**
Rhythm generator E/F	-50	-60	4.0 (calibrated)	0.35 (analog value)
Pattern-formation	-50	-60	0.3	0.15
Interneurons, commissurals	-50	-60	—	—
Afferent encoders	-52	-58	—	—
Renshaw	-52.5	-58	—	—

Spiking chemical synapses, in both the toolbox class used in the MuJoCo mirror and the Simulink `SpikingSynapse` block, carry a conductance that decays between events and jumps by a fixed increment per presynaptic spike, saturating at its maximum,

$$\dot{g} \;=\; -\,g/\tau_{\text{syn}} \;\text{ between spikes,} \qquad g \leftarrow \min(g_{\max},\, g + g_{\text{inc}}) \;\text{ per spike,} \tag{B4}$$

with $I_{\text{syn}} = g\,(E_{\text{syn}} - V_{\text{post}})$ as for the graded synapse of Equation (B2). The Simulink `HybridSpikingSynapse` implements the same dynamics with the spike detected from the presynaptic voltage crossing a detection threshold that must sit strictly between the presynaptic resting and firing potentials (a detection threshold equal to the firing potential fires nothing, because the presynaptic voltage resets at threshold). Reversal potentials are

0 mV (excitation, above rest) and -70 mV (inhibition, shunting at rest) in the spiking frame, and $+8/-5$ mV in the motor neurons' 0–5 mV frame. The graded-to-spiking command pathways (descending drive, posture, and bias surrogates) keep the graded synapse of Equation (B2).

B.4.2 Calibration of Record

Table B3 lists every calibration factor produced by `calibrate_spiking.py` (`spiking_calibration.json`), with the measured residual of each. The spike-to-motor-neuron factors are bisected so that a steady 40 Hz presynaptic train reproduces the non-spiking steady-state motor-neuron voltage; the spike-to-spiking factors are rate-calibrated (a 40 Hz source drives a relay cell at 40 Hz, and a calibrated inhibitory factor silences a 40 Hz-driven cell); the readout-tap factor maps a 50 Hz train to the full-scale analog level. The rhythm-generator threshold increment was set by an isolated-pair sweep followed by a full-network sweep over alternation quality and period target.

B.4.3 Verification Gates of the MuJoCo Mirror

Table B4 records the complete gate set of Section 7.2.5 with verdicts and headline numbers. The two failures are reported as failures: the constant-drive period misses its tolerance with the tuning knobs identified, and the air-regime cadence clause misses while every structural clause passes. Gate scripts, logs, and machine-readable outputs are archived in the campaign folder (`tools/`, `logs/`, and the `json` metric files respectively).

Table B3: Calibration of record for the spiking mirror (`spiking_calibration.json`). Residuals are the measured deviation at each factor's calibration target.

Factor	Value	Calibrated against	Residual
Graded saturation reference	0.9567	analog network measurement	—
Spike-to-MN, excitatory ($k_{sn,exc}$)	5.4087	MN steady voltage	1.0×10^{-4} n
Spike-to-MN, inhibitory ($k_{sn,inh}$)	1.2284	MN steady voltage	2.6×10^{-4} n
Graded-to-spiking drive (k_{ns})	0.35	relay firing rate	-0.67 Hz
Spike-to-spiking, excitation	1.8754	40 Hz rate transfer	0.0 Hz
Spike-to-spiking, inhibition	0.9996	silences 40 Hz cell	0.0 Hz
Readout tap (g_{inc})	1.7093	50 Hz → full scale	5.00 mV che
RG threshold increment	4.0 mV/spike	pair + full-net sweeps	period targ
Tonic bias current	16 nA	equilibrium -54 mV	—
Synaptic time constants	5 ms (exc), 20 ms (inh)	analog values	—

B.4.4 AnimatLab Conversion Variants

Table B5 catalogues every spiking copy built from the AnimatLab standalone exports. The conversion rules, applied by `tools/make_spiking.py` to copies (no original file touched), are: thresholds of neurons with unreachable (positive) thresholds move into the spiking regime, either -55 mV with -59 mV for motor-neuron-named cells in the v1 copies, or per-class calibrated values from the measured graded peaks in the v2 copies (rhythm-generator/pattern-formation/interneuron -53.5 mV, motor neurons -44 mV, Renshaw -44.5 mV); relative accommodation is zeroed; every graded chemical synapse type is re-emitted as a per-spike type with the same identifier and amplitude, 10 ms decay, and maximum conductance $\max(5,\ 2\times\text{amplitude})$; and each connection's own class tag is flipped to the spiking enum. That last rule is a documented format trap: the per-connection `<Type>` tag is stored independently of the synapse type, and converting only the type kills

Table B4: Complete verification-gate table of the spiking-mirror campaign. Baseline scores reproduce digit for digit; "split" denotes a gate whose clauses pass and fail in part.

Gate	Verdict	Headline numbers
Pipeline compile	PASS	runner, curriculum, parameters, and network builder all compile.
Defaults gate	PASS	(410, 376, 1186) populations, inputs, and connections after the selector edit; variant build (888, 382, 7430).
Non-spiking bit-identity	PASS	default baseline score -161.56754173676563 identical before and after the edit (all 17 digits); other baselines -144.46440311078322 and -200.37955834081458.
Topology mirror	PASS	directed edge multisets identical in three gain configurations (1,186 / 7,908 / 7,762); inputs and motor-neuron maps identical; readout taps 12 / 12 / 16.
Constant-drive rhythm	SPLIT	sustained bilateral antiphase (E 4.9 Hz, F 3.2 Hz per leg; swing 5.13 mV in every window) = PASS; period 0.621 s versus analog 0.399 s ($+55.6\,\% > 20\,\%$) = FAIL.
Perturbation basin	PASS	12/12 trials under 1 % jitter on every scalar; final swing minimum 4.58 mV (margin $0.89\times$); zero explosions.
Air-regime smoke	FAIL (cadence)	finite, 53 envelope rises, knee range 106.0° of twin's 117.9°, hip 50.4° of 71.1°; period 0.215 s versus 2.108 s.
Untuned ground evaluation	ran	score -236.35; bilateral (8 right / 9 left cycles); cycle knee minimum -79.8°; no non-finite states; stayed up. Single run, no statistics.
Simulink regression	PASS	all committed demonstrations and units tests reproduce exactly with the spiking blocks in the library (two-neuron test 6.07×10^{-4} mV).
AnimatLab baselines	PASS	five headless baseline runs, exit 0, periods and antiphase reproducing the records (0.443 s, 0.462 s, 1.305 s).

the run at load with no diagnostic detail. Structural validation confirms for every copy that neuron, connection, synapse-type, chart, and stimulus sets are identical to the original and that every class tag matches its synapse type.

Figure B1 shows the ablation result at the joint level for the two-layer family: with graded synapses kept, spiking-regime thresholds leave the joint oscillation essentially intact, while the full conversion pins the joints. This is the visual form of Table 7.6's conclusion that the synapse conversion is the breaking change.

The spiking copies are headless-run artifacts only: none has been opened in the AnimatLab graphical environment, and graphical verification — or an equivalent acceptance ruling — remains outstanding for every `.asim` listed in Table B5. The mirror work also scoped, but did not attempt, two further conversions: the full 410-population Simulink spinal network (whose every one of 1,392 gains is tuned for graded voltages, making a spiking conversion a full re-tune rather than a parameter port) and the spiking mirrors of the alternative MuJoCo network variants (separate architectures whose fresh tuning is future work).

B.5 Connectomes of the Spinal-Network Models

The connectome of the production gait2392 spinal network (the 410-neuron SNS-Toolbox instantiation of Section 8.4) is documented in Figures B2 and B3. Both schematics are generated by `Code/MuJoCo_SNS/spinal/draw_circuit.py` from the compiled network at the fitted walking configuration (`fitted_walk_params.json`, the IK/NNLS back-solve refit; 410 neurons, 201 per leg plus 8 shared, and 376 inputs; the eight shared cells are the

Table B5: AnimatLab spiking-conversion variants of the spiking-mirror campaign. "Connexions" counts flipped per-connection class tags; the ablation converts thresholds only. All copies validate structurally and run headless to their configured end; rhythm outcomes as in Table 7.6.

Variant	Thresholds	Neurons	Synapse types	Connexions	Rhythm outcome
Two-layer v1	−55/−59 mV	80	17	150	dead (2 spikes / 5 s)
Two-layer v2	calibrated	80	17	150	dead
Two-layer v3	v1, 30 ms decay	80	17	150	dead
Bilateral v1	−55/−59 mV	96	20	208	latch, one side 442 Hz–450 Hz
Bilateral v2	calibrated	96	20	208	latch
Project export v1	−55/−59 mV	122	50	243	synchronous 11.8 Hz co-burstir
Project export v2	calibrated	122	50	243	dead (passive settle)
Ablation	spiking, graded synapses	80	0	0	oscillates, near-baseline ranges
2023 reference (baseline)	spiking from birth	56	3 (spiking)	84	1.305 s bursts, antiphase +0.49

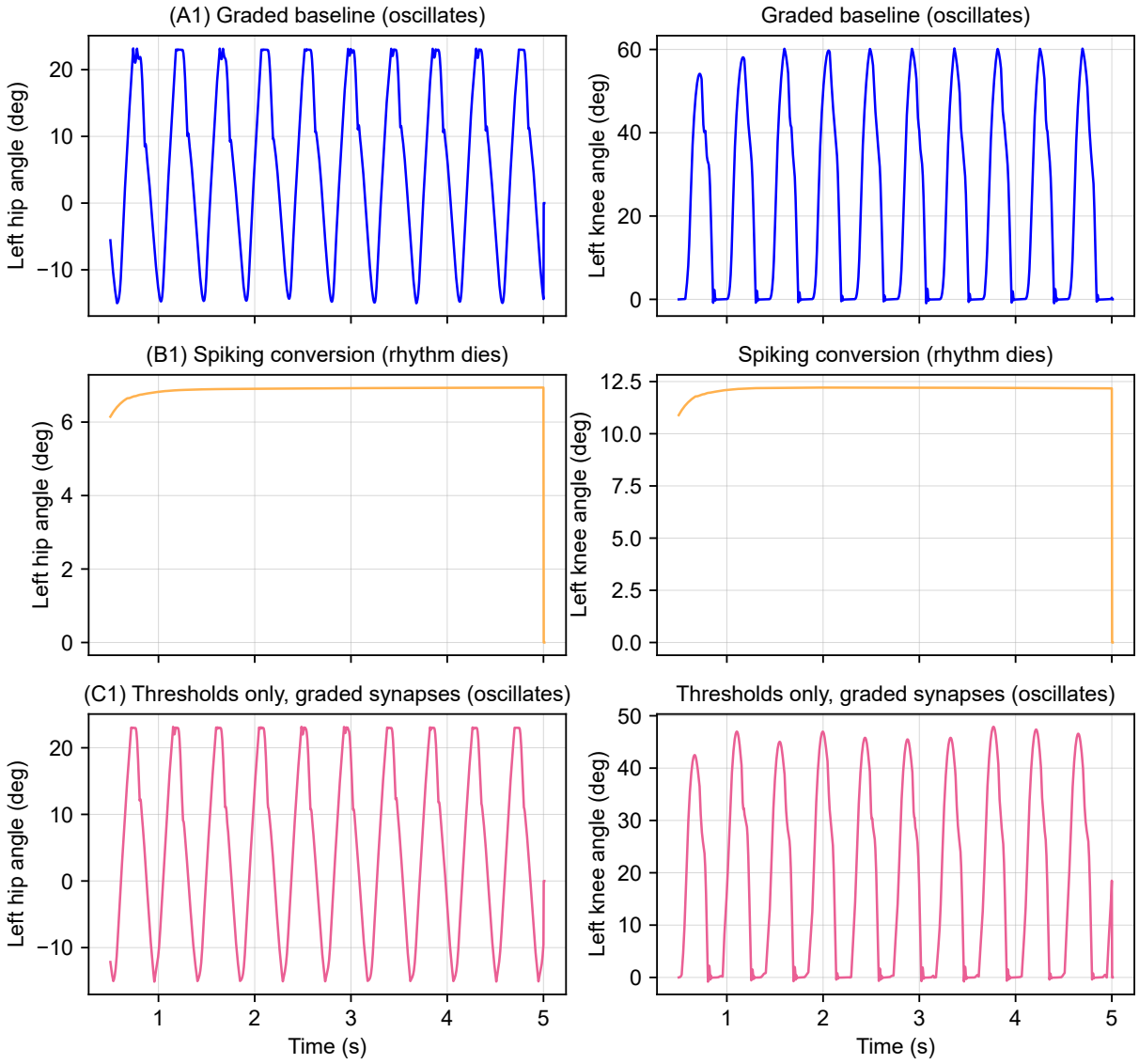


Figure B1: Left-leg joint angles of the AnimatLab two-layer walker family over the 0.5 s–5 s chart window: **(A, B)** graded baseline; **(C, D)** full spiking conversion (rhythm dies, joints pin); **(E, F)** thresholds-only ablation with graded synapses kept (oscillation preserved). Left column: hip angle. Right column: knee angle.

single DRIVE and POSTURE descending neurons and the six balance cells BAL_PF, BAL_DF, BAL_TRK_EXT, BAL_TRK_FLX, BAL_LAT_R, and BAL_LAT_L), with every motoneuron-pool membership pulled live from muscle_map.py rather than hardcoded, so the drawings cannot drift from the built network. The caption of Figure B2 and its alt-text sidecar repeat these population counts, so prose, caption, and alt text cannot drift apart either. Synapse markers follow the project convention throughout: open circles are neurons, white triangles mark excitatory synapses, and solid black dots mark inhibitory synapses.

Figure B2 draws both legs in full. A vertical dashed midline separates the sides and a pale central corridor gathers the interleg commissural inhibition. Each side is organized in labeled layer bands: descending DRIVE (MLR surrogate), POSTURE, and balance inputs; the rhythm-generator half-centers with their mutual inhibition and adaptation cells; the phase-window pattern-formation cells; the Ia, II, and Ib afferent populations with the stance-gated Ib load-sharing interneurons; and the ten motoneuron pools per side (HIP-E 7, HIP-F 4, KNEE-E 4, KNEE-F 4, ANK-PF 9, ANK-DF 3, HIP-AB 6, HIP-AD 6, TRK-E 1, and TRK-F 2, that is 46 motoneurons per leg) driving their muscles. Figure B3 shows the same network for one leg as a DRIVE/POSTURE to rhythm-generator to pattern-formation to motoneuron hierarchy, with the solved standing bias injected into every pool, the sensory column of Ia, II, and Ib afferents, and a boxed key listing every pool abbreviation with its member muscles. The spiking mirror of Section B.4 replicates this topology exactly (Table B4, topology-mirror gate), so these two figures also serve as the connectivity reference for that model.

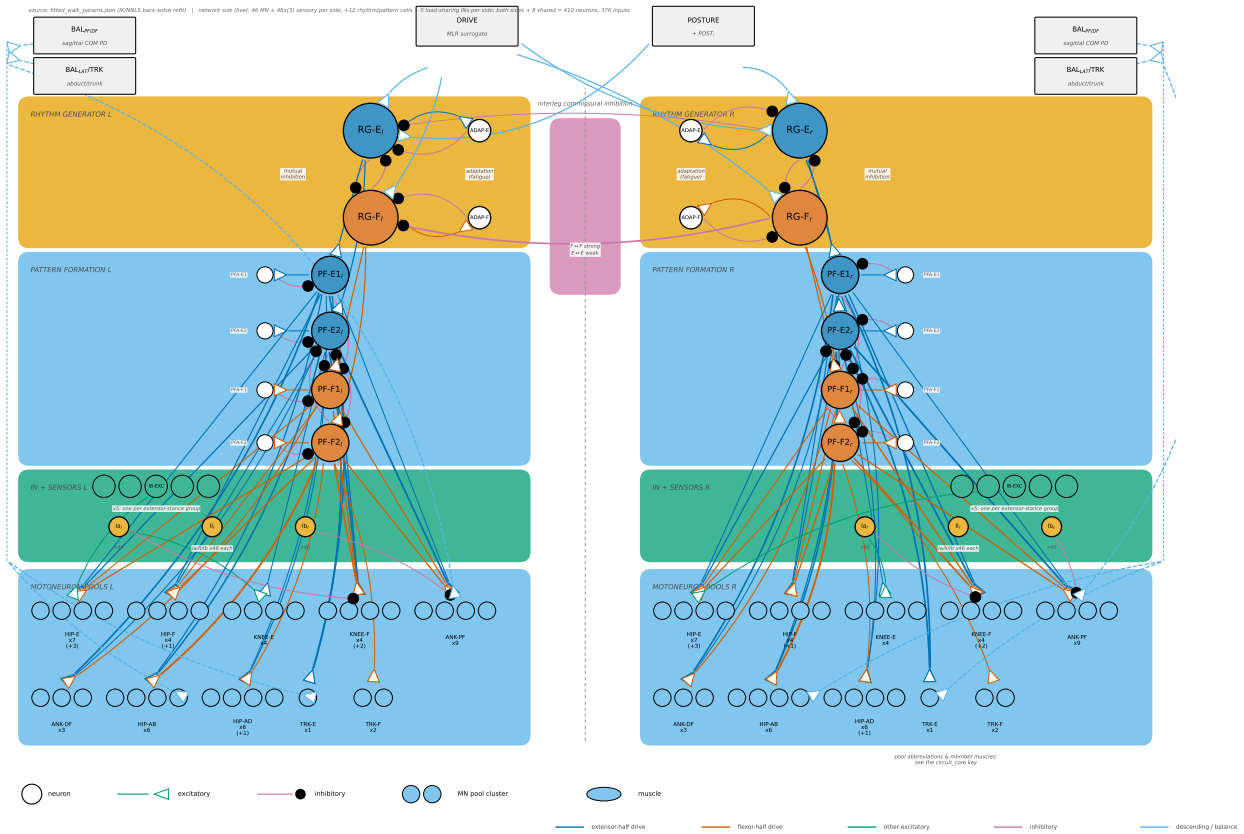


Figure B2: Bilateral connectome of the gait2392 spinal network rendered from the compiled network at the fitted walking configuration (410 neurons, 201 per side plus 8 shared; 376 inputs). Each side is organized in labeled layer bands: descending DRIVE (MLR surrogate), POSTURE, and balance inputs; rhythm-generator half-centers with mutual inhibition and adaptation cells; phase-window pattern-formation cells; Ia, II, and Ib afferents with the stance-gated Ib load-sharing interneurons; and the ten motoneuron pools per side driving their muscles. The central corridor gathers the interleg commissural inhibition across the dashed midline. Open circles are neurons, white triangles mark excitatory synapses, and solid black dots mark inhibitory synapses; the gray note names the source files.

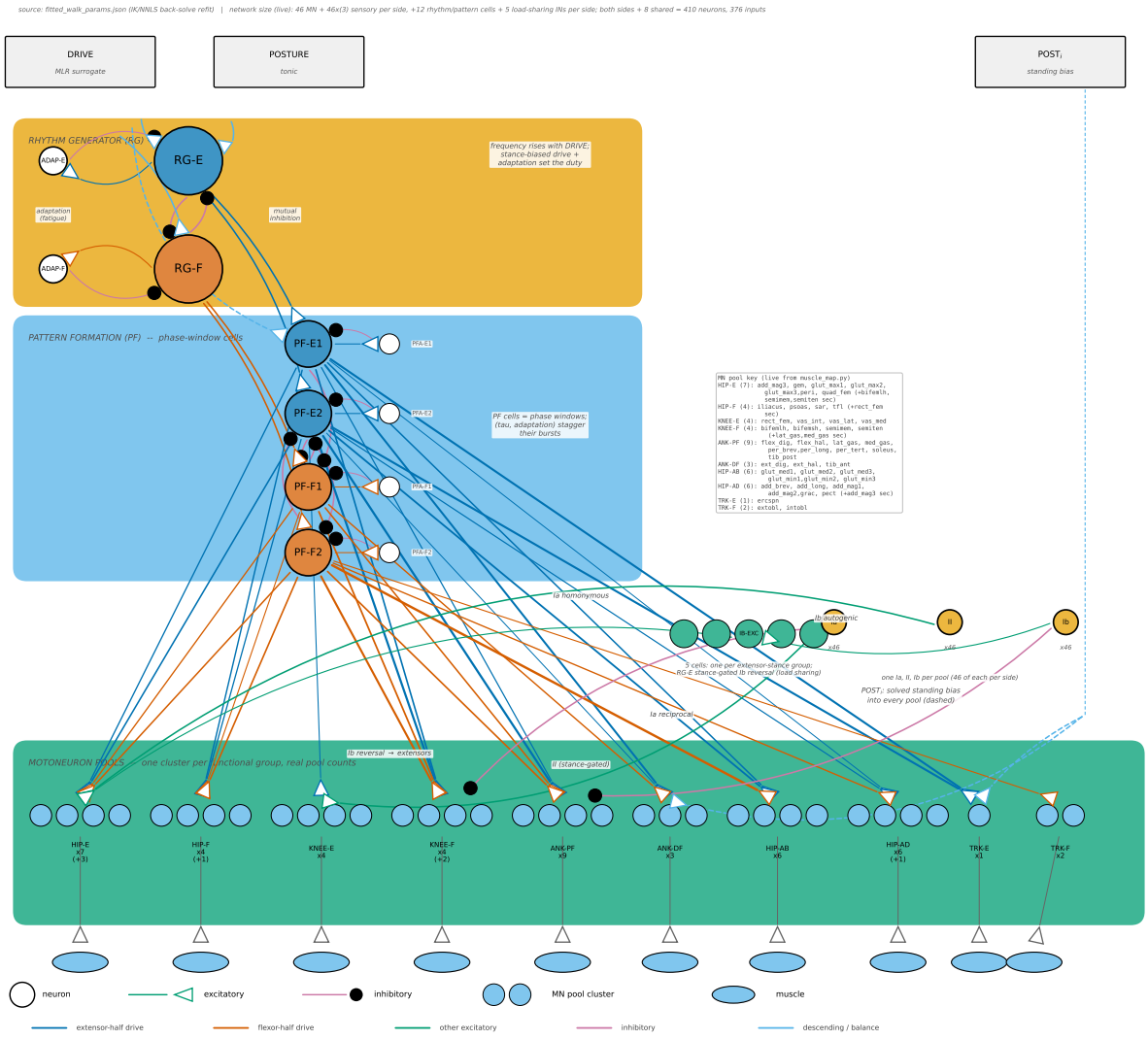


Figure B3: One-side hierarchy of the same gait2392 spinal network: descending DRIVE (MLR surrogate) and POSTURE inputs, the solved standing bias injected as dashed lines into every motoneuron pool, the persistent-sodium rhythm-generator half-centers, the phase-window pattern-formation cells, the Ia, II, and Ib afferent column, and the motoneuron pools (46 per leg) projecting to their muscles. A boxed key lists every pool abbreviation with its member muscles.

Appendix C: Model Geometry, Bracket Frames, and Compliance Parameters

This appendix documents the numeric geometry and compliance parameters behind the force and torque models of Chapters 3 and 4 as they exist in the project's MATLAB code, with file names given so that any value can be traced to the code that produced it. Route and bracket coordinates are coded in meters and reported here in millimeters; stiffnesses are coded and reported in newtons per meter. The bracket reference points are interpretation choices for the 3D-printed (Onyx FDM) attachment hardware, set from CAD by the author; moving a reference point changes the identified stiffness values, so the points are recorded here alongside the values they produced.

C.1 Static BPA Pathway Data

The static BPA model routes each actuator as a straight-segment polyline through via points stored in a per-test `Location` array of dimensions $M \times 3 \times N$, where N is the number of joint frames in the motion sweep and M the number of via points; the first row in the distal link frame is marked by a cross-point index. There are no binormal or surface-frame quantities: at each frame the force direction is the normalized vector from the cross-point row to the previous row transformed into the crossing frame,

$$\hat{\mathbf{u}} \propto \mathbf{R}^{-1}\,\mathbf{p}_{C-1} \;-\; \mathbf{p}_{C}, \tag{C1}$$

where $\mathbf{p}_C$ is the cross-point row, $\mathbf{p}_{C-1}$ the previous row, and $\mathbf{R}$ the frame-crossing rotation [Provenance: `minimizeFlxPin.m` lines 160 and 524–537 (`UD`) and 201–235 (`computeForceVector`); `minimizeExtX3.m` lines 687–700; BPA struct fields, `minimizeFlxPin.m` lines 28–35]. The pinned-knee test routes are archived as `FlxPinBPASet.mat` (5 flexor tests, $M = 2$, knee angle -120° to $+10^\circ$) and `ExtPinBPASet.mat` (9 extensor tests, $M = 7$, knee angle -125° to $+35^\circ$).

Table C1 lists the five pinned-knee flexor tests; their routes are two points, femur-frame origin $\mathbf{p}_1 = [-75.0,\ 100.0,\ 32.8]$ mm and tibia-frame insertion $\mathbf{p}_2 = [-50.1,\ -45.0,\ 32.3]$ mm (test 1; `FlxPinBPASet.mat`, kf(1), frame 1). Table C3 lists the extensor pinned-knee route of test 1 at $\theta_k = -125^\circ$; rows 1–6 are femur-frame and row 7 is tibia-frame, and as the knee extends, rows 2–6 release onto $\mathbf{p}_1$ (wrap released at $+35^\circ$). Table C4 lists the per-test constants of the nine extensor tests; the validation pool used for the correction terms of Section 4.4 is tests $\{1, 2, 5, 6, 7, 8\}$ [Provenance: `Dig_crossPredict.m` line 46].

Table C1: Pinned-knee flexor test BPA constants, read directly from `FlxPinBPASet.mat` (kf(1)–kf(5)). Labels are the laboratory shorthand used in the fitting logs (`minimizeFlxPin10mm_2brk.m` line 51); l_0, tendon, and pressure are the values used by the model. All tests use ∅10 mm BPAs; the 47 cm test's measured angles carry the $+5.3^\circ$ encoder correction at load time (`minimizeFlxPin.m` lines 77–80), and it was held out of the two-bracket identification as validation.

Test	**Label**	$\boldsymbol{l_0}$ (m)	Tendon (m)	P (kPa)	F_m (N)	**Use**
kf(1)	48 cm	0.485	0	605.7	443.4	training
kf(2)	46 cm	0.457	0	614.2	441.4	training
kf(3)	47 cm	0.479	0	623.2	443.0	held-out validation
kf(4)	40 cm-tendon	0.406	0.035	623.4	436.9	training
kf(5)	41 cm	0.410	0	615.4	437.3	training

The biomimetic flexor route (`buildKneeFlexorContext20mm.m`) is a three-point path

Table C2: Biomimetic knee, extensor 20 mm BPA route seed (femur frame: p1–p5; tibia frame: p6–p9; `buildKneeExtContext20mm.m` lines 431–440, the author's fcec-ray construction; the prior SolidWorks seed is kept commented at lines 421–430). Release schedule of the verified elimination, read from the redesign of record (`Vas_Pam_20mm_Result_20260925.mat`, `routeInfo.eliminatedAngleD`): p5 at -89.8°, p4 at -55.7°, p6 at -47.8°, p7 at -20.2°, p3 at $+6.1^\circ$, p8 at $+7.4^\circ$.

Point	x (mm)	y (mm)	z (mm)	**Frame**	**Role**
p1	40.00	35.00	0	femur	BPA origin (attachment)
p2	83.90	−274.76	0	femur	BPA contacts mounting base
p3	64.06	−426.80	0	femur	femoral contact
p4	52.83	−441.88	0	femur	femoral condyle contact
p5	37.77	−452.14	0	femur	femoral condyle contact
p6	69.85	34.33	0	tibia	tibial contact (upper)
p7	73.50	19.13	0	tibia	tibial contact (mid)
p8	71.98	−12.23	0	tibia	tibial contact (lower)
p9	34.06	−66.87	0	tibia	patellar ligament ring (distal, pEnd)

Table C3: Pinned-knee extensor BPA route of test 1 at $\theta_k = -125^\circ$ (`ExtPinBPASet.mat`, ke(1), frame 1; mm).

Point	x (mm)	y (mm)	z (mm)
p1	30.00	−50.00	0
p2	65.00	−350.00	0
p3	42.40	−408.50	0
p4	28.20	−421.60	0
p5	10.50	−425.20	0
p6	−8.60	−422.80	0
p7	42.50	−75.90	0

Table C4: Pinned-knee extensor test BPA constants, read directly from `ExtPinBPASet.mat` (ke(1)–ke(9)). Labels are the laboratory shorthand used in the fitting logs (`Dig_resultsTable.m` line 32); l_0, tendon, and pressure are the values used by the model; note that the shorthand rounds several lengths (e.g. "42 cm" is $l_0 = 0.415$ m and "43 cm-tendon" is $l_0 = 0.432$ m). All tests use ∅10 mm BPAs. The correction terms of Section 4.4 were identified on the validation pool $\{1, 2, 5, 6, 7, 8\}$; tests 3, 4, and 9 were excluded.

Test	**Label**	$\boldsymbol{l_0}$ (m)	Tendon (m)	P (kPa)	F_m (N)	**Use**
ke(1)	40 cm	0.400	0	624.9	436.3	pool
ke(2)	40 cm-tendon	0.400	0.040	618.4	436.3	pool
ke(3)	42 cm	0.415	0	605.2	437.8	excluded
ke(4)	42 cm-tendon	0.415	0.040	605.2	437.8	excluded
ke(5)	43 cm	0.436	0	621.8	439.7	pool
ke(6)	43 cm-tendon	0.432	0.022	622.9	439.3	pool
ke(7)	46 cm	0.457	0	605.9	441.4	pool
ke(8)	47 cm	0.465	0	605.9	442.0	pool
ke(9)	48 cm	0.480	0	605.9	443.1	excluded

(femur-frame origin $\mathbf{p}_1$, a moving wrap point $\mathbf{p}_{\text{wrap}}$ on the tibia, and tibia-frame insertion $\mathbf{p}_{\text{end}}$), with the wrap point held on a standoff surface of radius `bpaRb` about the joint and its z placed on the projected $\mathbf{p}_1$–$\mathbf{p}_{\text{end}}$ line; the optimizer seed is $\mathbf{p}_{1,0} = [-50.0,\ 35.0,\ 50.0]$ mm and $\mathbf{p}_{2,0} = [-12.24,\ -8.87,\ 27.87]$ mm with $l_0 = 415$ mm and 25 mm tendon [Provenance: `buildKneeFlexorContext20mm.m` lines 209–218; `buildKneeFlexorRoute20mm.m` lines 98–186]. The biomimetic extensor seed route is given in Table C2, including the release schedule of the route-elimination rule.

C.2 Joint Transformation Matrices

The knee transform is planar: every joint frame is a rotation about z by the knee angle θ_k composed with a translation fitted to robot CAD tables of the migrating instantaneous center of rotation. With $\mathbf{R} = \mathbf{R}_z(\theta_k)$, the robot (femur→ICR) transform is $\mathbf{T}_{\text{Pam}} = [\mathbf{R},\ [f_3(\theta_k),\ f_4(\theta_k),\ 0]^\top;\ \mathbf{0}^\top\ 1]$ with f_3, f_4 cubic-spline fits (ICR x: 23.30 to 10.47 mm; y: -416.65 to -395.66 mm over $\theta_k = 0.17$ to -2.62 rad), and the ICR→tibia transform is the pure translation $[f_{13}(\theta_k),\ f_{14}(\theta_k),\ 0]^\top$ (x: 29.66 to -7.58 mm; y: 25.97 to 22.33 mm), composed as $\mathbf{T}_{t1\to f} = \mathbf{T}_{t1\to\text{ICR}}\ \mathbf{T}_{\text{Pam}}^{-1}$ over 100 frames spanning $\theta_k \in [-120^\circ, +10^\circ]$ [Provenance: `buildKneeFlexorContext20mm.m` lines 34–80; `buildKneeExtContext20mm.m` lines 30–78; `Functions/ModernRobotics/RpToTrans.m`]. In the archived test sets the per-frame

transform maps tibia to femur/ICR and takes the form

$$\mathbf{T}_k(\theta_k) \;=\; \begin{bmatrix} \cos\theta_k & -\sin\theta_k & 0 & d_x(\theta_k) \\ \sin\theta_k & \cos\theta_k & 0 & d_y(\theta_k) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \tag{C2}$$

with rotation block $\mathbf{R}_z(\theta_k)$ and translation components d_x, d_y in meters. In the archived flexor test set the translation is constant across the sweep, $[d_x,\, d_y,\, 0] = [4.5,\, -395.8,\, 0]$ mm, and the rotation sweeps from $\mathbf{R}_z(-120^\circ)$ at full flexion to $\mathbf{R}_z(+10^\circ)$ at full extension (`FlxPinBPASet.mat`, kf(1)).

C.3 Bracket Reference Points and Frames

Table C5 lists the bracket reference points. Each compliant bracket carries an instantaneous body-fixed frame anchored at its reference point, constructed by two rotations (yaw about z, then pitch about the yawed y). With $\mathbf{p}_b$ the vector from the reference point to the bracket-to-BPA attachment target, the yaw and pitch angles are

$$\theta_{br} = \operatorname{atan2}(p_{by},\, p_{bx}), \qquad \phi_{br} = \operatorname{atan2}(p'_{bz},\, p'_{bx}) \quad \text{with} \quad \mathbf{p}'_b = \mathbf{R}_z(\theta_{br})^\top \mathbf{p}_b, \tag{C3}$$

and the bracket-frame pose is

$$\mathbf{R}_{kbr} = \mathbf{R}_z(\theta_{br})\, \mathbf{R}_y(\phi_{br})^\top, \qquad \mathbf{T}_{kbr} = [\mathbf{R}_{kbr},\; \mathbf{p}_{br};\; \mathbf{0}\;\; 1]\,, \tag{C4}$$

so the frame's x axis aims along the nominal load path. The angles are computed per configuration from the bracket geometry, not fixed [Provenance: `minimizeFlxPin2brk.m` lines 252–298; `minimizeExtX3.m` lines 461–483]. As executed, however, the flexor evaluator of record assembles *both* of its bracket frames from the yaw rotation alone (the pitch-only, one-transform pose $\mathbf{T}_{kbr} = [\mathbf{R}_z(\theta_{br}), \mathbf{p}_{br}; \mathbf{0}\ 1]$): insertion bracket at line 289 and origin bracket at line 305 of `minimizeFlxPin.m`, with the matching pitch-only point reconstructions at lines 318 and 331; the two-rotation assembly of Eq. (C4) is computed but left commented at lines 288, 304, and 330. For the pinned extensor the two-rotation assembly is live (transMode default `2trans`; `minimizeExtX3.m` lines 3–6 and 479–482). Force is transformed into the bracket frame by $\mathbf{RowVecTrans}(\mathbf{T}_{kbr}^{-1}, \mathbf{f}_k)$ and deformed points are mapped back by $\mathbf{RowVecTrans}(\mathbf{T}_{kbr}, \cdot)$. Because the insertion array $[X_1, X_2, X_1]$ is y-symmetric, the assembly choice is numerically immaterial there; for the origin bracket it is not (see Table C6). The pitch-only (yaw-only) variant that the earlier one-transform fit used survives in `minimizeFlxPinX3.m` lines 200–209.

Table C5: Bracket reference points (mm, as coded; divided by 1000 in code). All are author-set CAD interpretation points; frame denotes the frame the vector is expressed in.

Point	x	y	z	**Configuration / meaning**
Pbri	−48.11	−107.81	13.80	pinned-knee flexor, insertion bracket (knee-ICR frame)
Pbr2	−52.61	0	75.06	pinned-knee flexor, origin-side 2nd bracket (hip frame)
Pbr	−3.84	−46.44	62.50	pinned-knee extensor, rib midpoint, medial side (hip frame)
Pbr	8.38	20.75	25.10	biomimetic extensor 10 mm, cantilever centroid
Pbr	9.48	−33.38	33.90	biomimetic flexor 20 mm, bolt-pattern centroid
Pbr	−19.00	22.00	27.60	biomimetic flexor 10 mm, cantilever centroid

C.4 Stiffness Arrays and Compliance

Every configuration models bracket compliance as a diagonal Hooke-law spring in the bracket frame, $\mathbf{K}_{br} = \mathrm{diag}(k_1, k_2, k_3)$ newtons per meter. The scalar compliance along the force path and its series combinations are

$$c_b = \hat{\mathbf{u}}\,\mathbf{K}_{br}^{-1}\hat{\mathbf{u}}^{\top}, \qquad c_{\mathrm{eff}} = c_b + c_{\mathrm{ten}}\ (+\,c_{b2}\ \text{for the second bracket}), \tag{C5}$$

exactly as in Equations (4.8)–(4.18); when an artificial tendon is present its axial spring ($\times 2$ cables, effective steel area 1.51×10^{-6} m^2, $E = 193$ GPa) adds in series, and the force balance $F_{\max} F^*(\varepsilon^*(\delta), P^*) = k_{\mathrm{eff}}\,\delta$ is solved per frame by `fzero` on the contraction interval [Provenance: stiffness arrays: `minimizeFlxPin.m` lines 387 and 400; `minimizeExtX3.m` lines 575–584; `minimizeExt.m` line 352; `minimizeFlx.m` line 314. `fzero` force balance: `minimizeFlxPin.m` line 438; `minimizeExtX3.m` line 611; `minimizeExt.m` line 389; `minimizeFlx.m` line 352]. The per-configuration stiffness arrays and their series combinations are given in Table C6.

The wrapping-loss term X_3 (Equations 4.23–4.25) is evaluated as $X_3\,R\,|\Delta\theta|\,\zeta^2$ per contact arc. On the pinned extensor the two physical contact arcs use $R_1 = 12$ mm (27° to −23°) and $R_2 = 40$ mm, with arc-length normalizers $27.75/(ang_1 - ang_2)$ and $92.46/(ang_2 + 120^\circ)$; on the biomimetic configurations the bend measure is $w_{\mathrm{rap}}|\theta_{\mathrm{turn}}|$ with $w_{\mathrm{rap}} = 25$ mm [Provenance: `minimizeExtX3.m` lines 333–359; `buildKneeFlexorRoute20mm.m` line 268; `MonoPamDataExplicit_balanceX3.m` lines 489–494]. Table C7 records the optimizer parameterization and bounds of the adopted identification campaign: the two-bracket flexor fit solves all three terms, and the extensor refit locks (X_1, X_2) to the adopted flexor-front

Table C6: Bracket stiffness arrays by configuration ($K = [k_1, k_2, k_3]$ in bracket-frame axes, N/m; optimizer solves X_1, X_2 assigned as shown), as executed by the evaluators of record. The flexor evaluator has no frame switch: both brackets use the pitch-only assembly (see the Bracket Reference Points and Frames section above). The pinned-extensor evaluator carries a `transMode` argument (default `2trans`, env `EXTX3_TRANS`); its axis assignment was revised by the author, with the previous assignment kept commented in the file.

Configuration	$\boldsymbol{K}$ array	**Source**
Pinned flexor, single bracket (legacy)	$[X_1,\ X_2,\ X_1]$	`minimizeFlxPinX3.m` line 267
Pinned flexor, insertion bracket (pitch-only frame)	$[X_1,\ X_2,\ X_1]$	`minimizeFlxPin.m` line 387 (frame li
Pinned flexor, origin bracket (pitch-only frame)	$[X_1,\ X_1,\ X_2]$	`minimizeFlxPin.m` line 400 (frame li
Pinned extensor, 2trans (with X_3)	$[X_1,\ X_2,\ X_2]$	`minimizeExtX3.m` line 575
Biomimetic extensor	$[X_1,\ X_2,\ X_1]$	`minimizeExt.m` line 352
Biomimetic flexor	$[X_1,\ X_2,\ X_2]$	`minimizeFlx.m` line 314

values (the extensor front was generated with the pair fixed to flexor front row 77 and only (X_0, X_3) searched), enforcing a single stiffness pair across configurations.

Table C7: Optimizer parameterization and bounds of the adopted identification. Length offsets are carried in centimeters and stiffnesses on a base-10 logarithmic scale; the initial population of the flexor run was drawn from $X_0 \in [5, 15]$ mm, $X_1 \in [5 \times 10^4, 5 \times 10^5]$ N/m, $X_2 \in [7 \times 10^3, 1.5 \times 10^4]$ N/m.

Identification	**Solved terms**	**Bounds**
Two-bracket pinned flexor	X_0, X_1, X_2	$X_0 \in [0,\ 20]$ mm; $X_1 \in [3 \times 10^4,\ 10^6]$, $X_2 \in [5 \times 10^3,\ 2 \times 10^4]$
Pinned extensor refit	X_0, X_3	$X_0 \in [-20,\ +5]$ mm; $X_3 \in [0,\ 1]$

[Provenance: `minimizeFlxPin10mm_2brk.m` lines 82–83 (bounds) and 424–432 (parameterization); `minimizeExt10mmX3.m` lines 23–26 (lock load) and 67–68 (bounds); `Collect_ExtPinX3_sweep.m` lines 59–60.]

C.5 Adopted Correction-Term Values

Table C8 records the correction-term values adopted at the time of writing. The initial single-bracket identification reported with the results figures (Table 4.1) was superseded by the two-stage campaign described in Section 4.7: a two-bracket flexor identification over the four training tests of Table C1 (encoder-corrected, cross-validated), followed by an extensor refit that locks (X_1, X_2) to a member of the same flexor Pareto front and solves only the extensor-specific terms (X_0, X_3) on the extensor validation pool. The pinned-flexor data do not uniquely determine (X_1, X_2) (the cross-validation fronts contain a family of stiffness pairs of comparable fit quality), so the adopted flexor row (front row 77) and the locked extensor refit (flxr77 front row 32, generated with the pair fixed to that same row-77 pair) are two operating points tied to one and the same stiffness pair. The row-77 pair was adopted because it is the pair for which both redesign configurations meet the required 5 % torque-overshoot target simultaneously: in the adoption campaign the flexor redesign reached a minimum margin of $+5.00$ % and the locked extensor redesign $+5.043$ % at -20.2° (`Dig_out/frommat_flx_20260920.log` and `Dig_out/Opt_flxr77_20260920.log`); the refined redesigns of record improve this to $+5.01$ % (flexor) and $+5.02$ % (extensor).

Table C8: Adopted correction-term values of record. The flexor fit solves (χ_0, χ_1, χ_2); the extensor refit solves (χ_0, χ_3) with the stiffness pair locked to the flexor values.

	χ_0 (mm)	χ_1 (N/m)	χ_2 (N/m)	χ_3
Flexor	$+3.94$	3.998×10^4	1.473×10^4	
Extensor	-6.38	3.998×10^4	1.473×10^4	0.158

The flexor values are pooled-front row 77 of `minimizeFlxPin10_results_20260908_2brkt_2trans_noT`

(212 members); the extensor values are row 32 of `minimizeExt10mmX3_results_20260920_flxr77.mat`, a front generated with (χ_1, χ_2) locked to that same row-77 pair (mat fields `LOCKSRC`, `xi1lock`, `xi2lock`, `flxRow`) with only (χ_0, χ_3) searched. The redesign context builders load exactly these values: `buildKneeFlexorContext20mm.m` pick 77 at lines 171–182, with χ_3 from flxr77 row 32 at lines 188–191, and `buildKneeExtContext20mm.m` pick 32 at lines 164–171.

Table C9 reports the per-test goodness of fit at the adopted values. The corrected model improves on the uncorrected rigid model in every extensor pool test, and the fraction of variance unexplained stays below unity on every pool test except the 43 cm-tendon test (FVU 1.33, against a rigid-baseline FVU of 4.73). The three excluded tests (3, 4, 9) are not part of the fit of record. On the biomimetic extensor test ($l_0 = 51.8$ cm) the same values give RMSE 1.17 N m, FVU 0.67, and maximum residual 2.21 N m, against a rigid baseline of RMSE 3.03 N m, FVU 4.49, and maximum residual 5.59 N m; the shared-pair correction now explains the bulk of that configuration's measured torque variance as well as its level [Provenance: `Dig_out/xi_audit_tmp/reeval_gof_20260926.m` and `reeval_bio_20260926.m`, evaluator calls of record; extensor rigid baseline from the `baselineScores` field of the extensor results mat].

C.6 Redesigned Extensor Route Geometry

This section documents the geometry of the redesigned extensor route of record (Section 4.7) as saved by the optimization run of record (`Vas_Pam_20mm_Result_20260925.mat`, the run cited in Table C2). Figure C1 reproduces the optimizer's tiled route-geometry view

Table C9: Per-test torque-prediction goodness of fit at the adopted correction terms of Table C8: RMSE and FVU of the corrected model, with the uncorrected rigid-model RMSE for the extensor pool. The 47 cm flexor test is pure held-out validation.

Configuration	**Test**	**RMSE** (N m)	**FVU**	**Rigid RMSE** (N m)
Pinned flexor	48 cm	2.16	0.17	
	46 cm	1.37	0.05	
	47 cm (held out)	2.29	0.16	
	40 cm-tendon	1.61	0.04	
	41 cm	1.08	0.03	
Pinned extensor pool	40 cm	1.24	0.13	3.09
	40 cm-tendon	1.35	0.89	3.62
	43 cm	0.98	0.16	2.44
	43 cm-tendon	1.52	1.33	2.88
	46 cm	0.49	0.05	2.18
	47 cm	0.70	0.17	0.95

from that results file: eight pose panels sweep the 100-frame solution, drawing full flexion ($\theta_k = -120.0^\circ$), the solver frame immediately before each of the six contact-elimination events, and full extension ($+10.0^\circ$). The panel angles are the last poses at which the released row is still active; the release angles of Table C2 are the first poses at which each row is off, one pose spacing of 1.31° later. Beyond the release schedule, the results file also carries a top-level `removedText` field reading '`-p8`'. This field is a plotting leftover rather than a route statement: the plotting block of `Opt_run_Ext.m` (lines 730–736) computes `removedText` only as part of each interior pose-panel title, naming the point released in the next frame, and the value left in the results file is the one from the last interior panel, the p8 release-event panel at $+6.06^\circ$ in Figure C1. The route of record does contain p8: like p3, released at $+6.1^\circ$, p8 stays active from full flexion through 97 of the 100 design-grid frames until its release at $+7.4^\circ$ near full extension (Table C2), so the seed table and release

schedule document the route as run. Each panel draws the active p1:p9 route in the femur frame (tibia-side rows transformed through the ICR) over the femur-cylinder, femur-wall, corrected-condyle, and tibia clearance envelopes of the route-elimination rule, with the local bend-radius checks at p2 and p8 drawn dashed. The optimized endpoint values of this design are $\mathbf{p}_1 = [40.0,\ 137.0,\ 0]$ mm (femur frame; the square marker of Figure C1) and $\mathbf{p}_{\mathrm{End}} = [34.4,\ -197.1,\ 0]$ mm (tibia frame; the diamond marker), with resting length 827.7 mm and 36.8 mm of tendon.

Figure C2 places the same optimized route on the OpenSim bone-surface point clouds (`Open_Sim_Bone_Geometry`) at exactly zero knee angle: the full skeleton (femur, tibia, pelvis and hip, sacrum, lumbar spine, and foot) is drawn behind the three human vastus reference paths in light orange, the optimized p1-to-pEnd route is drawn in magenta with its endpoints circled, and the optimized route carries a total BPA-plus-tendon path length of 0.784 m at this pose. Animated sweeps of both redesigned routes over the full flexion range are archived with the optimization results (`Knee_Extensor_20mm.gif`, 2.3 MB, and `Knee_Flexor_20mm.gif`, 4.7 MB, in `Code/Matlab/Mesh_Optimization/Results`).

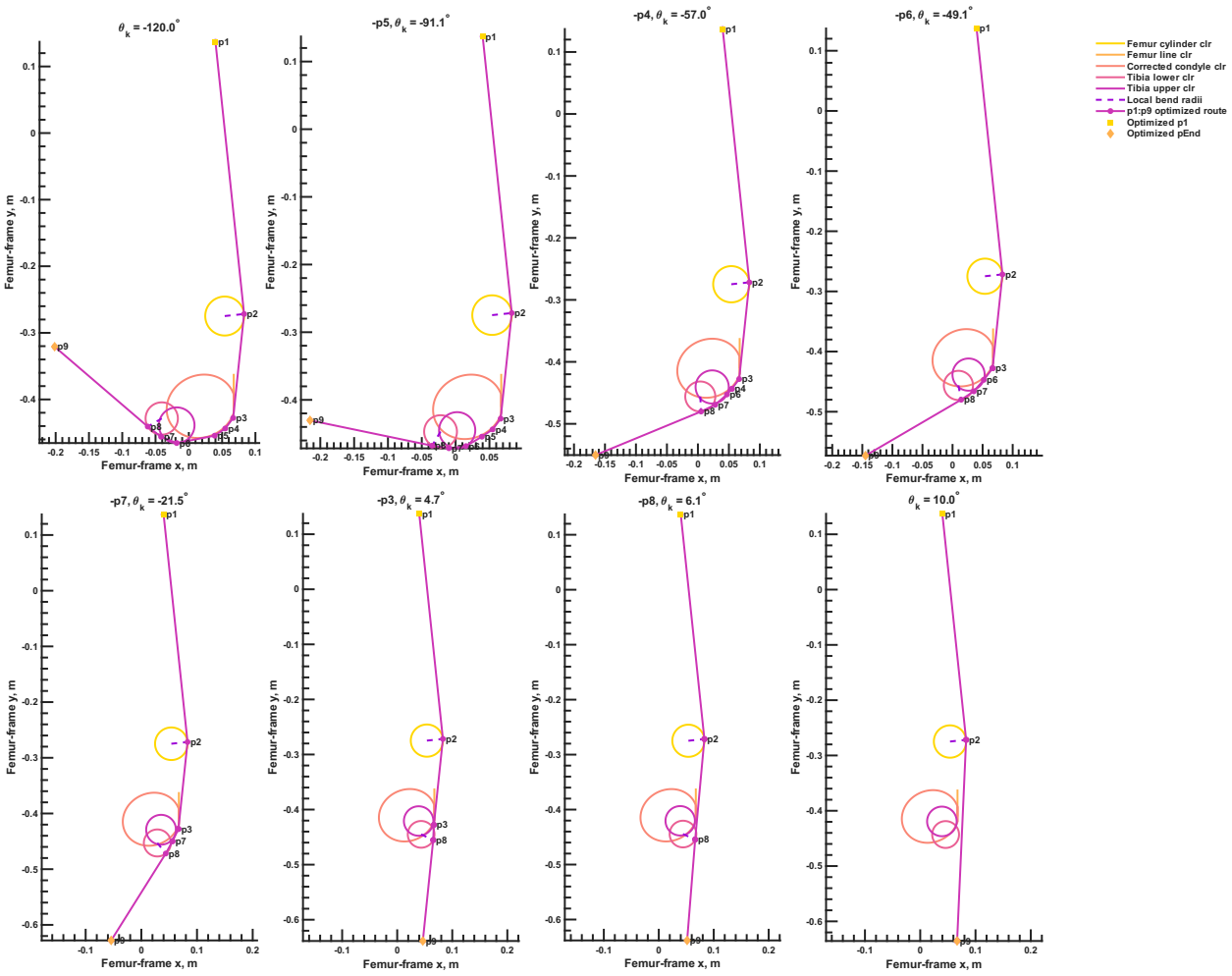


Figure C1: Eight-panel route-geometry view of the optimized 9-point extensor route of record (`Vas_Pam_20mm_Result_20260925.mat`), drawn in the femur frame. Regenerated by `Export_AppendixRouteGeo.m`, which ports the plotting block of `Opt_run_Ext.m` verbatim.

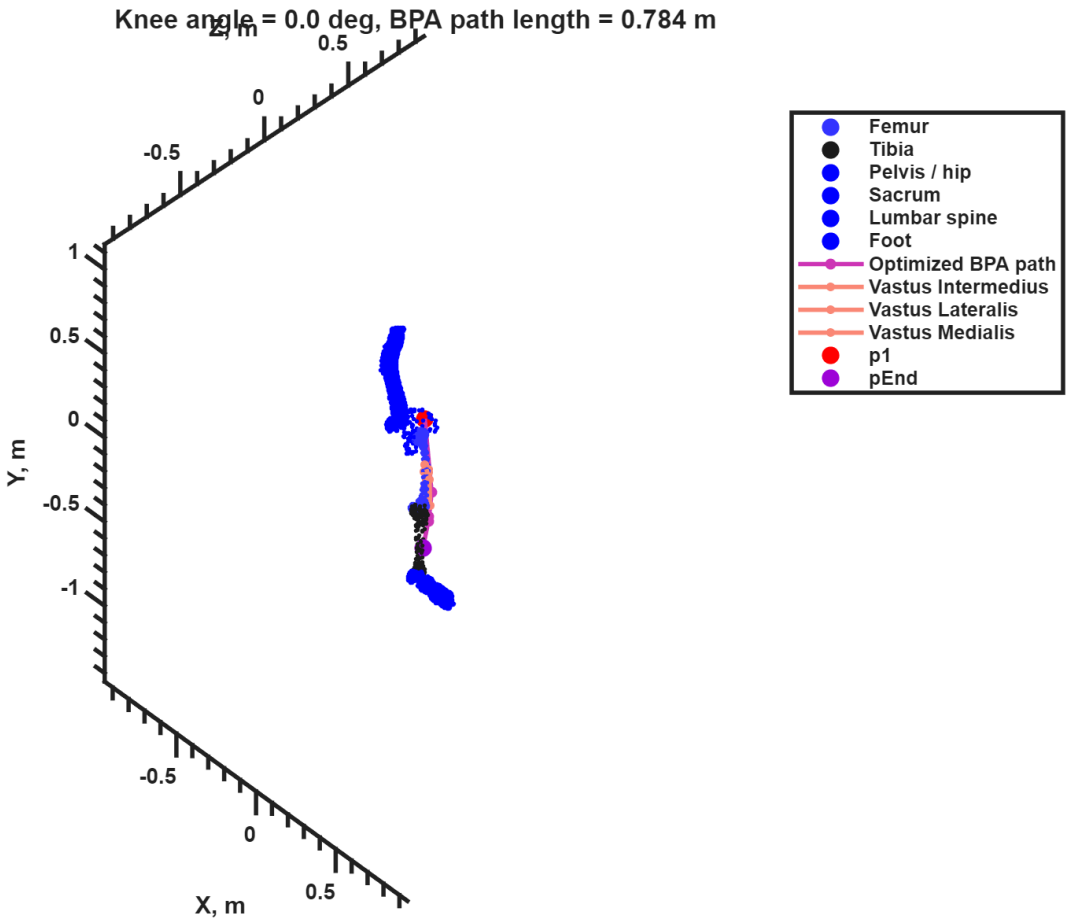


Figure C2: Static bone-mesh view of the optimized 20 mm extensor route at $\theta_k = 0^\circ$, rendered by `Export_AppendixBoneMesh.m`.